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DEPARTMENT OF MECHANICAL ENGINEERING
Accredited by NBA (Tier-I)

PROJECT- PHASE 2 (22RAP41)
On

DESIGN AND DEVELOPMENT OF DEEP LEARNING ASSISTED INTELLIGENT AUTOMATED ARECA SORTING SYSTEM

Submitted in partial fulfilment of M. Tech Degree
"ROBOTICS AND ARTIFICIAL INTELLIGENCE"

SUBMITTED BY

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UNDER THE GUIDANCE OF

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AY 2024-2025

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Certificate

This is to Certify that the project entitled **"DESIGN AND DEVELOPMENT OF DEEP LEARNING ASSISTED INTELLIGENT AUTOMATED ARECA SORTING SYSTEM"** submitted by **Mr. Sujith Kumar P H (1NT23RAI12)** bonafide student of **Nitte Meenakshi Institute of Technology, Bengaluru** in partial fulfilment for the award of MTech Degree in **Robotics and Artificial Intelligence** during the academic year **2024-25** is an authentic work carried out by him under our supervision and guidance. It is certified that all corrections/suggestions indicated by us have been incorporated in the report deposited in the department library. The project work report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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ABSTRACT

The project outlined the design and development of deep learning assisted intelligent automated areca sorting system for arecanut which integrated computer vision, deep learning and precise actuation. Automated Areca Sorting System utilized the Raspberry Pi 4 as a device to send images to a webcam that could acquire pictures in real time. The Workflow involved Gaussian noise filtering followed by a set of classifications between ripe and unripe. Developing a deep learning model and deployment on a local trust model using transfer learning with a VGG16 model which queued the images into produce categories with a resulting accuracy of 98.4%. Once classified, the classification data downstairs were passed onto an ESP32 microcontroller which actuated a three-servo mechanical sorting system consisting of a flap controller and two directional pushers for the ripe and unripe categories. The Automated system operated at machine speed, with minimal lag shown during human sorting processes thus outperforming human partners in terms of operational speed in operate a consistent drive that would eventually be able to work continuously without the requirement for fatigue management. Experimental results provided confidence that mechanically sorted produce with AI based vision processing can work in an agricultural setting and provide an efficient scalable and cost - effective solution to classification and processing of arecanuts.

Keywords: Raspberry Pi 4, ESP32, CNN, VGG16, OpenCV, Deep Learning, Gaussian noise filtering, Servo Motor, Arecanut Classification.

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Finally, I would like to express my deep appreciation to my classmates and my indebtedness to my parents for their moral support and encouragement.

SUJITH KUMAR P H

DECLARATION

I hereby declare that the project entitled **“DESIGN AND DEVELOPMENT OF DEEP LEARNING ASSISTED INTELLIGENT AUTOMATED ARECA SORTING SYSTEM”** submitted by Mr. Sujith Kumar P H to **Nitte Meenakshi Institute of Technology, Bangalore** in partial fulfilment of the requirement for the award of the degree of Master of Technology in **Robotics and Artificial Intelligence** is a record of Bonafide project work being carried out by me under the guidance of **Dr. Kiran M C**. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any degree or diploma in this institute or any other university.

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DATE:

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CHAPTER 1

INTRODUCTION

1.1 Overview

The arecanut is regarded as a seed and is not a true nut. It is classified as a fruit such as berry. Arecanut, often referred to as betelnut. When freshly harvested, the fruits outer husk is green and the seed inside is soft enough to be cut with an ordinary blade. As the fruit matures, the husk changes to a yellow or orange shade and during the drying process, the seed hardens to a sense, wood-like texture. For centuries, arecanut cultivation has been closely tied to the cultural traditions and social practices of communities across the world

Figure 1.1: Areca Plantation

Ancient Indian and Chinese manuscripts demonstrate the use of Arecanut is more than two thousand years old. The practice of chewing Arecanut with betel leaves, lime, and other flavoring agents have developed into a seemingly solid feature of ritual and social practices. The figure 1.1 is our own Areca Plantation in Sorab, Shimoga district. The arecanut is used around the world by humans for its many uses in numerous industries. The consumption of the raw arecanut is primarily conducted by people chewing the nut for its stimulating effects, which include boiling, drying or fermentation. Both Ayurvedic medicine and other traditional systems of medicine affirm that arecanut be used to aid digestion and also has a medicinal usage as well. The industries using areca nut extraction use herbal products for human consumption and also in textile dyes and natural tanneries. The wellbeing of millions of farmers, traders and workers across the global regions of India, and Southeast Asia entirely rests on areca nut cultivation and production. [120].

Areca nut was initially cultivated within the Kyasanuru Seeme area of Shimoga district, Karnataka, India [120]. Currently in the Malnad region the farmers used to cultivate the same variety of arecanut, which gives high yielding production and more economic support to the cultivators. The arecanut crop holds the significant markets trends, specifically in the region of Coastal like Dakshina Kannada, Udupi and Goa. Southern part of India and Karnataka such as Shimoga, Chikmagalur, Kodagu, Uttar Kannada, Hassan and some areas. Northeastern states like Nagaland, Assam, Meghalaya, where it major plantation yield. In worldwide, India is the leading seller and purchaser of arecanuts. Seventy to eighty percent of India's arecanut production comes from Karnataka, Where Karnataka only contributing ten lakh tonnes of arecanut.

Figure 1.2: Field View of Arecanut Plantation from our Cultivated Land

Figure 1.3: Unripe Arecanut in Plant

Convolutional neural networks (CNNs) along with deep learning models perform feature extraction and classification tasks which allow instantaneous detection of quality parameters and statistical quality control to streamline and improves the arecanut sorting process. A convolution neural network (CNN) is used in this approach to extract the color and texture features. Color-related characteristics are essential for identifying maturation levels and general quality categories in raw arecanut sorting. Accurate classification and dependable sorting performance are made feasible by deep learning algorithms and tailored image processing techniques.

To improve quality control, arecanut recognition and sorting incorporates machine vision, image processing, and learning. The automated approach uses Convolution Neural Networks to analyze arecanut photos and determine their quality. This makes it possible to intervene promptly and maintain sorting accuracy throughout the process.

1.2 An Overview of Arecanut: Economic Value and Production Statistics

Arecanut has a very high commercial value, primarily in India and Southeast-Asia. The contribution of arecanut to the social and economic livelihoods of millions of farmers, traders and workers is immense in the form of multi-thousand crore rupee industries that include cultivation, processing, packaging, marketing and export. The below figure 1.4 of areca nut production and production volume by nation in 2022 shows India dominates global production, having taken almost 6799% of world production at 167 billion tonnes. This places India in a lead position ahead of all other producers. Next is Bangladesh with approximately 13.62% of world production approximately 265,000 million tonnes and Myanmar with approximately 657% of the world production (156,000 million tonnes). Indonesia with 236,750 million tonnes and Sri Lanka with 333,720 million tonnes are substantial production levels. Thailand, Nepal, Bhutan, and Malaysia also produce almost <1% quantity of world production which are all negligible production quantities. Areca nut production by country provides evidence that India is the major element of the areca nut in international market, as it supports the distinction of being a backbone of the areca nut production in both quantity and position.

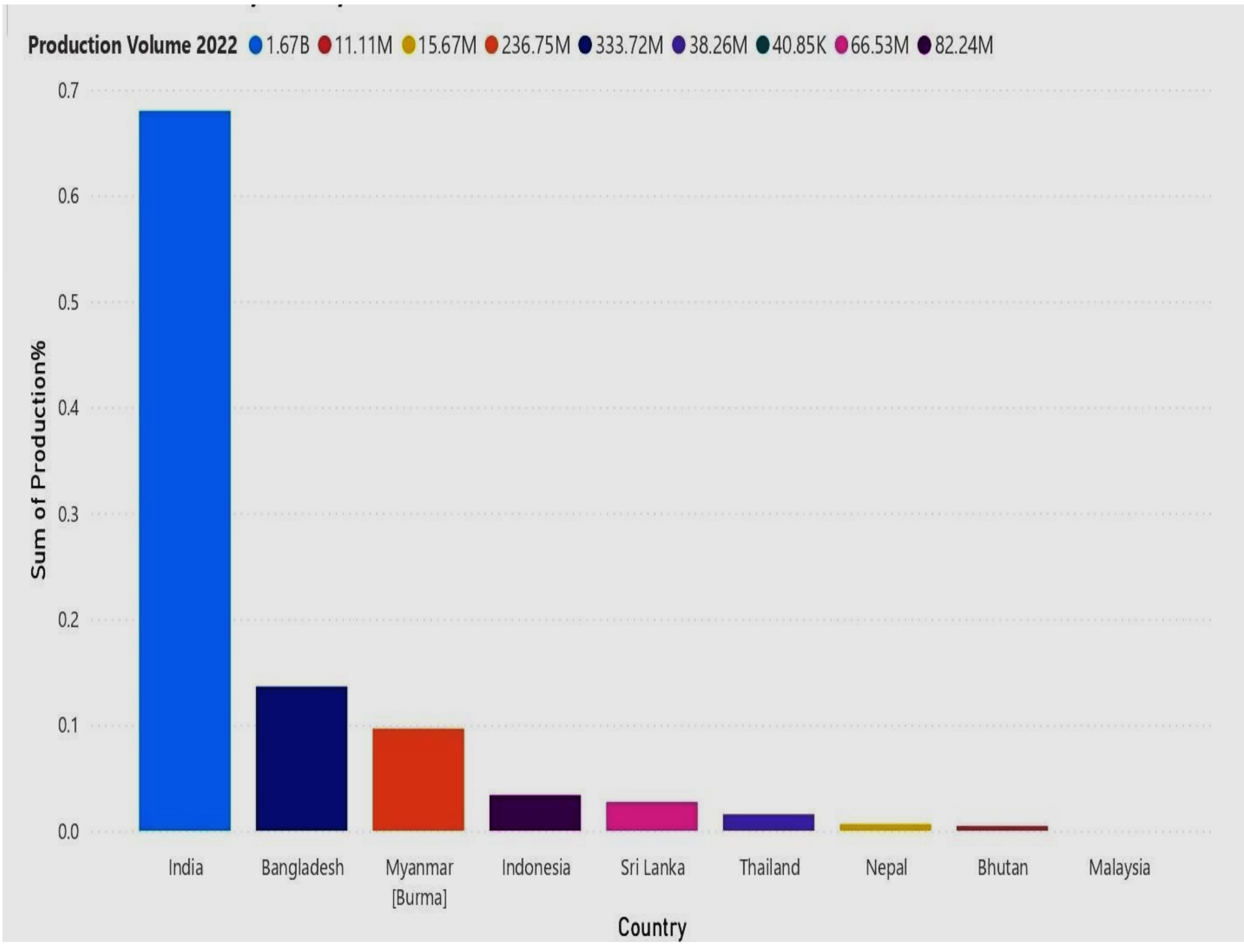


Figure 1.4: Arecanut Global Production Share by Country

1.3 Global Markets: Arecanut Imports and Exports

In 2023-2024, India exported nearly 11,800 metric tons of areca nuts, which was an increase from around 6,000 metric tons exported in the previous year. This approximately ₹415 crores export value. The leading importing countries of Indian arecanuts are Vietnam, Nepal, United Arab Emirates, Bhutan and Sri Lanka. India exports two varieties of areca nuts like red arecanuts produced by boiling green nuts and hull it and white arecanuts made from sun- dried fully mature nuts. Although India is one of the largest producers, areca nuts imports are being made to help fulfill the domestic demand. In 2024, India imported about 30,271 metric tonnes which was a 60%

decrease over the previous year, with total imports with around ₹1000 crores. The major exporting countries to India were Bangladesh, Indonesia, Myanmar, Vietnam, and Sri Lanka.

Figure 1.5: International Arecanut Trade - Import and Export by country

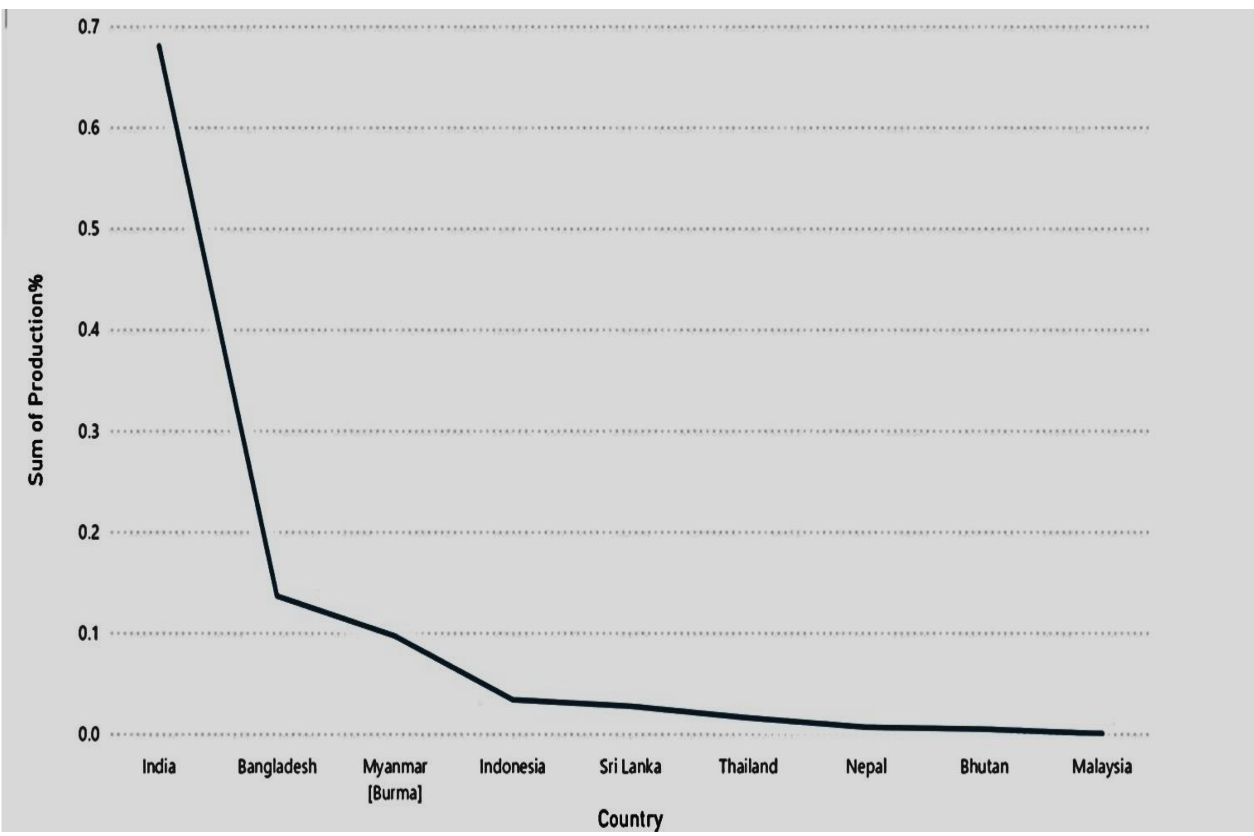


Figure 1.6: Country-wise Arecanuts Share of Total Production (2023-2024)

1.4 An Overview of Arecanut Cultivation Areas and Genetic Varieties

Areca nut is the most dominant crop in Karnataka followed by Kerala and Assam. The areca nut crop is largely endemic to the Malnad and coastal areas where the agroclimatic conditions are highly favorable.

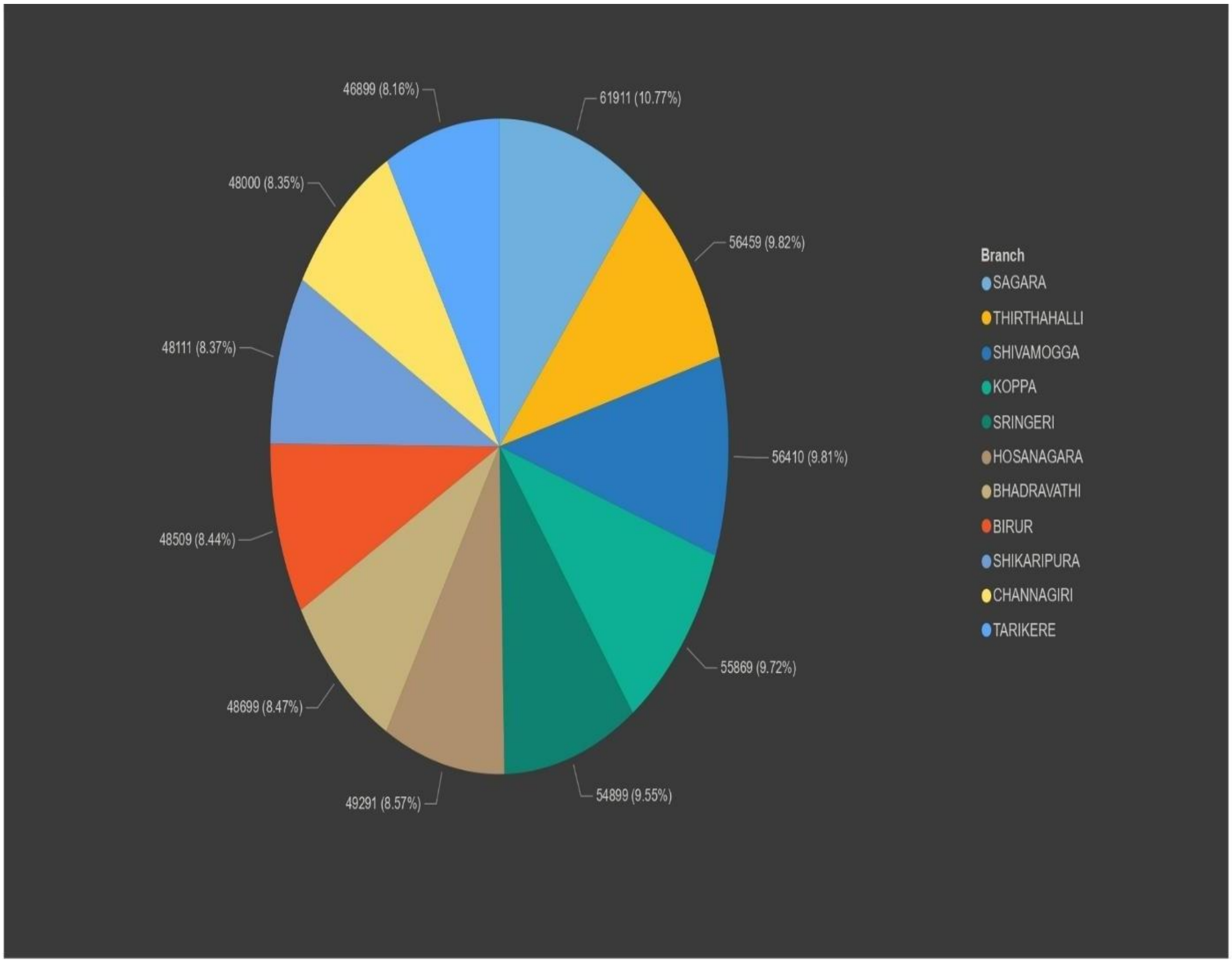


Figure 1.7: Branch-wise Average Price of Arecanuts

Considering Shivamogga and Chikmagalur districts markets centers mentioned in figure 1.7 above statistical chart which depicts the total average price of areca nuts by branch for selected market centers. The Shivamogga branch had the largest share of average price distribution at about 29.68%, next is Sagara at 11.88%, Tirthahalli at 11.06%, Sringeri at 10.24%, Koppa at 10.15%, and Hosanagar at 5.67%. Channagiri at 5.4%, Birur at 5.26%, then Bhadravati, Shikaripura and Tarikere were less than 5% each. The distributed share of average price illustrates how certain market branches by price location activity can influence the overall average trend for arecanut prices in Karnataka. Major producing districts for arecanut in Karnataka are Shivamogga, Chikmagalur, Uttara Kannada, Dakshina Kannada and Kodagu. Areca palm can be cultivated profusely in Karnataka where the temperature varies between 15°C and 38°C. The figure 1.8 describes Areca palm thrives in laterite soil or red loamy soils where there is high organic matter and good drainage. The major cultivated varieties of areca nut in Karnataka are MangalaSaemi, Sumangala, South local, Sreemangala, Mohitnagar, SAS-I, Thirthahalli, Sreevardhana.

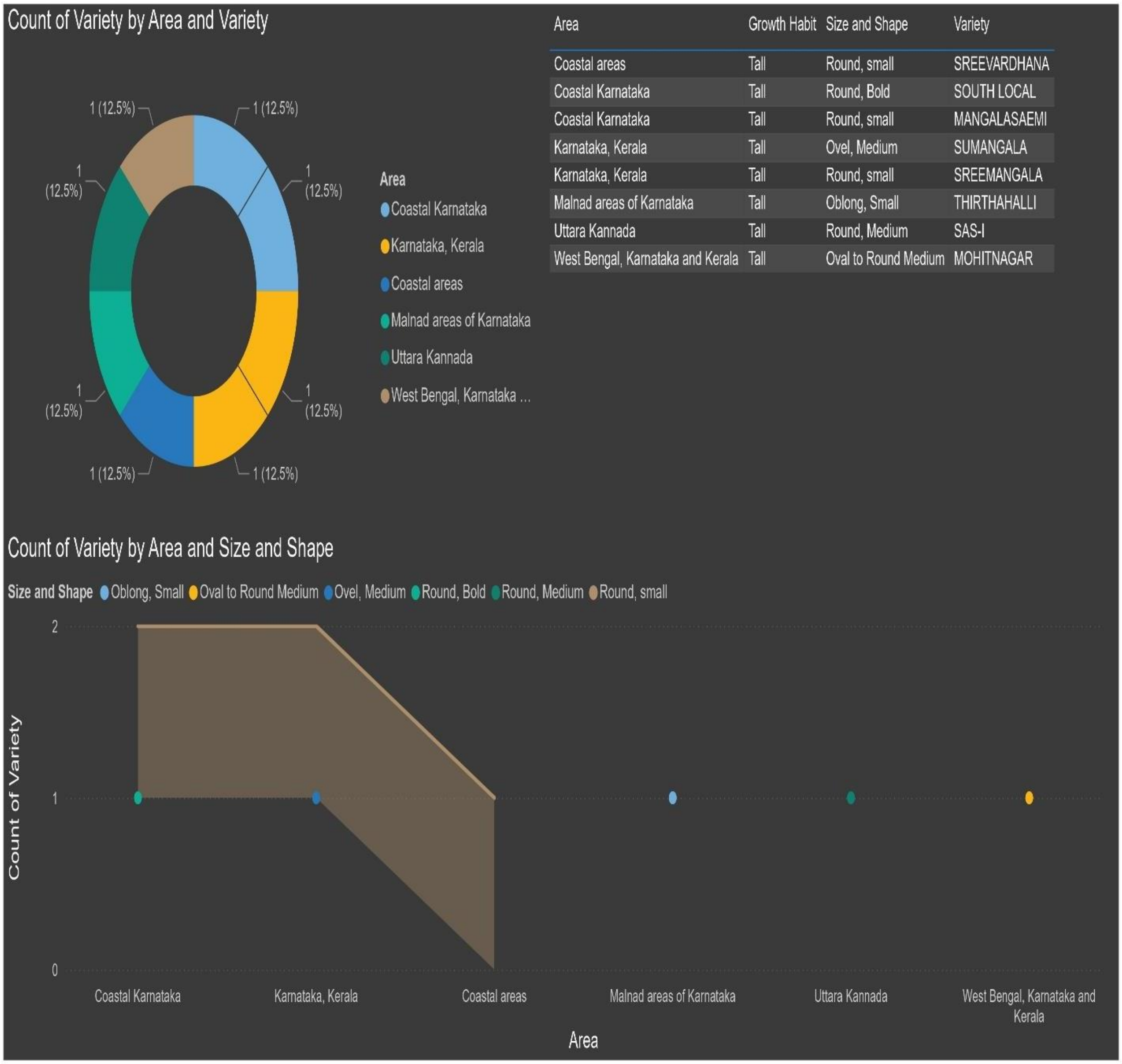


Figure 1.8: Distribution of Arecanuts Varieties Categorized by Area, Size and Shape

Statistical market pricing survey in 2024, The figure 1.9 provides comprehensive information on the Sum of Average Prices, Maxtrade Prices, and Mintrade Prices of areca nuts at different market branches in Karnataka. Shivamogga has the highest average price of ₹41,843,724 with maxtrade and mintrade amounts of ₹44,742,428 and ₹34,624,179 respectively. Sagara has an average price of ₹16,600,540, maxtrade of ₹17,642,503 and mintrade of ₹15,619,573. Other larger branches such as Tirthahalli, Sringeri and Koppa all have similar strong average price levels ranging from about

₹14 to ₹15 million, while their respective maxtrade and mintrade prices were consistent with volumes throughout the production year shown in below figure 1.9. Branches, such as Hosanagara, Channagiri, Birur, and Bhadravathi have lower price levels which indicates regional variation in either market demand or quality. In summation, the data reflects price stability and competing trade volume across principal arecanut markets.

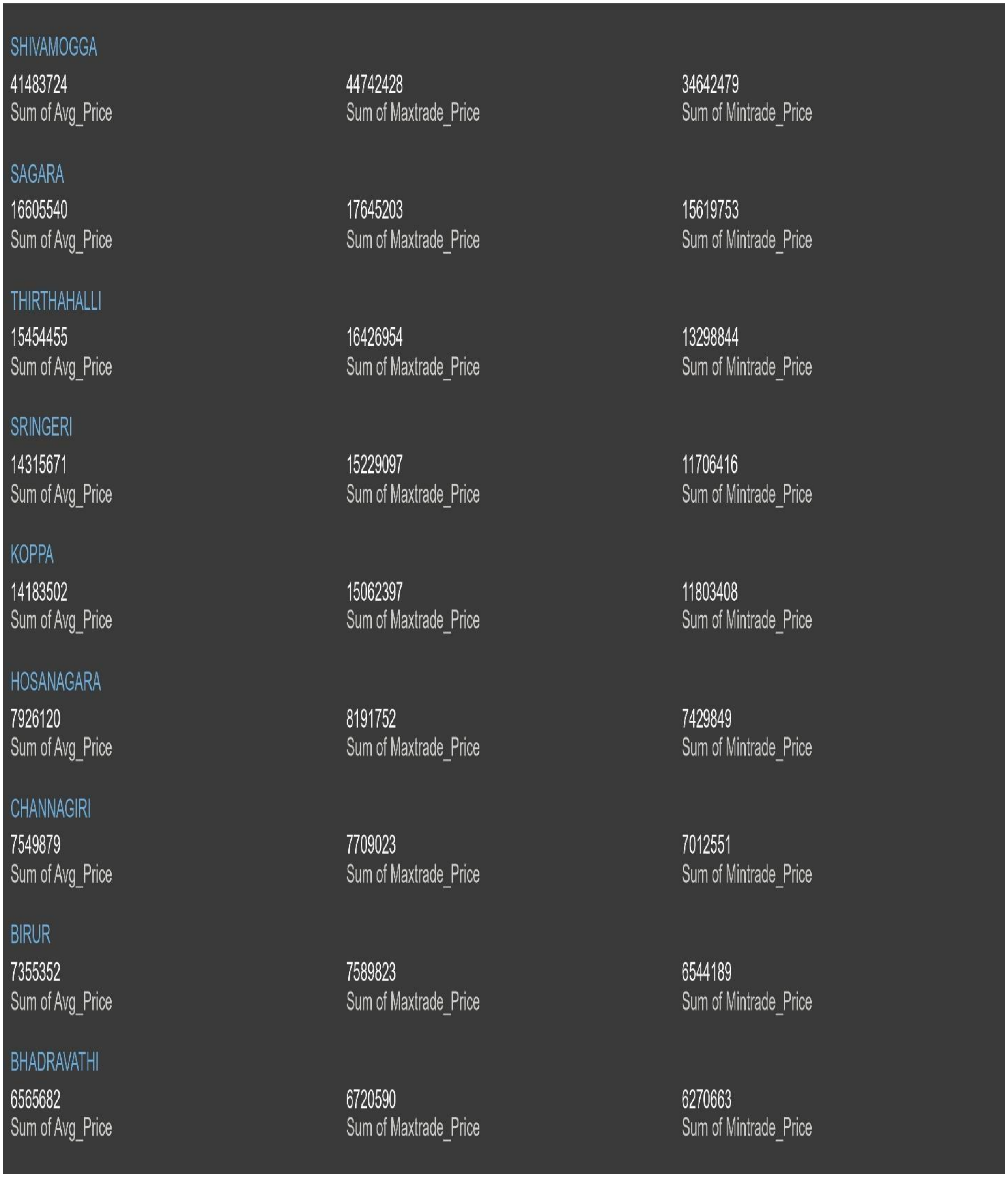


Figure 1.9: Comparative Statistical Analysis of Market Prices in Shivamogga and Chikmagalur

1.5 Manual Sorting and Chopping of Arecanut

In the manual segregation process, freshly harvested areca nuts are typically placed on a flat table or working surface and skilled workers visually inspect each nut and segregate based on quality aspects of maturity stage, husk color, size and signs of fungal infection/insect damage. Totally manual and human effort which relies on operator experience, ability to visually discriminate among nuts and physical endurance. Additionally, Manual sorting is a slow process and can span several days with batches of nuts harvested mostly during harvest season. A skilled worker typically works at a rate of 15 - 25 kg/hour and lower numbers can be expected when nuts are dirty, together or mixed in maturity. For a batch of between 100 kg, the amount of time required to process by two to three workers, not accounting for breaks and setup time is typically 4 - 6 hours. The workers chop the nut manually with hand-driven cutters or small bench-mounted blades. Because of the hardness of the areca nut, each cut requires some effort. This effort usually slows the workers down and contributes to fatigue. The chopping rate is typically at a range of 10– 12 kg/hour however, that chopping 100 kg of nuts could take 6–8 hours with 2–3 worker. There is also an increased risk for repetitive strain injuries and accidental cuts when participating in work with long hours.

1.6 Automatic Arecanut Sorting and Chopping

The automatic system replaces the process of manual inspection using a machine vision system that consists of a Raspberry Pi or industrial computer, coupled with an AI- trained camera module. Areca nuts are transported on a conveyor belt in a single layer where the camera takes high- resolution images of the nuts. The system incorporates a deep learning classification model to identify features like color (maturity) or visible damages so that system can classify the nuts instantaneously. Pneumatic pistons (or mechanisms such as servo-driven gates) redirect each nut into the appropriate container bin for further handling. The system negates human error associated with subjective judgement and fatigue and overall classification accuracy is 90-98%. The system can handle 1-2 nuts per second. Sorting a 10 kg sample of Areca nuts in a batch can take only 15 to 30 minutes with only one operator for monitoring and supervision only.

For automatic chopping, the nuts are introduced into a high-speed motor-driven cutting system with adjustable slicing thickness. The system operates on precision timing to align the nuts with the blade and provide uniform cuts at 200-300 kg/hour. It takes only 20-30 minutes to chop a 100 kg load and without fatigue for the operator. More importantly, uniform slicing is a benefit for drying efficiency and limiting mold-related complications, while leading to an even more uniform product for sale. Safety concerns are also considerably reduced as operators will not need direct contact with blades.

1.7 Comparison of Manual Versus Automatic Systems

The Figure 1.10 explains differences between manual and automatic systems are significant and related to speed, accuracy, labor cost and safety. For instance, under a manual system, sorting and chopping 100 kg of nuts could take 8 - 10 total processing hours with multiple workers. An automated system would require less than one hour, including both sorting and chopping, under a fast timeline with very little labor associated with processing and no need for supervision. The accuracy of manual methods is 70 - 80%, due primarily to human error and fatigue, while the accuracy for artificial intelligence vision system methods would be closer to 90 - 95%. Ultimately, more accuracy and uniformity mean a higher value product which is understand the buyer of processed nut products needs to be willing to pay for speeding the machine versus manual processes. Regarding labor costs, automating your operation significantly reduces recurring labor expenses. Humans require staff while a machine is a one-time investment. The payback period in labor savings on the machine is short for most large-scale operations, often occurring because of higher throughput and less waste. Worker safety is improved with a reduced permutation of worker tools, and we generated no strain on the worker with repetitive tasking.

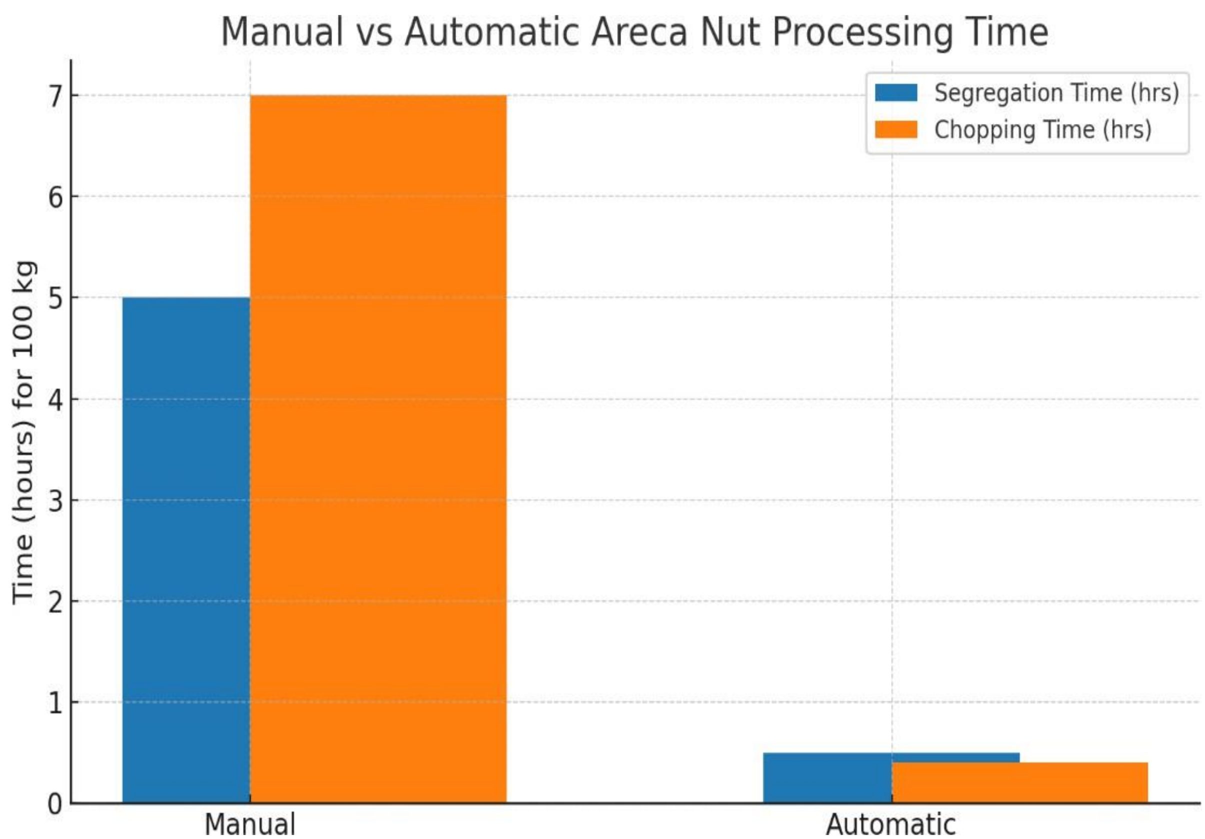


Figure 1.10: Comparative Analysis of Processing Time - Manual and Automatic Arecanut Methods

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Literature Review

Arjun et al. [1] In the present investigation, Ground robotic technologies, in particular, have advanced significant to improve the weed management in agriculture, countering the limitations of conventional herbicide applications. Also, infrared and thermal imaging use the heat signature of weeds to detect them. AI navigation systems are, therefore, also being utilized in autonomous and unmanned ground vehicles for selective weeding.

Khaled et al. [2] The modelling techniques for bell-pepper sorting uses machine vision combined with advanced image processing and artificial intelligence to boost both accuracy and efficiency in automation sorting systems. This method uses high-resolution cameras and multispectral imaging to collect comprehensive visual information about bell peppers including their color, size, shape, and surface defects. Convolutional neural networks along with deep learning models perform feature extraction and classification tasks which allow instantaneous detection of quality parameters.

Ajit et al. [3] In the present investigation, the combination of Three Sigma controls and Support Vector Machine classification integrates machine learning with statistical quality control to automate and optimize arecanut sorting. Using the Three Sigma control method quality thresholds are established through analysis of size, shape texture and color variations in arecanuts to maintain consistent grading standards.

Yonghua et al. [4] The paper combines deep learning techniques to enhance real time fruit classification and grading. The convolutional neural network (CNN) extracts spatial features such as color, texture, and defects from citrus images, while network processes sequential data to improve classification accuracy by considering temporal dependencies.

Jiezheng et al. [5] In the present investigation, A weak allele of TGW5 has been identified as a key factor in enhancing seed propagation and enabling efficient size-based seed sorting for hybrid rice production. Advanced imaging and machine learning-based sorting techniques leverage this genetic trait to achieve high precision in seed grading.

Hengquan et al. [6] The transfer learning- based betel nut classification method leverages a pre-trained deep learning model to improve the accuracy and efficiency of automatic classification. By using a convolutional neural network pre- trained on a large dataset, the model extracts key features such as texture, color, and surface defects from betel nut images.

Siddesha et al. [7] In the present investigation, In the classification of raw arecanut images, features pertaining to color are of prime importance in quality grade discrimination. These include color histograms, mean and standard deviation of RGB or HSV color spaces and texture-based color variation which could be used in defect identification, ripeness determination, and overall quality assessment.

Anthony et al. [8] In the present investigation, as per the main features in the concept robots for new sustainable crop production systems, these include automation, precision, and resource efficiency. Thus, modern agricultural robots would incorporate AI, machine learning, and advanced sensors in real-time monitoring of crops, assessing their health, and intervening precisely in case of need through special applications such as precision seeding, weeding, and harvesting.

Emmanuel et al. [9] Deep reinforcement learning uses a lightweight vision model that enhances efficiency in sequential robotic object sorting through adaptive decision- making under low computational burden. The lightweight vision model, most likely built upon efficient CNN architectures, guarantees presenting and detecting objects in real time while keeping processing latency at a minimum. Deep reinforcement learning will optimize sorting strategies through interaction with the environment.

Agustami et al. [10] In the present investigation, the goals of designing and fabricating a cutting machine for fresh betel nut are efficiency, accuracy, and user safety. The machine incorporates sharp stainless-steel blades which are effective in reducing clean and uniform cutting, manual efforts, and time. A motor or a manual mechanism ensures smooth operation of the machine which does not damage the nut.

Akshay et al. [11] Identification and categorization of areca nut diseases focus on the symptoms of discoloration, spotting or fungal infections, and are achieved through specialized image processing and machine learning techniques. Important aspects are high resolution images capture, contrast enhancement in preprocessing, and color, texture, and shape-based feature extraction.

Madhu et al. [12] In the present investigation, identifying infections by symptom on the leaf and nut for areca nut is done using Convolutional Neural Networks, which utilizes deep learning for precise classification of infections. The model is trained on labelled datasets that help improve accuracy as well as identify different types of diseases.

Kuo et al. [13] In this work, the identification and categorization of areca nuts combines machine vision, image processing, and AI to automate quality control. Its key features are high- res imaging, effective noise filtering in the preprocessing stage, and color, texture, and shape-based feature extraction. SVM and CNNs classify nuts considering their sizes, maturity, and defects, which allows grading without variations in quality.

Prathibha et al. [14] The development of a rapid, efficient, and cost-effective screening technique for testing arecanut against Phytophthora meadii, the causative agent of fruit rot disease, aims to enhance disease management and resistance breeding. Technique should provide quick and reliable detection of pathogen infection, allowing early intervention.

Agustami et al. [15] The development of a chopper for areca nut fronds with performance optimization using Response Surface Methodology focuses on enhancing efficiency, productivity, and cost-effectiveness. The chopper is designed to process fronds into smaller, uniform pieces for easier disposal or use as organic mulch. RSM is employed to analyze and optimize key parameters such as blade speed, cutting angle, and feed rate to achieve maximum chopping efficiency with minimal energy consumption.

Sunil et al. [16] The development of a software interface for AI-driven weed control in robotic vehicles focuses on real- time weed detection, classification, and targeted removal. The system integrates advanced computer vision and machine learning algorithms to differentiate weeds from crops with high accuracy. A user-friendly interface enables monitoring and control, providing data visualization and performance analytics.

Manohar et al. [17] The development of a deep learning-based disease detection system for areca nut aims to provide accurate and automated identification of infections such as fruit rot, leaf spot, and bud rot. Using convolutional neural networks (CNNs), the system analyzes images of areca nut plants to classify and detect disease symptoms at an early stage. The model is trained on a diverse dataset to ensure robustness across different environmental conditions.

Arun et al. [18] Disease detection in arecanut using Convolutional Neural Networks (CNNs) leverages deep learning for accurate and automated identification of plant diseases. The system processes images of arecanut leaves, fruits, and stems to detect diseases like fruit rot, leaf spot, and bud rot with high precision. CNN models extract key features from images, enabling early diagnosis and reducing dependency on manual inspections.

Dhanush et al. [19] Enhancing arecanut farming profits through technological advancements involves a CNN-based approach for efficient grading and sorting. The deep learning system automates the classification of arecanut based on quality parameters such as size, shape and color, ensuring uniform grading. Convolutional Neural Networks (CNNs) accurately analyze images of arecanuts, reducing manual labor and subjectivity in quality assessment.

William et al. [20] Field-based multispecies weed and crop detection using ground robots and advanced YOLO models employs a data and model-centric approach for precision agriculture.

System integrates real-time image acquisition and deep learning-based object detection to accurately distinguish multiple weed and crop species in diverse field conditions.

Yuchen et al. [21] Identification and sorting of impurities in tea using spectral vision leverages advanced imaging techniques to enhance quality control in tea processing. The system uses hyperspectral or multispectral imaging to detect impurities such as stems, dust and foreign particles based on their spectral signatures.

Ellen et al. [22] Machine vision-based algorithms for detecting sunburn in pomegranates enable automated sorting in post-harvest processing. Using advanced image processing and deep learning techniques, the system analyzes color, texture, and surface irregularities to accurately identify sunburned fruits.

Parnian et al. [23] The study explores consumer behavior in plastic food packaging waste sorting through focus group discussions in Germany. Examines awareness, attitudes and challenges faced by consumers in correctly sorting plastic waste. Key insights include the impact of labelling clarity, recycling knowledge, and convenience on sorting behaviours.

Bhojaraja et al. [24] Mapping age-wise discrimination of arecanut crop water requirements using hyperspectral remote sensing enables precise irrigation management and resource optimization. The above technique utilizes hyperspectral imaging to capture crop reflectance variations, assessing water stress levels across different growth stages.

Adnan et al. [25] A multi-scale and multi-receptive field-based feature fusion approach enhances the robust segmentation of plant diseases and fruits using agricultural images. The indicated method integrates features extracted at different scales and receptive fields, improving the model's ability to detect variations in texture, color, and shape.

Asha et al. [26] Multispectral sorting of maize kernels based on visibly high-risk kernels sourced from another country effectively reduces fumonisin contamination and toxigenic Fusarium presence. Indicated technique utilizes multispectral imaging to detect and remove infected or damaged kernels based on their spectral signatures.

Parichat et al. [27] Near- Infrared Spectroscopy (NIRS) offers a non-destructive and efficient alternative for sorting table grapes based on Seedlessness. The technology enables rapid, accurate, and automated sorting without damaging the fruit, improving processing efficiency and reducing manual labor.

Bharadwaj et al. [28] The paper discusses the extraction of color and texture features from arecanut images, utilizing techniques such as HSV color space conversion and Gabor transforms to capture essential characteristics. Lastly, the paper emphasizes the potential of these automated systems to enhance grading efficiency and accuracy, reducing reliance on manual labor and minimizing subjectivity.

Abhin et al. [29] The paper employs the Biomod2 ensemble platform to assess the impact of climate change on arecanut cultivation across India. Projections indicate a potential decline in areas classified as 'very high' and 'high' suitability, particularly in Karnataka, which currently accounts for over 50% of arecanut cultivation.

Mansour et al. [30] The paper on non-destructive sorting techniques for pepper using odor parameters explores the use of volatile compound analysis for quality assessment. The study evaluates the effectiveness of odor-based sorting in distinguishing between different ripeness levels and quality grades.

Nishat et al. [31] The study integrates deep learning with Explainable AI (XAI) using Convolutional Neural Networks (CNNs) and the LIME framework to detect chicken meat freshness. A robotic arm sorts meat in real-time based on freshness predictions, processing over 1,000 samples efficiently.

Shanuka et al. [32] The for mentioned paper explores advancements in automating the sorting of construction and demolition waste using artificial intelligence (AI) and robotics. Key insights include the emphasis on enhancing AI for precise waste categorization, addressing challenges such as the scarcity of publicly available datasets through techniques like data augmentation and transfer learning.

Dhanesh et al. [33] The paper explores the method to segment arecanut bunches by leveraging the Hue, Saturation, and Value (HSV) color space. The above approach aims to facilitate the automatic determination of arecanut maturity levels, which is crucial for optimizing harvest timing and maximizing profitability.

Ajit et al. [34] The paper introduces a method that utilizes color space transformation, statistical modelling, and classification techniques to enhance arecanut sorting. The study converts RGB images into the color space to better isolate relevant color components for segmentation.

Deepthi et al. [35] The study presents a sustainable approach to synthesizing activated carbon from arecanut husk biomass. This method leverages the inherent potassium enrichment in arecanut husks, facilitating in-situ activation during carbonization.

Siddesha et al. [36] The paper investigates the use of texture analysis for classifying arecanuts, employing methods such as Gabor wavelet transformations and Local Binary Patterns for feature extraction. Classification tasks like Support Vector Machine and K- Nearest Neighbors are utilized, achieving classification rates up to 91.43%.

Tomasz et al. [37] The paper explores the integration of machine vision and deep learning to enhance the accuracy and efficiency of scrap sorting processes. Using an RGB camera and a convolutional neural network, the system detects prohibited and hazardous elements-based metal scrap, ensuring the quality and safety of recycling operations.

Franklin et al. [38] The paper introduces a novel approach to estimating the volume and mass of baobab fruits using computer vision and machine learning techniques. By capturing single-view images with readily available cameras, such as those in smartphones, the study achieved over 97% accuracy in detecting and segmenting the fruits.

Pallavi et al. [39] The paper presents a methodology for assessing dragon fruit quality through machine learning. A Raspberry Pi is utilized to count the total number of fruits in a collection and to categorize them based on maturity levels using the aforementioned machine learning algorithms.

Abdul et al. [40] The paper introduces a comprehensive collection of 3,004 pre-processed images encompassing four distinct date fruit varieties: Aseel, Fasli Toto, Gajar, and Kupro. This structured dataset is organized into eighteen directories, facilitating detailed analysis and the development of intelligent grading systems.

Wenbo et al. [41] The paper introduces an innovative approach to classifying various vegetables and fruits by integrating diffusion maps with machine learning algorithms. Initially, images undergo preprocessing steps, including Gaussian filtering, grayscale conversion, and binarization, to enhance image quality. Subsequently, statistical texture features are extracted from these processed images.

Marcelo et al. [42] The paper presents a non-destructive approach to evaluating blueberry quality by integrating image processing with machine learning techniques. High- resolution RGB images of blueberries were captured using a mobile device, and spectral bands were extracted from these images.

Munirah et al. [43] The paper presents an innovative approach to enhancing the accuracy and efficiency of oil palm fruit grading. The study employs Fuzzy C-Means clustering for image segmentation, allowing for precise identification of fruit regions by assigning membership degrees to each pixel.

Seung et al. [44] The paper on deep learning-based early detection of Botrytis cinerea infection in strawberries using hyperspectral fluorescence imaging focuses on leveraging spectral and spatial features for accurate classification. Key aspects include preprocessing hyperspectral fluorescence images to enhance relevant features, applying feature extraction techniques.

Wenchang et al. [45] The paper on the design and development of a high-speed sorting system based on machine vision guiding focuses on integrating real-time image processing with automation for efficient object sorting. Key features include high-speed cameras for capturing images of objects in motion, image preprocessing techniques like filtering and edge detection, and machine vision algorithms for object recognition and classification.

Bankole et al. [46] The paper on the model design and simulation of an automatic sorting machine using a proximity sensor highlights the integration of sensor-based detection with automated sorting mechanisms. Key features include the use of proximity sensors to detect and classify objects based on material or size, a microcontroller or PLC for real-time decision-making, and an actuator mechanism for sorting.

Schutter et al. [47] Key features include algorithm-based scheduling to minimize bottlenecks, optimized conveyor and sorting mechanism design for high-speed processing, and real-time tracking using sensors and machine vision. The study explores load balancing techniques, predictive maintenance strategies, and automation improvements to reduce downtime.

Zeinab et al. [48] The paper on machine learning-based automated waste sorting in the construction industry focuses on enhancing efficiency and sustainability through AI-driven classification. Key features include the use of machine learning algorithms to identify and categorize different types of construction waste, sensor integration for precise material detection, and automation mechanisms for real-time sorting. The study presents a comparative case analysis, evaluating the competitiveness of machine learning-based sorting against traditional methods in terms of accuracy, speed, and cost- effectiveness.

Calliane et al. [49] The paper on machine learning methods applied online for early jam detection in a waste sorting plant focuses on real-time monitoring and predictive maintenance to prevent system disruptions. Key features include sensor data collection from conveyors and sorting mechanisms, machine learning algorithms to identify early signs of jams, and automated alerts or corrective actions to minimize downtime. The system processes data continuously, enabling proactive intervention and optimizing operational efficiency.

Ainara et al. [50] The paper on Raman spectroscopy integrated with machine learning for industrial sorting of Waste Electric and Electronic Equipment plastics highlights advanced material identification for efficient recycling. Key features include the for non-destructive chemical analysis, machine learning algorithms for precise classification of different plastic types, and real-time sorting automation to enhance recycling efficiency.

Meizhen et al. [51] The paper on the carbon footprint impact of waste sorting in municipal household waste treatment system analyzes how effective sorting reduces environmental impact.

Key features include life cycle assessment to evaluate carbon emissions at different waste management stages, comparison of sorted versus unsorted waste treatment in terms of energy consumption and emissions, and the role of recycling and waste-to-energy processes in reducing the carbon footprint.

Abderrahim et al. [52] Key features include AI-driven waste classification using machine learning and computer vision, IoT-enabled smart bins with real-time fill level monitoring, and data-driven route optimization for waste collection to reduce costs and emissions. It also highlights automated sorting using robotics and sensor-based detection, predictive maintenance for waste management systems, and the integration of cloud-based analytics for improved decision-making. The study emphasizes how AI and IoT can enhance sustainability, efficiency, and environmental impact in urban waste management.

Aberge et al. [53] The paper on the prototype of an AI-powered assistance system for the digitalization of manual waste sorting focuses on enhancing efficiency and accuracy in waste classification. Key features include AI-driven image recognition using deep learning models, real-time feedback for workers through augmented reality or display interfaces, and sensor integration to improve sorting accuracy. The system assists manual sorting by providing instant classification and guidance, reducing human error and increasing sorting efficiency.

Howard et al. [54] The paper on sorting plastic waste for a circular economy using lanthanide luminescent markers explores an advanced tagging system for efficient recycling. Key features include the integration of lanthanide-based luminescent markers to enable precise identification of different plastic types, spectroscopic detection techniques for automated sorting, and machine learning algorithms to enhance classification accuracy.

Daichi et al. [55] The paper on a single- axis tracking interferometer for measuring two-dimensional error motions of machine tools and industrial robots presents a high-precision metrology system for motion analysis. Key features include interferometric tracking for real-time measurement of linear and angular displacements, high-resolution laser-based detection for sub-micron accuracy, and compensation algorithms to correct positioning errors.

Lackner et al. [56] The paper on 3D printing software for 6-axis industrial robots, introduces an advanced tool for additive manufacturing. Key features include adaptive slicing algorithms optimized for multi-axis printing, collision avoidance strategies for complex geometries, and toolpath generation tailored to robotic kinematics. The software enables greater design flexibility by utilizing non-planar and curved layer deposition, reducing the need for support structures.

Mathias et al. [57] The paper presents a method for the energetic comparison of 6-axis industrial robots, focusing on energy consumption analysis for resource-efficient plant design. Key features include energy modelling and measurement techniques to evaluate power usage during different motion tasks, optimization strategies for trajectory planning to reduce energy waste, and comparative analysis of various robot configurations. The study highlights potential improvements in robotic system efficiency, emphasizing sustainable manufacturing by minimizing energy consumption.

Melchiorri et al. [58] The paper on performance and sealing material evaluation in 6-axis force-torque sensors for underwater robotics focuses on ensuring sensor reliability in submerged environments. Key features include material analysis for waterproof sealing, evaluation of sensor accuracy under varying pressure conditions, and testing for long-term durability against corrosion and water ingress.

Hossam et al. [59] The paper on the development and evaluation of a dual-purpose machine for chopping and crushing forage crops presents an efficient system for livestock feed processing. Key features include a hybrid cutting and crushing mechanism for handling different crop types, an adjustable blade system to control output size, and a high-efficiency motor for optimized energy consumption. The study evaluates the machine's performance in terms of chopping uniformity, crushing effectiveness, and throughput capacity. The findings highlight its potential to enhance fodder preparation, reduce manual labor, and improve feed digestibility for livestock farming.

Ramayanty et al. [60] The paper explores the development of an autonomous chess robot system utilizing computer vision and deep learning for real-time gameplay. Paper discusses how image processing techniques detect and recognize chess pieces and board positions, while deep learning models assist in strategic decision-making. Key methodologies include object detection, move prediction using neural networks, and robotic arm manipulation for physical piece movement.

Truong et al. [61] The paper explores how Edge Computing and Computer Vision enhance Industrial IoT through the visual digitization of analog gauges. The study discusses how real-time image processing and deep learning algorithms accurately read and interpret gauge values, reducing the need for manual monitoring. Key methodologies include object detection, OCR-based digit recognition, and edge-based AI inference to ensure low-latency data processing.

Michail et al. [62] The paper explores the YOLOv8 model. It discusses how YOLOv8's advanced object detection and segmentation capabilities identify and classify faults in industrial and infrastructure systems. Key methodologies include real-time image analysis, deep learning-based anomaly detection, and transfer learning to improve model adaptability across different environments.

Ibrahim et al. [63] The paper presents a computer vision-based solution for analyzing ballast degradation using deep learning and data fusion techniques. Explores how high-resolution imaging, combined with AI-driven feature extraction, detects wear, contamination, and fragmentation in railway ballast. Key methodologies include convolutional neural networks, multispectral imaging, and sensor data fusion for precise condition assessment.

Kelin et al. [64] The paper explores the use of optical MRI imaging combined with computer vision for extracting and analyzing morphological features of renal tumors. The present work discusses how advanced image processing techniques, including deep learning and feature extraction algorithms, enhance tumor detection, segmentation, and classification. Key methodologies involve edge detection, texture analysis and shape modelling to improve diagnostic accuracy and treatment planning.

Deng et al. [65] The paper explores smart structural health monitoring using computer vision and edge computing. The work discusses how real-time image processing and AI-driven analytics detect structural defects, such as cracks, deformations, and material degradation. By leveraging edge computing, the system processes data locally, reducing latency and enabling faster decision-making. The study highlights how this approach enhances infrastructure safety, minimizes maintenance costs and enables proactive monitoring in bridges, buildings, and other critical structures.

Zhen et al. [66] The paper explores the use of thermal imaging and computer vision for detecting and quantifying voids in precast grouted structures. Discusses how infrared thermography, combined with image processing algorithms, identifies temperature variations that indicate hidden voids. The study highlights how this approach enhances structural integrity assessment, improves quality control, and reduces inspection time in construction and civil engineering applications.

Varun et al. [67] The paper discusses the integration of machine learning and computer vision to automate processes in advanced logistics within the framework of the Integrated Logistic Platform 4.0. It explores how AI-driven image analysis and predictive algorithms enhance inventory management, real-time tracking, and automated quality control. The study highlights how ILP 4.0 leverages smart automation to reduce errors, improve decision-making, and enhance overall logistics performance.

Capuaa et al. [68] The paper explores the use of interactive AI and computer vision for optimizing waste handling in construction and demolition. The paper introduces prompt-guided segmentation techniques that enable precise identification and classification of various waste materials. By leveraging deep learning models and real-time image analysis, the system enhances sorting efficiency, reduces manual effort, and improves recycling processes.

Nilakshan et al. [69] The paper explores the use of interactive AI and computer vision for optimizing waste handling in construction and demolition. Introduces prompt-guided segmentation techniques that enable precise identification and classification of various waste materials. Real-time image analysis, the system enhances sorting efficiency, reduces manual effort, and improves recycling processes.

Tahmid et al. [70] The paper explores the synchronization evaluation of a Digital Twin for a robotic assembly system using computer vision. It discusses how real-time image processing and AI-based vision techniques monitor and compare the physical and virtual models of the assembly process. The study highlights how integrating computer vision with Digital Twin enhances system performance, improves predictive maintenance, and optimizes automation in robotic assembly lines.

Yao et al. [71] The paper examines the application of computer vision algorithms in analyzing and predicting ceramic surface textures. The present work explores techniques such as image processing, feature extraction and machine learning to assess surface quality, detect defects, and classify textures. The study highlights how computer vision improves efficiency, reduces manual inspection efforts, and enhances product consistency in the ceramics industry.

Sumaira et al. [72] The paper explores the role of computer vision in smart agriculture and precision farming, highlighting key techniques and applications. Discusses how image processing, deep learning, and machine learning algorithms are used for crop monitoring, disease detection, yield estimation, and automated irrigation. The study emphasizes how computer vision enhances decision-making, optimizes resource usage, and improves agricultural productivity, making farming more efficient and sustainable.

Zhang et al. [73] The paper investigates the use of intelligent image processing for recognizing and optimizing urban public sports facilities. It explores how computer vision and AI-based techniques analyze satellite and street-level images to assess facility distribution, usage patterns, and maintenance needs. The study emphasizes how intelligent image processing can aid urban planners in optimizing facility placement, improving accessibility, and enhancing resource allocation for better public sports infrastructure.

Varucha et al. [74] The paper explores the use of image processing techniques for precision disease diagnosis in sugar beet agriculture. Current work discusses how advanced computer vision and machine learning algorithms can analyze leaf images to detect and classify diseases accurately. Key methodologies include color and texture analysis, feature extraction, and deep learning-based classification. The study highlights the potential of automated disease detection systems to improve early diagnosis, reduce reliance on manual inspections, and enhance crop management. Jorge et al. [75] The paper examines the integration of Robotic Process Automation and Artificial Intelligence within the context of Industry 4.0. The authors highlight that combining RPA with AI techniques, such as Artificial Neural Networks, Text Mining, and Natural Language Processing, enhances the accuracy and execution of automated processes.

Alexej et al. [76] The paper explores the application of AI-based algorithms to automate assembly processes, particularly focusing on the detection of bores with varying geometries and internal structures. In high-mix low-volume production environments, where product diversity is high and quantities are low, traditional automation faces challenges due to frequent changeovers and complex part variations. The study compares the performance of two prevalent AI object detection algorithms, Single Shot Detection (SSD) against conventional detection methods like gradient filters.

Ahmed et al. [77] The paper details the development of an affordable delta robot designed for efficient pick-and-place tasks, enhanced by an integrated vision system. Delta robots are renowned for their high-speed capabilities in such operations due to their overhead installation and parallel arm structure. Zero minimize costs, the design utilizes stepper motors in place of traditional AC servo motors and employs accessible materials and fabrication techniques like laser cutting and 3D printing.

Phan et al. [78] The paper introduces a meticulously curated dataset aimed at enhancing 6D pose estimation algorithms crucial for robotic grasping tasks in industrial settings. The researchers captured high-quality RGB and depth images, annotating each with precise 6D pose data, encompassing both rotation and translation of various objects. The system incorporates a measurement frame fitted with 16 CCD laser displacement sensors, which measure the relative pose of the TO within the EF's coordinate system.

Molengraft et al. [79] The paper introduces an innovative control strategy aimed at minimizing oscillations in products handled by suction during robotic pick-and-place tasks. The authors developed a control method that significantly reduces swinging motions, enhancing the precision and speed of these operations. The system assesses grippers based on criteria such as adaptability to varying shapes and sizes, grasping force, cycle time, and energy efficiency. This comprehensive evaluation framework aims to identify the most suitable grippers for efficient and effective plastic recycling processes.

Haider et al. [80] The paper presents the creation of an automated robotic arm aimed at assisting small scale farmers in hydroponic systems. The design features a 1.3-meter-tall robotic arm constructed from aluminum and mild steel, offering four degrees of freedom and a gripper adaptable to objects ranging from 5.5 to 12 cm. Weighing 60 kg, the arm can handle payloads between 8 and 10 kg and is mounted on a mobile platform to extend its operational reach while conserving energy.

Fengchun et al. [81] The paper introduces an innovative end effector (EF) tailored for accurate manipulation of square-shaped items. The EF features a square configuration equipped with eight CCD laser displacement sensors, strategically positioned to monitor deviations along its edges. Complementing this design, a frame structure, also integrated with eight sensors, is utilized to assess the pose deviations of target objects relative to the EF.

Nicole et al. [82] The paper introduces a comprehensive framework designed to assess the effectiveness of various robotic grippers in handling post-consumer plastic materials. Benchmarking system evaluates grippers based on key performance indicators such as grasping reliability, adaptability to diverse plastic geometries, and operational efficiency under realistic sorting conditions.

Liangliang et al. [83] The paper introduces an automated system designed to assist in the harvesting of heavy fruits like pumpkins, addressing labor challenges faced by aging farming populations. The system integrates robotic arm and a specially designed end-effector for grasping pumpkins. which are processed using various versions of the YOLO object detection algorithm to accurately identify and locate pumpkins.

Alexander et al. [84] The paper presents a method to optimize the positioning of a robot's base to enhance efficiency in pick-and-place tasks. Approach employs a point-to-point trajectory planner that evaluates potential placements by considering the robot's kinematic limits, collision possibilities, and specific task constraints. By systematically analyzing these factors, the method identifies optimal base locations that improve operational efficiency and adaptability in industrial settings.

Truong et al. [85] The paper presents an innovative approach to creating a robotic system capable of autonomously playing chess against human opponents.

Vivek et al. [86] The systematic review examines advancements in robotic grippers designed for handling rigid and porous fabric materials, such as those used in automotive acoustic applications. The study highlights the unique challenges posed by these materials, including their stiffness and permeability, which necessitate specialized gripping solutions.

Daniel et al. [87] Image enhancement is a critical process in machine vision and industrial image processing, aiming to improve image quality for more accurate analysis and decision-making.

Techniques such as contrast adjustment, noise reduction and edge enhancement are commonly employed to highlight essential features while suppressing irrelevant information.

Jacopo et al. [88] The paper explores the techniques for supporting the selection of robot grippers typically involves evaluating key parameters such as object size, shape, weight, material, and required precision.

Guohang et al. [89] The study introduces an innovative, unpowered double-buffering flexible crating system designed to minimize mechanical damage to tomatoes during robotic harvesting. The findings offer valuable insights into enhancing post-harvest handling efficiency and preserving fruit quality in automated agricultural systems.

Pakruddin et al. [90] The paper addresses the scarcity of publicly available, high-quality datasets for training deep learning models in the context of pomegranate disease detection. to develop and refine machine learning or deep learning models for pomegranate disease detection, thereby contributing to improved crop yield and quality.

Petia et al. [91] The study presents an innovative approach to classifying carrot quality to enhance waste management and control. The researchers developed an improved convolutional neural network (CNN) that combines average and max pooling functions, resulting in a classification accuracy of 99.43% for grading carrots.

Zeshan et al. [92] The paper proposes a novel convolutional neural network model that integrates enhanced attention mechanisms with explainable artificial intelligence techniques to improve the accuracy and interpretability of fruit and vegetable classification.

Francisca et al. [93] The study introduces a non-destructive method for detecting seed spoilage in mangoes. By employing X-ray imaging, the internal structure of mango seeds was captured, allowing for the identification of spoilage without damaging the fruit. These X-ray images were then analyzed using convolutional neural networks (CNNs), which effectively classified the seeds as either healthy or spoiled.

Sonal et al. [94] The study presents a novel, non-destructive prototype device designed to classify custard apples based on their maturity stages. Utilizing image processing techniques, the device captures and analyzes the skin color of fruit areoles to determine five distinct maturity levels: 0%, 25%, 50%, 75%, and 100% areole opening. The system integrates a Raspberry Pi board, camera module and LCD touch screen, enabling real-time, accurate classification of fruit maturity. Benjamin et al. [95] The offers a comprehensive examination of the evolution and current state of machine vision technologies in fruit sorting and grading. The systematic review highlights significant technological advancements, which have enhanced the accuracy and efficiency of fruit quality assessment.

Dhanalakshmi et al. [96] The paper introduces an innovative approach to enhancing fruit image classification accuracy by integrating multiple convolutional neural network architectures. The proposed model combines features from different CNNs to capture diverse image, thereby improving classification performance.

Arya et al. [97] The study investigates the relationship between weather factors and the prevalence of leaf spot and fruit rot diseases in eggplants, alongside the development of an AI-based detection system. The research identifies that minimum temperature negatively correlates with leaf spot disease incidence, while fruit rot diseases disease incidence shows a positive correlation with factors such as wind speed, maximum relative humidity and rainfall.

Sandra et al. [98] The study introduces a non-destructive approach to identify internal blackheart disease in pomegranates. This technique enhances quality control in the pomegranate supply chain by effectively identifying affected fruits without causing any damage.

Schlüter et al. [99] A developed an AI-based method for recognition, inspection, and sorting of components in reverse logistics for remanufacturing. The study reveals the potential for intelligent automation to improve efficiency, accuracy, and adaptability in remanufacturing endeavors.

Sheth et al. [100] The researchers generated an ensemble model that was made of Inception and CNN architectures to classify images of natural disasters. The ensemble model obtained an accuracy of 92.79%, which had stronger performance than a stand-alone network, and the configuration helped facilitate generalizing multi-class disaster recognition from a dataset open to researchers.

Griepentrog et al. [101] A free deep reinforcement learning system which uses a lightweight vision backbone with fully convolutional networks to learn pixel-based Q-values for pushing, grasping, and placing tasks. In terms of cluttered environments, the dual transfer-learning with DenseNet121 had the best sorting performance.

Agustami et al. [102] The study presents the use of deep learning for the study of disease detection in areca nuts using image-based classification. A convolutional neural network (CNN) model was built in order to correctly identify affected nuts. Using this approach will enable farmers to detect affected areca nuts earlier, supporting precision agriculture and quality assurance.

Olivier et al. [103] The research implements an online machine learning system in a waste sorting facility. This system detects and detects conveyor jams before they become serious issues. The system will develop and monitor operational data of the sorting plant in real time.

Yin et al. [104] The research explore the carbon footprint of a community in Hangzhou's household waste sorting process through life cycle assessment. The sorted system produced more carbon reduction and more energy recovery.

Palli et al. [105] The study provides an experimental evaluation of sealing materials for an optoelectronic 6-axis force/torque sensor designed specifically for underwater robotics. while discussing the properties and usability of each rubber.

Bui et al. [106] The project builds a cost-effective autonomous chess robot that utilizes computer vision and deep learning for processing visual input, a 4- degree of freedom SCARA arm for movement, and Stockfish to generate moves.

Michail et al. [107] The paper proposes an end-to-end approach for digitizing analog industrial gauges and forming part of an IoT system - an edge computing part and a computer vision part.

Diani et al. [108] The work presented a prompt-guided computer vision workflow for construction and demolition waste (CDW) segmentation that takes into account bounding-box, point, and text prompts in cluttered waste scenes. It achieved 70 % class-wise accuracy 9 % greater than standard methods.

Fengchun et al. [109] The paper discusses a robotic tailored end effector integrated with a measurement system which allows for high-precision pick- and-place operations of square objects.

Nianzu et al. [110] The research presented a multiscale finite element model for simulating the dynamic impact and deformation of tomatoes crane following robotic post-picking crating.

Pakruddin et al. [111] The study provides a publicly available, standardized dataset of 5,099 pomegranate fruit images classified into five categories Healthy, Bacterial Blight, Anthracnose, Cercospora, and Alternaria.

Francisca et al. [112] The research presents a fast and non-destructive approach to locate spoilage in mango seeds using X- ray imaging and a VGG-16 CNN model. The model resulted in an overall accuracy of 97.66 % AUC=0.988, shown in Grad-CAM saliency maps.

Mojtaba et al. [113] The research discussion proposed a stereoscopic video-based computer-vision system utilizing ANN with PSO and BA based optimization to discriminate rice and two weed species under both natural and controlled lighting.

Akram et al. [114] The study proposes a combined computer vision-based prototype system that integrates fruit detection and robotic picking, together with automated quality grading for the fruit harvested.

Francisca et al. [115] An innovative approach to creating a robotic system capable of autonomously playing chess against human opponents. and a robotic manipulator for physically moving the pieces to affect the moves. It Has a closed-loop architecture that checks for the state of the game after each step to ensure that the system is correct and reliable. Such systems develop to integrate accurate perception, autonomous decision making and safe human-robot interaction for the best complete experience for playing against a human opponent.

Hemavathy et al. [116] The review underscores the importance of tailoring gripper designs to specific material properties to optimize handling processes in industrial applications, an extensive standardized dataset with information about multiple pomegranate fruit diseases was created to assist in deep learning research. The dataset has quality images with annotations of various disease classes taken under multiple conditions for efficient training and validation of models.

Yuan et al. [117] The literature review study presents the analysis of mechanical damage to tomatoes from dynamic impacts from post picking handling. The paper uses a multiscale finite element model to simulate a variety of impact scenarios for developing effective crating options for use in robotic harvesting. The study proposes a flexible crating system with unpowered double-buffering to mitigate damage and maintain quality of the fruit.

Dadashzadeh et al. [118] introduced a stereoscopic video-based computer vision system to facilitate weed discrimination in rice paddies in natural and controlled lighting environments. The study examined the extent machine learning algorithms could utilize the combined depth and color characteristics to distinguish between weeds and crops in rice paddies.

Umair et al. [119]. The paper describes a computer vision-based prototype robotic system for fruit picking and grading. A robotic manipulator and computer vision image analysis algorithms that will identify, harvest, and grade fruit according to pre-defined quality factors. This approach demonstrates a new way to save labor, improve grading accuracy, and do so more efficiently after harvesting.

2.2 Literature Review Summary

Nowadays, the research community defines a hybrid of agricultural and automation, all combined with machine vision systems, image processing technologies, and artificial and robotic intelligence. Nevertheless, some researchers, used CNN networks with YOLO and SVM models to classify areca nuts, and by extension classify images of pomegranates and citrus. Also enhanced model accuracy with image enhancement processes along with preprocessing processes, improved disease and defect detection using multispectral and hyperspectral imaging technologies. Robots are used primarily for sorting automation through innovations in gripper design, because of 6-axis arms, and delta robots which are highly precise and efficient. Decision making in the field is now conceivable with the proliferation of embedded AI technologies such as Raspberry Pi and edge-computing technology. Smart technology has thereby been used to sort agricultural by-products, track livestock, and short-run lots that collectively regulate manufacturing quality. Future research and methods of use employ transfer learning for enhancing classification sustainability, feature fusion models, and reinforcement learning to balance classification correctness with performance speed. The mechanical system must integrate characteristics of energy use where solar energy is used and parts must be durable and rehabilitated for sustainability. The Combining of these advancements leads to integrated smart agriculture systems, and therefore will benefit from higher productivity levels, less labor use, and assurance of better quality because of automated, smart systems. The tabular column explains the technologies used in the previous studies, so how I can implement the algorithms to present work. Literature Review help me to get a wide Knowledge about the sorting process, grading process.

2.3 Research Gap

1. Currently, areca sorting is performed using traditional methods involving manual labor.
2. Most existing research primarily focuses on image recognition, without integrating it into a fully automated system.
3. There has been limited work on implementing real-time, end-to-end automated sorting specifically for areca.

Table 2.1: Overall Comparative Analysis of Literature Review

Sl. No	Title	Authors	Algorithm	Salient Feature
1	A vision system based on CNN-LSTM for robotic citrus sorting	Yonghua et al. [19]	CNN and LSTM	Combining CNN with LSTM, the vision system can learn not only spatial features such as color and texture from citrus images, but also learns in a sequence to determine ripe, unripe with a great degree of accuracy.
2	Automatic detection of pomegranate fruit affected by blackheart disease using X-ray imaging	Sandra et al. [38]	Machine Learning	Uses non-destructive X-ray imaging for detecting internal blackheart disease in pomegranates allowing for early and accurate detection.

3	Detection Of Diseases in Arecanut Using Convolutional Neural Network	Akshay et al. [20]	CNN	Utilizes the Convolution Neural Network to classify and detect diseases occurring in arecanut correctly by employing visual symptoms taken from video imagery data.
4	Grading and sorting technique of dragon fruits using machine learning algorithms	Pallavi et al. [92]	Machine Learning	The system operates through machine learning to process dragon fruit features for automated sorting and grading which includes size and color and surface texture attributes.
5	Betel Nut Classification Method Based on Transfer Learning	Hengquan et al. [86]	Transfer Learning	Transfer learning-type framework for next-level classification accuracy of betel nuts for limited training data.
6	Possible Approaches to Arecanut Sorting / Grading using Computer Vision:	Haider et al. [54]	Computer Vision	Studies the different computer vision methods for the grading and sorting of the arecanuts through color, size and texture of the arecanut visual features.
7	Image-based and ML-driven analysis for assessing blueberry fruit quality	Marcelo et al.[61]	Machine Learning	Focusing on various dimensions from the related to color, texture, and size, concerning on machine learning and image processing methods for quality measures of blueberries.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

3.1 Problem Statement

The manual sorting of unhusked areca is a labor-demanding, time-consuming and results in inconsistent quality, reduced efficiency and severe economic loss, hence problem statement is intended to develop fully automated system and deep learning assisted areca sorting system for improved effectiveness

3.2 Objectives

1. Design and development of an automated sorting machine for unhusked areca after harvest.
2. Data collection and processing using Gaussian noise filtering.
3. Development of AI assisted automated system using convolution neural network and TensorFlow for automated sorting of unhusked areca.
4. Integration of developed algorithm to the automation system and process optimization.

3.3 Methodology Overview

The methodology to develop the AI-powered arecanut sorting system was implemented in stages, beginning with the selection and classification of arecanuts by ripeness stage, Ripe and Unripe, for supervised learning as ground truth for labels. A comprehensive literature review will document the scope of AI in agricultural automation to include existing post-harvest sorting technologies. High-resolution color images of arecanuts were obtained in controlled light. Upon taking the pictures, preprocessing of the images occurred in Gaussian noise filtering and background removal using rembg library, a white background was introduced using OpenCV masking to standardize the data set for processing. After processing, the images were resized to 224 × 224 then augmented by rotating, zooming, shifting, flipping to improve the model's generalization across the dataset. Datasets were split by training, validation, testing. A custom Convolutional Neural Network (CNN) was designed with three convolutional layers, pooling layers max-polling, L2 regularization, drop-out layers at each convolutional layer, and lastly a sigmoid activation function to classify as a binary class (Ripe or Unripe). The model settings included the Adam optimizer to compile the model using binary cross- entropy loss with callbacks for Early Stopping and Model Checkpoint to improve the training process. Training process and model performance were documented and evaluated using accuracy plots and confusion matrices.

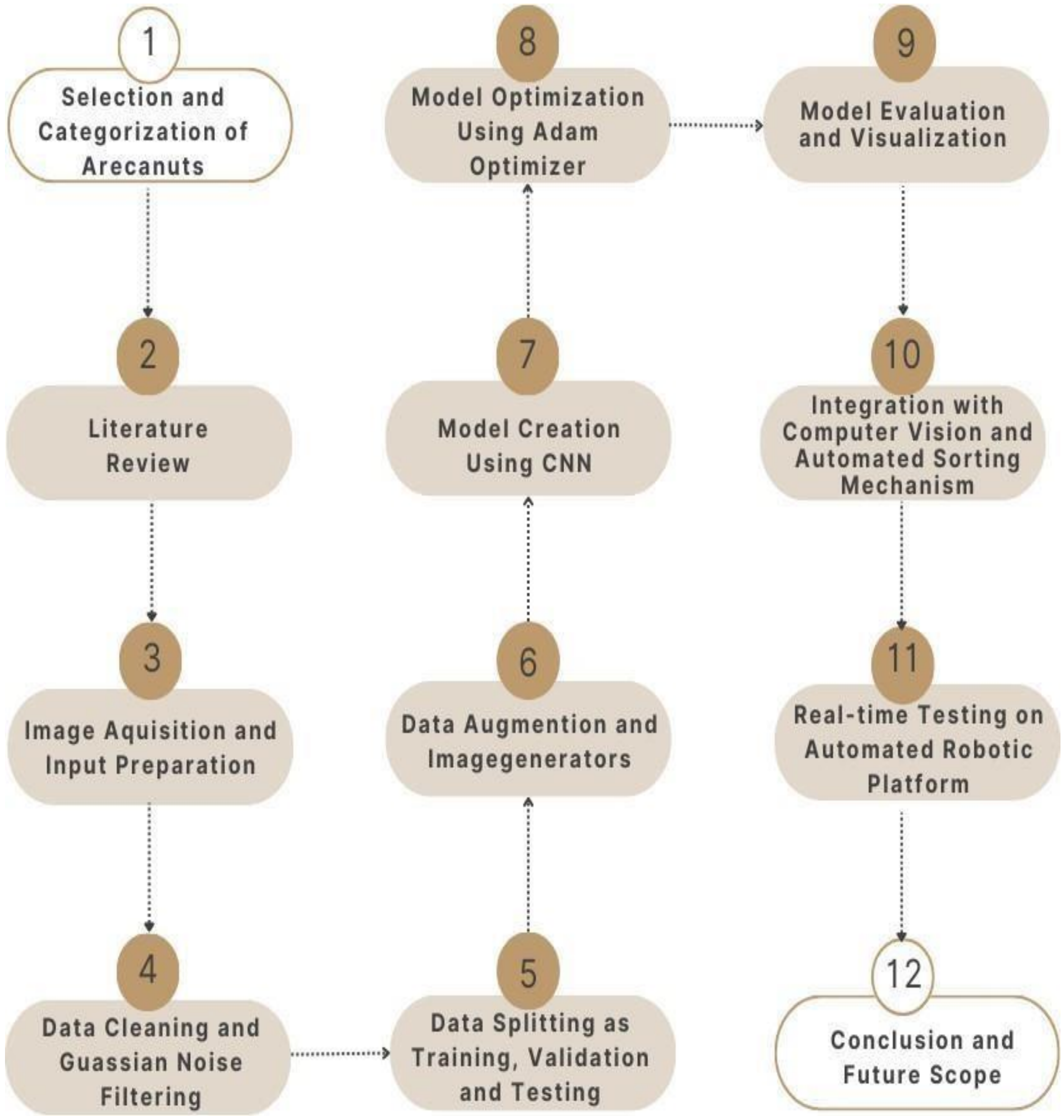


Figure 3.1 Methodology for the Design and Development of Automated Arecanut Sorting System

3.3.1 Image Acquisition and Input Preparation

To create a strong dataset for the classification of arecanuts, careful image acquisition was important. High quality images of ripe and unripe arecanuts were taken in controlled lighting conditions for comparison, minimizing shadows and reflective noise. The background had to be a neutral white color, so visually inconsistent images were eliminated and avoid complications when separating the objects in the preprocessing workflow. As shown in Figure 3.1, images were taken at multiple angles and positions when capturing the arecanuts, to exemplify variability and real- world settings. The image acquisition allows for lot quality and lot variability in the images that can help define a strong basis for preprocessing, noise filtering and training of the model. The trained model was implemented into a computer vision pipeline built with OpenCV for real time detection using a webcam. Ripe and unripe arecanuts were identified and classified by prediction scores and moved via a robotic platform on a servo motor driven sorting feature. Lastly, a final validation test proved accuracy, speed, and efficiency against manual sorting and evaluating the project finds considerable promise for next steps including incorporating with TensorFlow Lite, IoT integrating and expanding the model for multi-class ripeness detection.

(a)

(b)

Figure 3.2: Sample images of ripe and unripe arecanut with a neutral background.

(a) Original Ripe and Unripe Arecanuts

(b) After removing the background using rembg

(c) After Gaussian noise removal

Figure 3.3: Preprocessing result: (a) Original Ripe and Unripe Arecanuts, (b) After removing the background using rembg, (c) After Gaussian noise removal.

3.3.2 Data Augmentation and Image Generators

To strengthen generalization of the CNN model and increase the chance of avoiding overfitting, data augmentation was also heavily applied to the training dataset. Because real-life arecanuts may be presented in many varying rotations, lighting, and positions, data augmentation technique was frequently used to synthetically simulate this variability. Data augmentation was done using Keras' ImageDataGenerator to apply real-time data augmentation while model training occurs and without modifying the original training dataset. The parameters applied for data augmentation included, random rotation (up to 20°), random zoom (up to 30%), horizontal flip, width and height shift, and shear transformations. These transformation applications were diverse in order to create a variety of training data, by including several modified versions of each image, thereby providing a better opportunity for the model to learn more distinguishably invariant aspects of the original images. Additionally, the generator assisted in rescaling the image pixel values to fall within the [0, 1] pixel value range. This sped up the model's training process by facilitating convergence. The CNN was able to develop a generic representation of both ripe and unripe arecanuts while taking into consideration a number of potential scenarios thanks to the enhanced dataset. An image dataset can be enhanced with data using the Image Data Generator function, which can represent many real-time conditions, including foggy, desert, green, and watery ones. The following is a definition of the Sequential model's augmentation actions

1. The picture was flipped horizontally at random. When an object's orientation in an image has no bearing on its classification.
2. A random rotation method is used to the image, which adds consideration and helps guarantee model accuracy.
3. Adding zoom to the image facilitates the creation of images with different scales. Image generators are used to apply data augmentation to image datasets in the context of TensorFlow and deep learning. They also load batches of augmented images as fallows during model training. Prior to being fed into the model, all input photos are shrunk to the target size, which is a set size. Specifies the type of labels use categorical for one encoded label.

3.3.3 Model Optimization Using Adam Optimizer

The Convolutional Neural Network was trained using Adam (Adaptive Moment Estimation) optimization algorithms. Adam is computationally efficient while adaptive, which combines the benefits of Momentum and RMS algorithms by keeping exponentially decaying average, thus forming an average over past gradients and over squared gradients. Learning rate adaptation is also performed by Adam based on each parameter, making it perform well in high-dimensional parameter space, which makes it particularly useful for deep learning purposes. In this project, Adam was used with the Binary Cross entropy Loss Function, which is given as follows:[76]

$$\mathcal{L} = -[y.log(y^\wedge) + (1 - y) \cdot log\,(1 - y^\wedge)] \qquad [1]$$

Where,

y is a label denoted and y^ is the predicted probability. For ripe and unripe arecanuts, binary classification problems can be solved using the binary class entropy in this instance. The training Process achieved stability, quicker convergence time, and improved generalization by employing the Adam optimizer in combination with Early Stopping and Model Checkpoint.

The Adam optimizer, well-suited for multiclass classification tasks, was utilized with accuracy as the primary performance matrix. The model was trained 15 and 20 epochs using separate training and validation datasets. During backpropagation, Adam optimizer the model parameters, including weights and biases, to minimize the effectively Monitoring Training Progress The code logs and visualizes training and validation metrics during all of training, then plotted through training to evaluate performance and check for overfitting.

3.3.4 Model Development with Convolutional Neural Networks

Custom Convolutional Neural Network (CNN) was created from scratch to classify arecanuts to be ripe or unripe based on visual attributes. The CNN architecture was built to use separate layers to extract spatial features from input images in a hierarchy of Convolution layers. Each of the models are similar to start with the input layer taking the images and followed by three convolutional layers which have ReLU activation functions with 'same' padding. The 'same' padding allows the network to capture features at the edges.

A MaxPooling layer follows all three-layer blocks of convolution to reduce the spatial dimension and for mitigating the likelihood of overfitting. L2 regularization was applied to all convolutional and dense layers to reduce the prospect of overfitting and Dropout (0.5) was used as part of the model to randomly drop out neurons, in any one feedforward network, or training result.

The subsequent flattened output from the convolutional stack is subsequently fed into a fully connected dense layer, and also then finally followed by one sigmoid activated output neuron that is used to calculate a probability value for a binary classification. The binary cross entropy was applied as the loss function appropriate for two class prediction problems. The architecture itself was light and extremely fast, but it was still quite successful in resolving complex patterns in the pre-processed arecanut images.

The transformative impact of CNNs on computer vision is evident, as they find extensive utility across diverse domains, encompassing image classification, object detection, and image generation. The current study harnesses transfer learning with VGG16, employing pre-trained Arecanuts images to effectively detect and classify ripe and unripe. The augmented dataset, consisting of 2847 images categorized into Two classes (Ripe and Unripe) is strategically partitioned with 80% are marked for training, 10% for validation, 10% for testing.

Performance matrices such as precision, recall, accuracy, F1- Score to assess the predictions. The F1- Score is the harmonic mean of precision and recall. F1-score is calculated using below formula:

$$\mathrm{F1} - \mathrm{score} = \mathrm{precision} + \mathrm{recall2} \div (\mathrm{precision\ recall}) \qquad [2]$$

3.3.4.1 ReLU Activation Function (Rectified Linear Unit)

The Rectified Linear Unit adds a nonlinear component to the neural network and allows the variation of patterns in the datasets. e.g. small differences between arecanuts according to their color or textures and ripe versus unripe arecanuts. The ReLU function applies on an element-by-element basis, where all intermediate feature maps are provided from the convolution layer.
Mathematical expression [76]

$$\mathrm{f(x)} = \max(0, \mathrm{x}) \qquad [3]$$

where,

X is the input from the previous convolution layer- if x is negative, it is set to zero,
if x is non-negative x is unchanged.
In arecanut classification, ReLU ensures the deep learning model learns the activated features that are all positive, like the patterns of distinguishable color tones or surface patterns found in ripe or unripe areca nuts, which improves feature quality and reduces training time.

3.3.4.2 Max Pooling Layer

Max pooling is commonly used, which selects the maximum value from each pooling region. Mathematical Expression [76]

$$\mathrm{Yi, j} = \max(\mathrm{x}\,(\mathrm{i} * \mathrm{h}) : (\mathrm{i} * \mathrm{h} + \mathrm{h} - 1), (\mathrm{j} * \mathrm{w}) : (\mathrm{j} * \mathrm{w} + \mathrm{w} - 1)) \qquad [4]$$

Where

yi, j is the output of the pooling operation.
$x(i*h) :(i*h+h-1), (j*w) :(j*w+w-1)$ is the input region.
In arecanut sorting, max pooling is a good function to use since it highlights dominant textures or patterns (like dark spots or signs of ripening) and rejects less important variance like background noise, while also reducing the number of parameters we need to optimize and computation time.

3.3.4.3 Fully Connected Layer

The fully connected layer employed for classification. These layers connect every neuron in one layer to another neuron in the next layer.
Mathematical expression [76]

$$y = f(Wx + b) \quad [5]$$

where

W is the weight matrix.

b is the bias vector.

X is the input vector.

f is the activation function.

With arecanut, the CNN uses the fully connected layer to consolidate higher-level features and assign probabilities of being in certain classes (ripe or unripe) by assigning the highest probability to a class label that seems to have the highest chance of being correct.

3.3.4.4 The SoftMax Activation Function

The SoftMax activation function is the typical activation function used in the convolutional neural network output layer to convert raw class scores (also known as logits) into normalized probability values. The normalized output values will be within the (0, 1) range and added together will equal 1 representing a probability distribution on the predicted classes. In the arecanut classifier, the goal is either ripe or unripe, so the SoftMax function converts the output logits into a probabilistic score for each class, especially for the last layer to make a final decision.

3.3.5 CNN Model Architecture Design

The CNN architecture encompasses 2847 arecanut images categorized into two classes (Ripe and Unripe) with a strategic allocation of 80% for training, 10% for validation, and 10% for testing. In the design phase, critical decisions involve determining the number and types of layers (convolutional, pooling, fully connected), along with their specific configurations as shown in Figure 3.4. The process extends to Hyperparameter Tuning, focusing on optimizing parameters like batch size, learning rate, epochs and weight initialization methods. Additionally, the iterative training of the model is executed with varying data iterations, including 15, contributing to the robust prediction and assessment of model accuracy

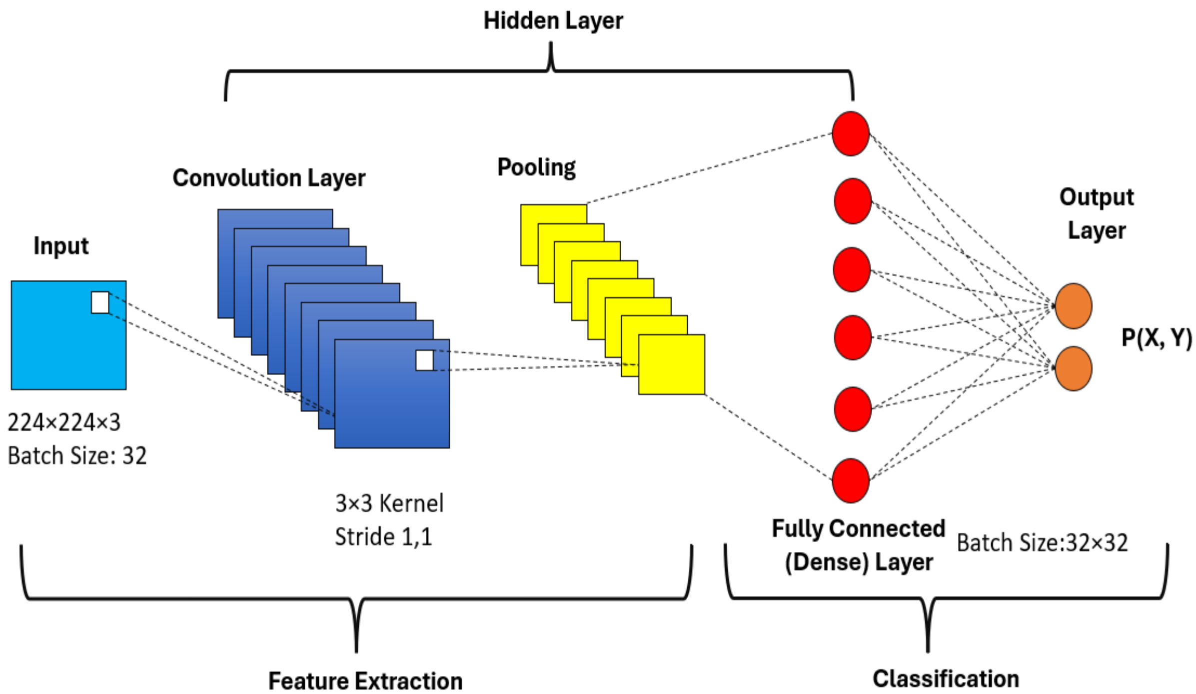


Figure 3.4: Convolutional Neural Network (CNN) Architecture

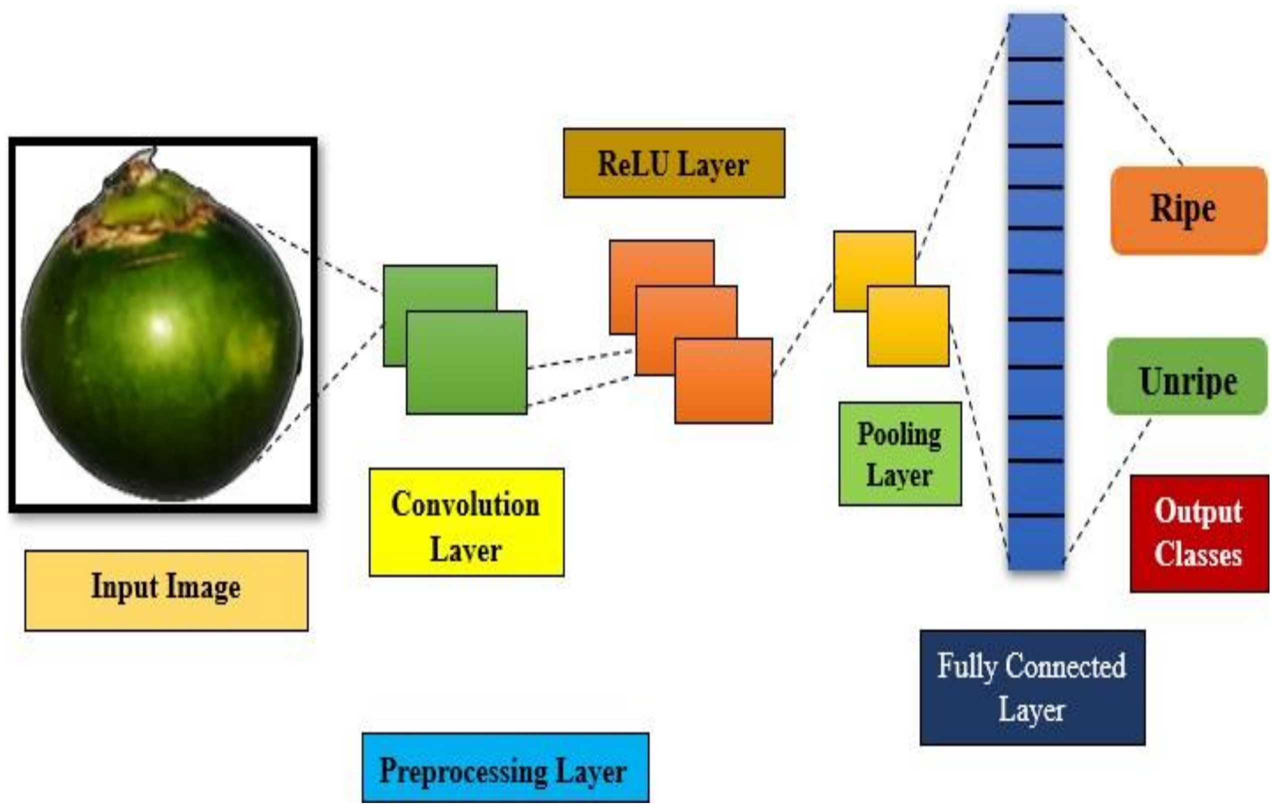


Figure 3.5: CNN Architecture for Arecanut Ripeness Classification

The structured Convolutional Neural Network (CNN) architecture developed for categorizing the arecanut as either ripe or unripe. The process starts with an unprocessed arecanut image depicted in Figure 3.2. The process of preprocessing the image will include background removal, resizing, and Gaussian noise filtering just to retain the important properties in the image such as color and surface textures that will allow the model to perform better on. The image, post-preprocessing arecanut image was presented in figure 3.5, will now pass through the convolutional layers, where the useful features in the images will be extracted using learnable filters.

A ReLU activation function will be used to the convolutional layer's output layer, adding non-linearity to make it easier for the network to learn challenging representations. While max pooling down samples the feature maps, reducing overfitting and retaining key features. Then softmax layer outputs class probabilities, predicting ripe and unripe arecanuts in real time for the automated areca sorting system.

3.3.6 Visual Geometry Group16 Architecture

The VGG16 is a convolutional neural network architecture with 16 layers containing trainable constraints, considered for image classification tasks. Visual Geometry Group16 is a computer vision model that learns to detect images by passing through 16 layers, gradually extracting significant features for classification. Figure 3.6 illustrates the architecture is fine-tuned specifically for the arecanut classification task, leveraging transfer learning to adapt the pre- trained model to the specific ripe and unripe images. By retraining the final layers of VGG16 on the arecanut dataset, the model retains its ability to detect complex patterns while becoming highly attuned to the unique characteristics of arecanut. The inclusion of VGG16 provides a robust framework for identifying intricate patterns in arecanut, ensuring high accuracy in distinguishing between Ripe and Unripe categories. Its deep convolutional layers effectively capture subtle differences in nut texture, color and shape, which are crucial for accurate classification. The combination of custom convolution neural network layers for initial feature extraction and the pre-trained VGG16 model for advanced feature recognition forms a comprehensive approach. This dual-layer strategy significantly enhances the precision of Arecanut Ripeness Classification, leading to more reliable and accurate monitoring. The results are further validated through confusion matrix and performance metrics, demonstrating the model’s efficiency in real-world applications.

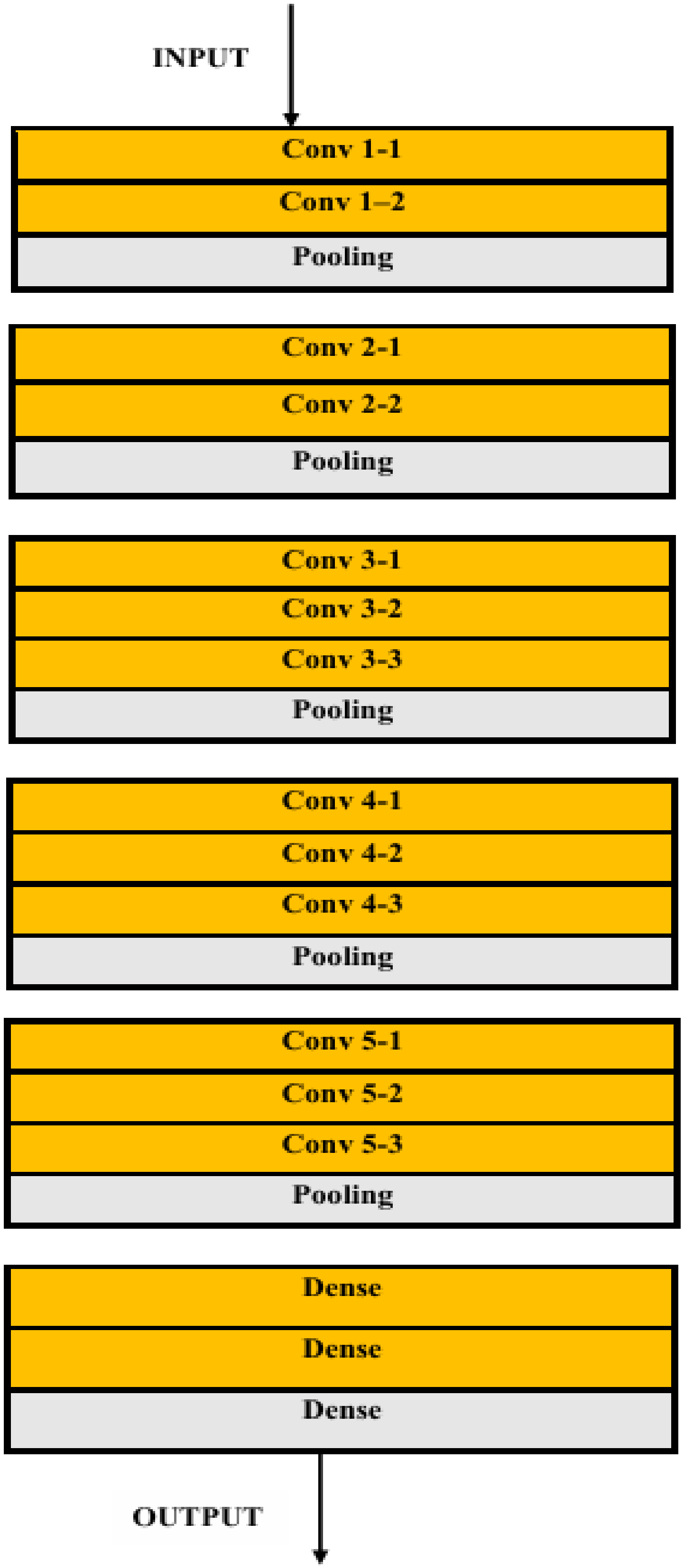


Figure 3.6: Architecture of Visual Geometry Group (16)

3.3.7 Confusion Matrix

A confusion matrix is tabular summary of a model ability to classify arecanut (Ripe and Unripe) in images. It assesses by presenting true positives, true negatives and false positives, false negatives count. Figure 3.7 illustrates the concise representation aids in evaluating the model's performance in disease detection.

True Positives (TP): Instances where the model correctly predicts the positive class. Correctly identifying a ripe nut in the final classification stage as a final stage.

True Negatives (TN): Instances where the model correctly predicts the negative class. Correctly recognizing a Ripe and Unripe as not belonging to the beginning or final stage.

False Positives (FP): Instances where the model incorrectly predicts the positive class. Incorrectly classifying a Arecanut as being in the beginning or final stage.

False Negatives (FN): Instances where the model fails to identify a condition when it is actually present. Incorrectly classifying a arecanut in the final disease stage as Ripe or Unripe in the beginning stage, missing the arecanut detection

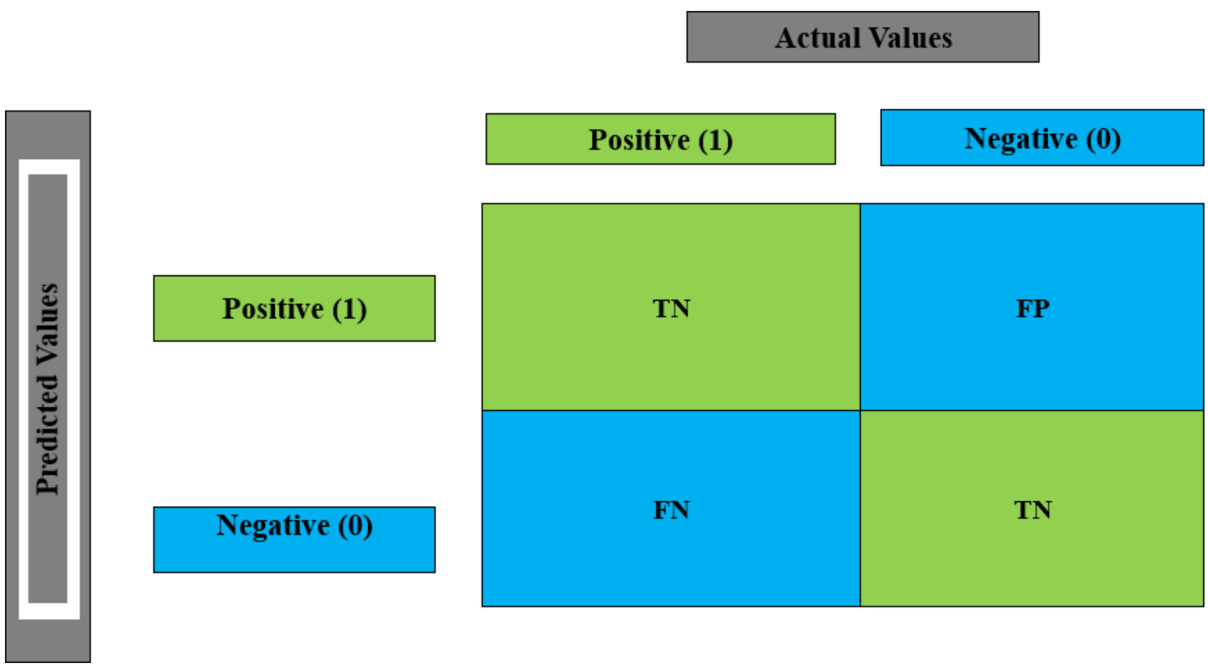


Figure 3.7: General Representation of Confusion Matrix

CHAPTER 4

MECHANICAL DESIGN OF THE AUTOMATED ARECA SORTING SYSTEM

4.1 Design Overview

The Design and Fabrication stage is essential in creating the physical infrastructure of the Automated Arecanut Sorting System. This stage encompasses the design, 3D modeling, and building of a highly functional, durable and accurate mechanical structure that holds the entire sorting process. The primary goal of the mechanical framework is to house an AI-based classification system and physical component sorting in one integrated development, while also remaining modular, functional and manufacturable.

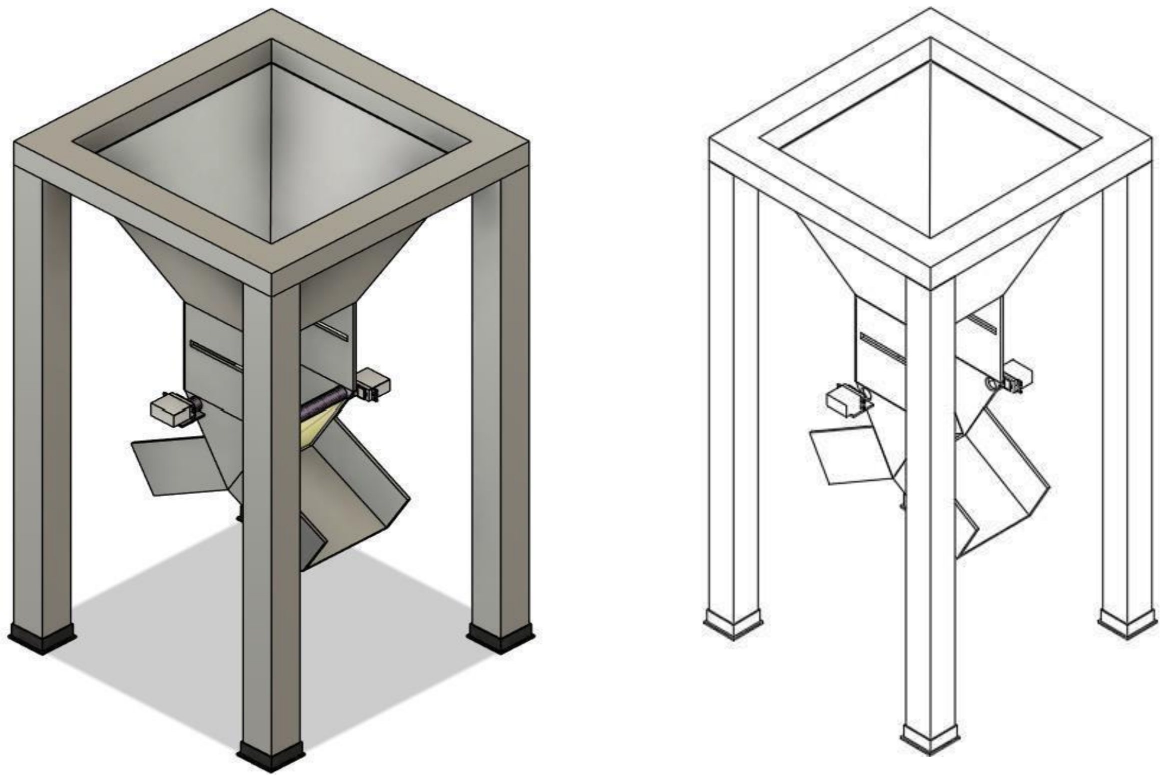
To achieve high levels of design accuracy and component compatibility, the entire setup was modelled in Autodesk Fusion 360, a high-level computer-aided design (CAD) software that can be employed for 3D modeling, simulation and designing for mechanical applications. Using Autodesk Fusion 360 also provides us with the ability to use parametric modeling on all components of the system, including the hopper frame, support legs, discharge guide channel, servo mounting and Discharge flap assembly.

Utilizing Autodesk Fusion 360 as the design source allowed for the development of a working model and visualization. The isometric exploded view created by Autodesk Fusion 360 provides an easy indication of the mechanical relationship and assembly of the main aspects of the system. The invented model has a hopper mount made of a Mild-steel frame and has four Mild-steel legs and rubber feet to dampen vibration.

The shape of the hopper is funnel-shaped to control the movement of the arecanuts towards the discharge channel where a flap mechanism operated by a servo will carry out the sorting process based on the output of the AI classification. Each component custom produced, such as the servo mounting plate, splined shaft, and discharge flap were modelled to provide exact tolerances that would allow it to move freely and align properly. Each mechanical component was selected and designed not only for function but for agriculture sector and fabricating capabilities that are accessible for manufacturing.

The CAD-based design to fabrication process was efficient in generating the design to fabrication with minimal waste and little chance of error prototypes. The final assembly will ensure function, structural integrity, and operational performance for a usable system to nestle in the sensors.

4.2 Automated Areca Sorting System Design



(a) (b)

Figure 4.1: 3D (a) and 2D (b) Isometric Representation of Automated Areca Sorting System (mm)

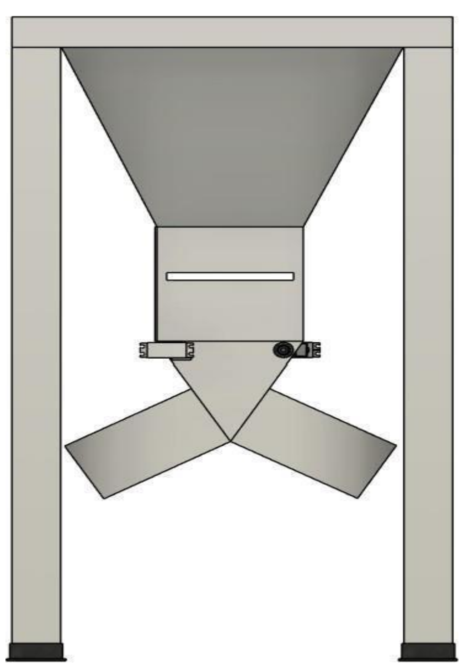


Figure 4.2: 3D Representation Front View

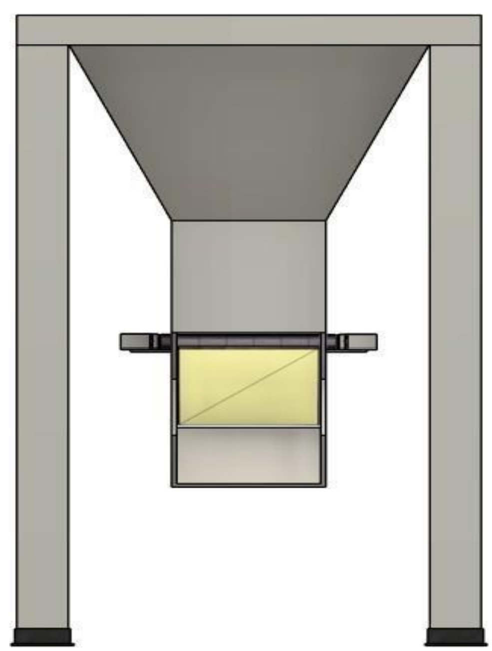


Figure 4.3: 3D Representation Side View

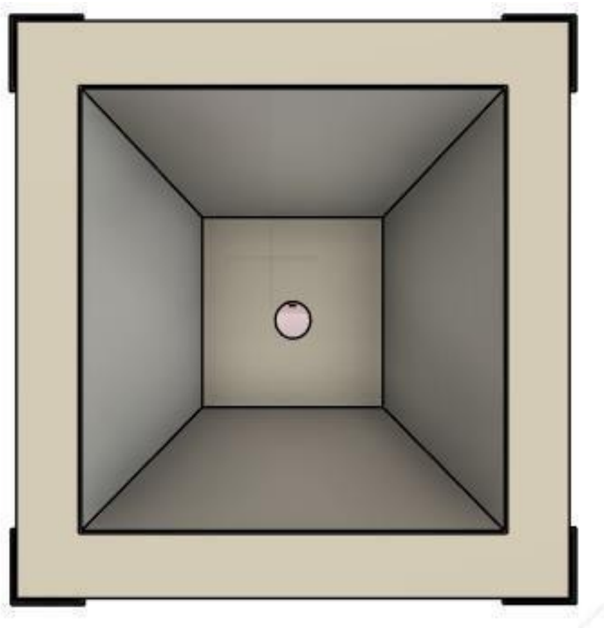


Figure 4.4: 3D Top View

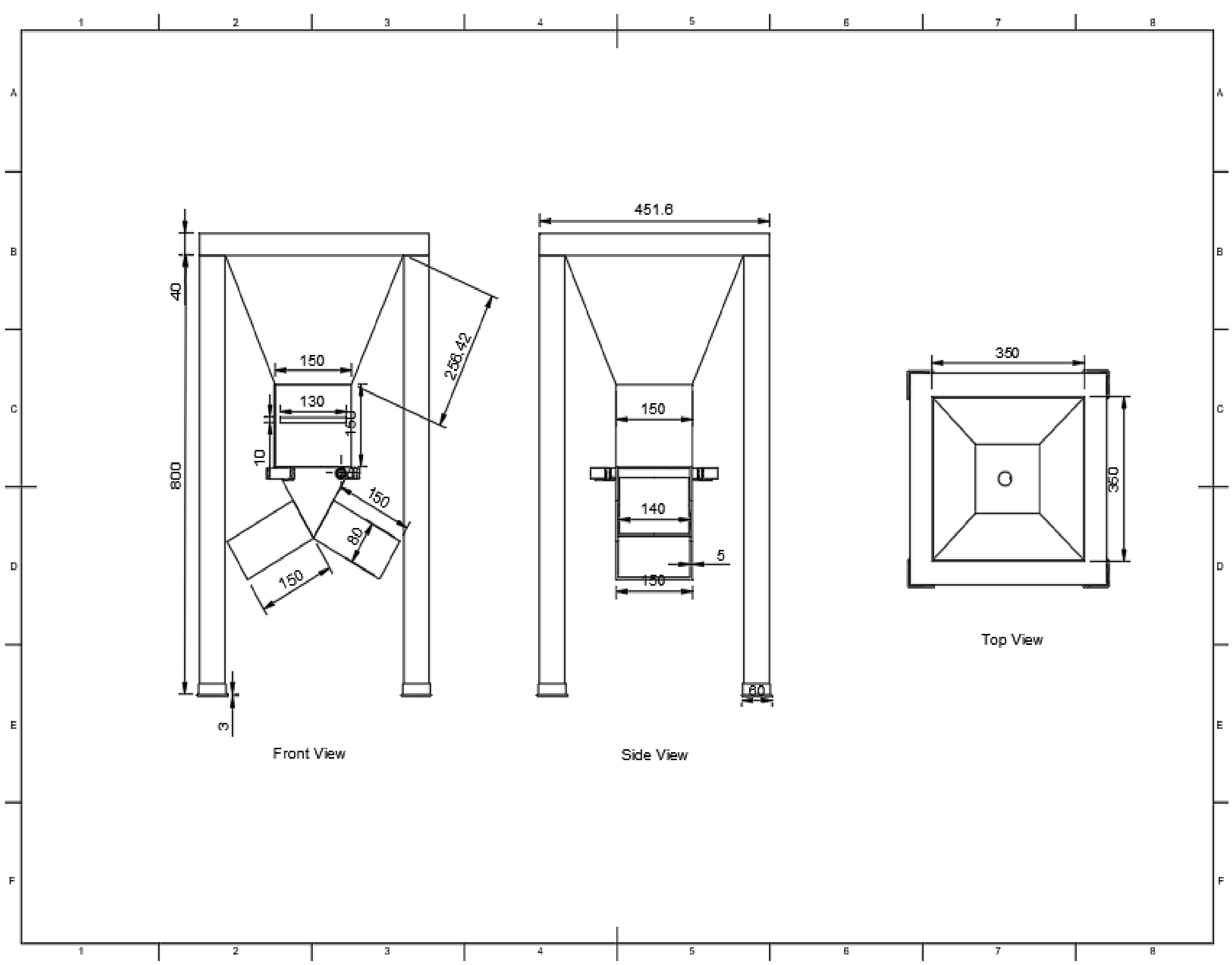
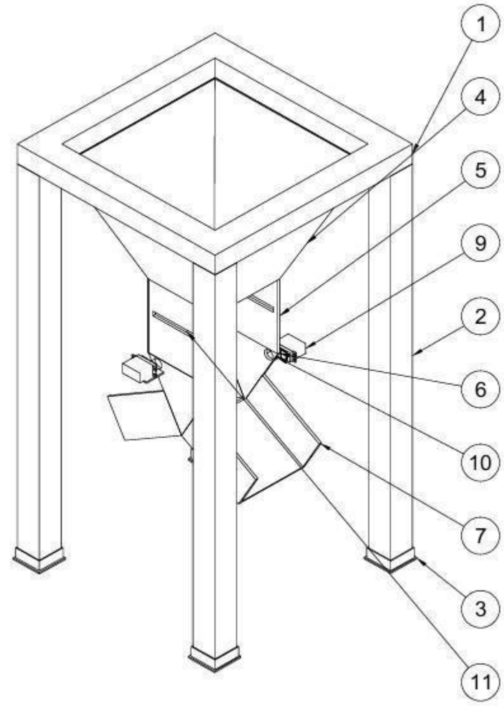


Figure 4.5: 2D Orthographic Projection of the Automated Areca Sorting System

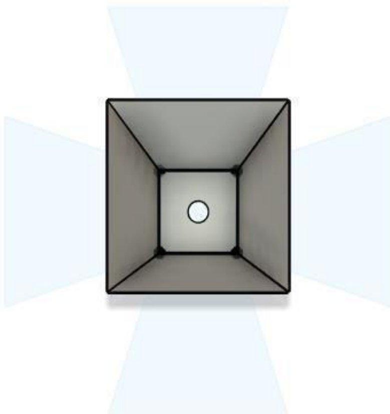


Parts List			
Item	Qty	Part Number	Material
1	1	Hopper Mounting Frame	Steel
2	4	Hopper Support Leg	Steel
3	4	Rubber foot	Rubber
4	1	Hopper	Steel
5	2	Rectangular Slotted Panel	Steel
6	2	Triangular Mounting Bracket	Steel
7	2	Discharge Guide Channel	Steel
8	2	Splined drive shaft	ABS Plastic
9	3	MG995	鋼
10	2	Servo Mounting Plate	Steel
11	1	Hopper Side Cover Plate	Steel
12	2	Discharge Flap	ABS Plastic

Figure 4.6: Bill of Material of Automated Areca Sorting System

The Automated Areca Sorting System is carefully designed to bring together mechanical, electronic and computing parts into a single, compact and efficient system to sort arecanuts based on ripeness. An overview of the full system can be seen in Figure 4.1, which shows both a 3D (a) and a 2D (b) isometric view of the system, which represent the system as a whole in one view along with the mechanical structure of the system. The front 3D view of the system in Figure 4.2 displays the hopper, image acquisition zone and actuator module in a vertical workflow. The 3D side view of the system in Figure 4.3 shows the internal alignment of certain parts of the system such as the camera, servo mechanism and discharge channels in the spatial orientation of the system. The top view of the system in figure 4.4, integrally represents the trajectory of mass first from the hopper to the classification exit and sorting exit. The ability to support a successful manufacturing path was aided by the 2D orthographic projection in Figure 4.5 which detailed technically correct dimensionality's, locations in space, and part alignments. Figure 4.6 provides a Bill of Materials (BOM), which details all parts used from the PLA 3D printed parts and mild steel frame to the MG995 servo motor and ESP32 microcontroller as per the quantity, material and specifications, for ease of replicating and fabricating.

4.3 Hopper



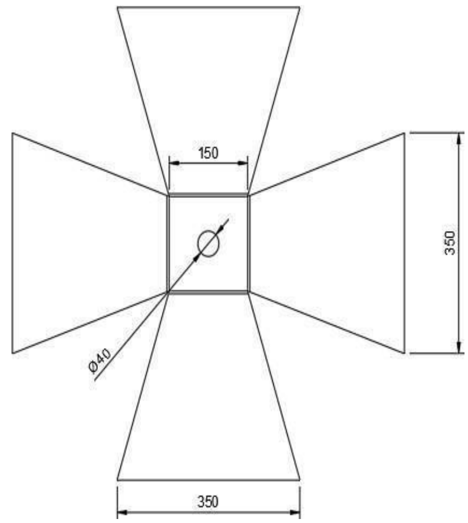
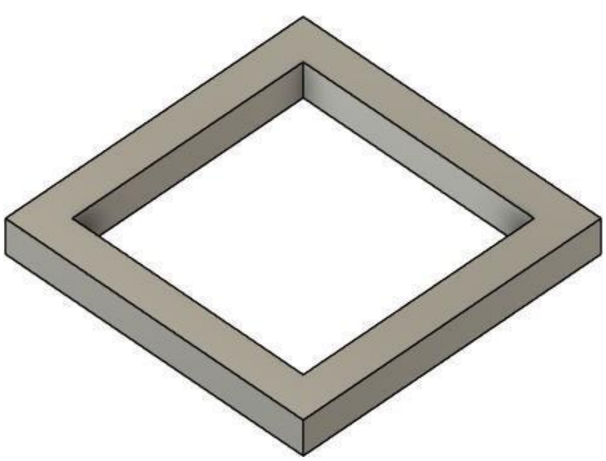


Figure 4.7: 3D Representation of Hopper

Figure 4.8: 2D Representation of Hopper

The hopper is the first feeder and is the part of the automated sorting system that holds raw arecanuts to help control the release of nuts onto the discharge channel and categorization. The flow of nuts from the hopper to the discharge channel is both continuous and controlled, The structural layout of the hopper and the dimensions of the hopper can be seen in Figures 4.7 (3D view) and 4.8 (2D view) to show the functional geometry and overall relationship to the system.

4.4 Hopper Frame



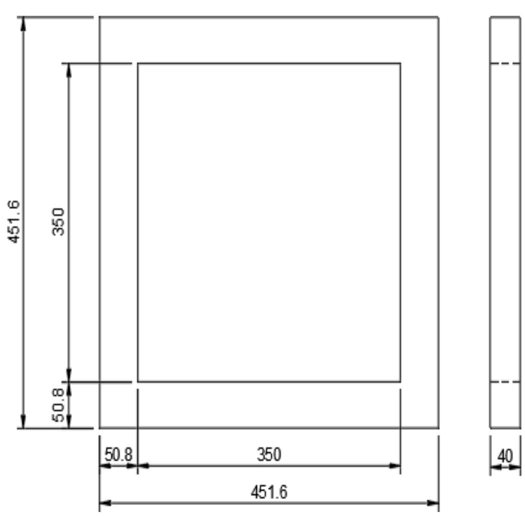
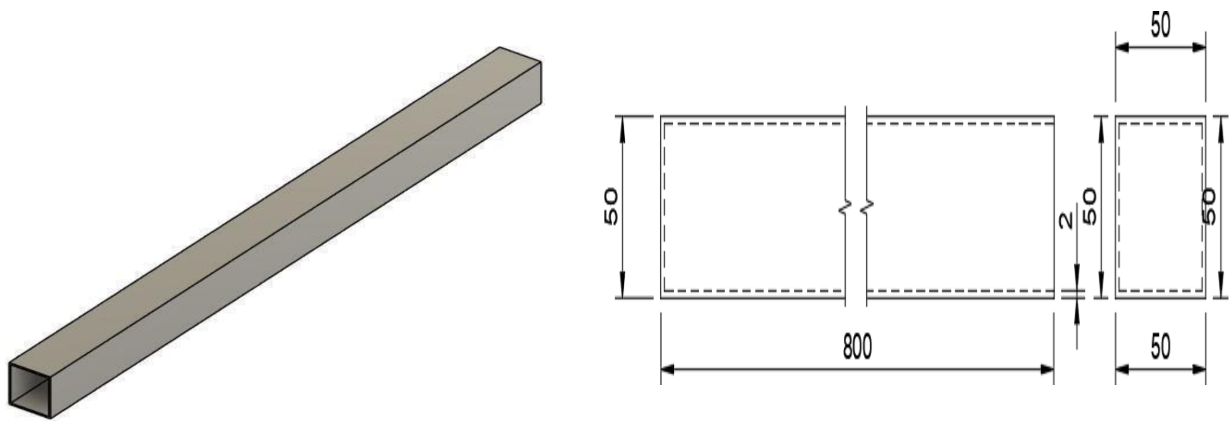


Figure 4.9: 3D Representation of Hopper Frame

Figure 4.10: 2D Representation Hopper Frame

The hopper frame is an important support structure for holding the hopper securely in place to provide stability when feeding arecanuts onto the system. The hopper frame structure is designed to absorb the dynamic loads and vibrations that occur when dispensing nuts, which enables the system to operate safely and reliably. The layout and design of the hopper frame are illustrated for 3D and 2D representations in Figure 4.9 and Figure 4.10 respectively as attached to the machine base.

4.5 Rectangular Bar Legs



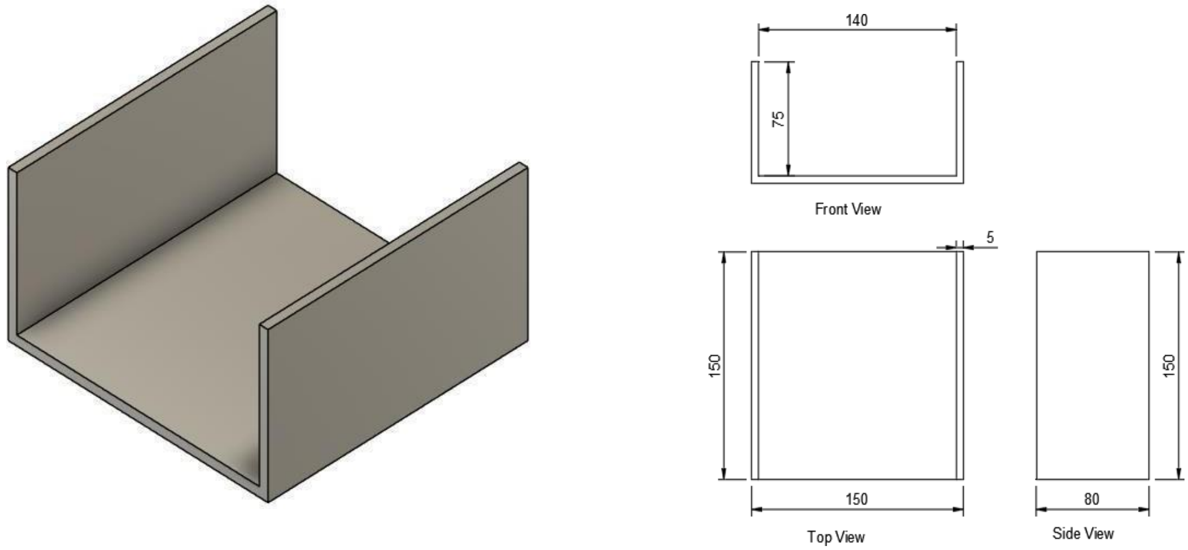
(a) (b)

Figure 4.11: 3D and 2D Representation of Rectangular Bar Leg (a-b)

The rectangular bar legs are the main support elements for the entire automated areca sorting system and provide stiffness as well as height to an ergonomic working height. The legs shown in Figure 4.11 (3D) and (2D) were materialized from robust materials to withstand the mechanical forces of all the subsystems mounted above. The structural configuration and design detailing will be evident in future drawings as mounting means and dimensions are shown.

4.6 Discharge Channel

The discharge channel serves as a crucial pathway that allows every arecanut to flow from the hopper smoothly into the image acquisition zone, sustaining a single nut flow for proper classification. It is sloped using servo actuation in order to limit the overlap or possible jamming of the arecanuts after arecanuts left the hopper. The structure and arrangement of the discharge channel is displayed in Figures 4.12 (3D) and (2D) where the orientation and location with respect to the other adjacent subsystems is shown.



(a) (b)

Figure 4.12: 3D and 2D Representation of Discharge Channel (a-b)

4.7 Rubber Foot

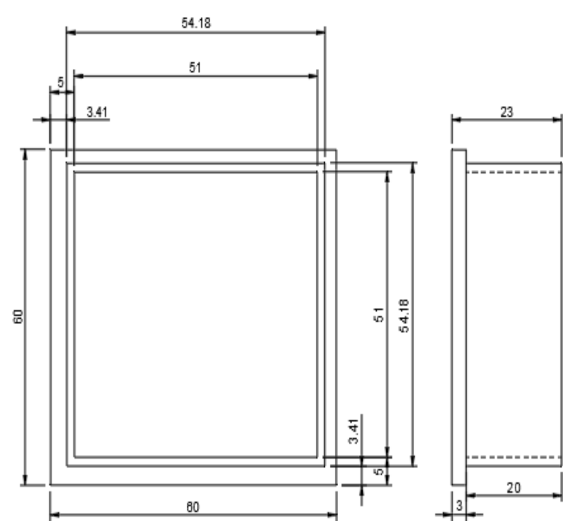
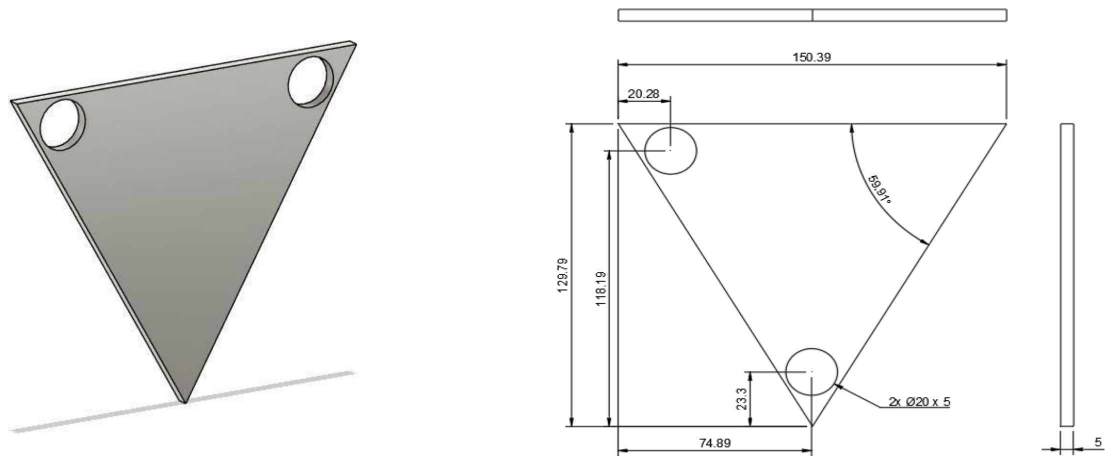


Figure 4.13: 3D Representation of Rubber Foot Figure 4.14: 2D Representation of Rubber Foot

A rubber foot has been provided on the bottom of each leg to improve stability and dampen vibrations during the operation of the system, which protects internal components and extend durability. The rubber foot also prevents the system from sliding on smooth surfaces, allowing the system to remain stationary. The rubber foot installation designs and locations are detailed in Figure 4.13 (3D view) and Figure 4.14 (2D view).

4.8 Triangular Bracket

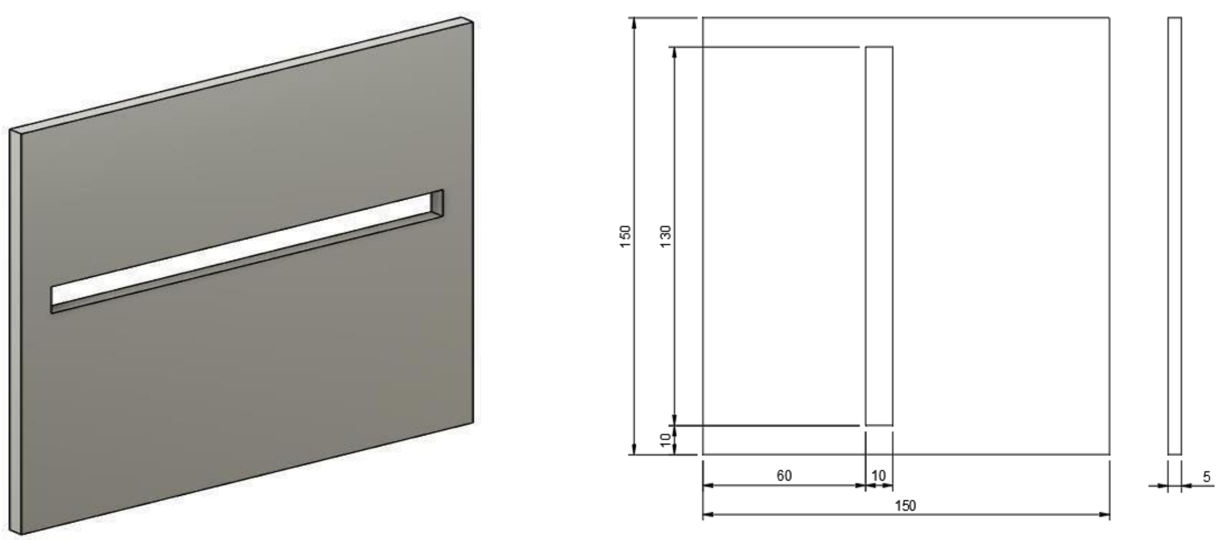


(a) (b)

Figure 4.15: 3D and 2D Representation of Triangular Bracket (a-b)

The triangular bracket is a key structural element used to reinforce the joints between the vertical legs and horizontal supports, ensuring rigidity and stability of the entire frame. It helps distribute mechanical stresses evenly during operation. The design details are clearly shown in Figure 4.15 (3D view) and (2D view).

4.9 Rectangular Slotted Panel

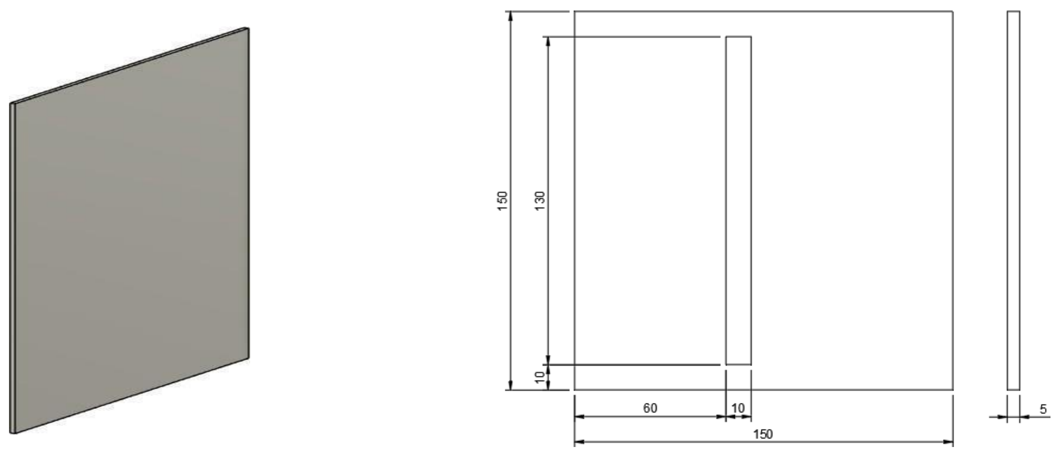


(a) (b)

Figure 4.16: 3D (a) and 2D (b) Representation of Rectangular Slotted Panel

The rectangular slotted panel serves as a mounting or supporting surface within the frame, often used for holding components or facilitating cable routing. Its slot design allows for flexible positioning of fixtures and improves airflow or material passage. The structural and dimensional details are depicted in Figure 4.16(a) for the 3D view and (b) for the 2D representation.

4.10 Side Cover Plate

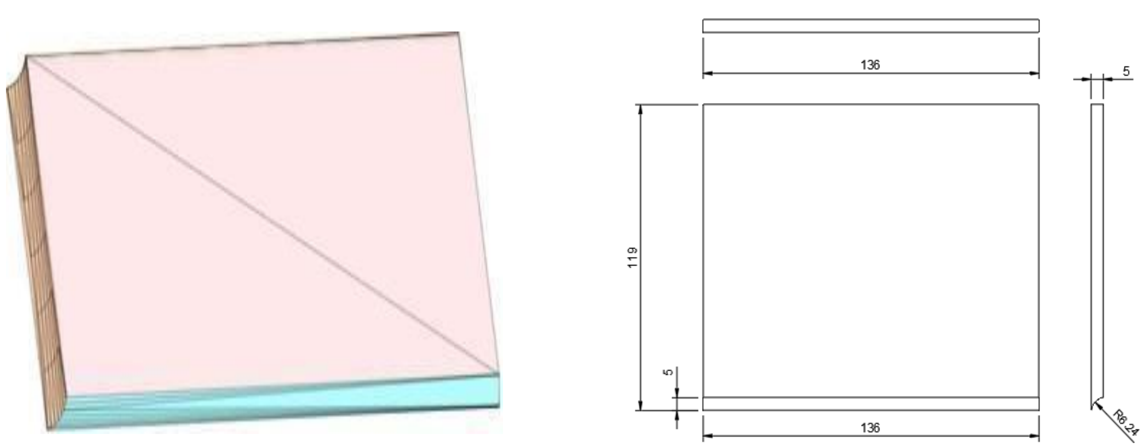


(a) (b)

Figure 4.17: 3D (a) and 2D (b) Representation of Side Cover Plate

The side cover plate is used to enclose and protect the internal components of the automated areca sorting system, ensuring safety and containing dust ingress. It is most often affixed to the sides of the frame structure and contributes to the overall rigidity as well as aesthetically finishes the system. Displayed in Figure 4.17 (a) is the 3D view of the component, while its precise dimensions and layout are displayed in Figure 4.17 (b).

4.11 Discharge Flap

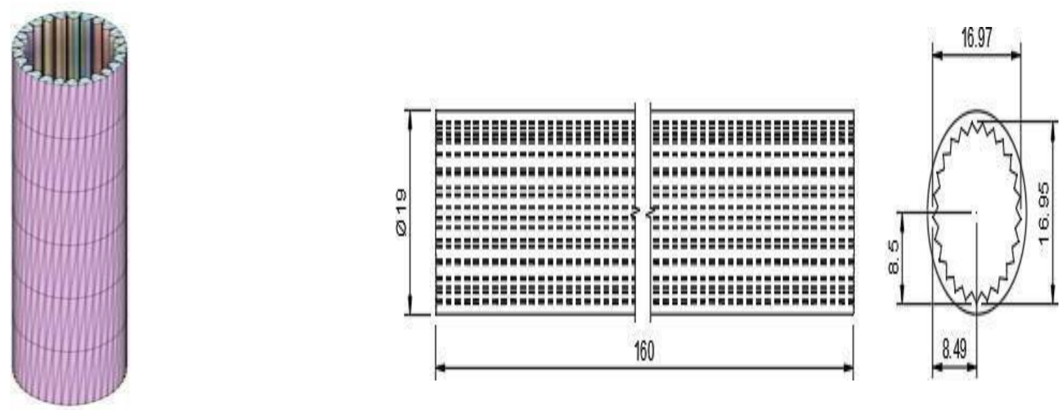


(a) (b)

Figure 4.18: 3D (a) and 2D (b) Representation of Discharge Flap

The discharge flap is a mechanical diverter that diverts the classified arecanuts into the correct bin (ripe or unripe) based on AI predicted classification. The flap is mechanically-operated by a servo motor and intended to rotate quickly with low resistance to allow for proper sorting. The structural design of the flap is seen in Figure 4.18 (a) and the dimensional views are provided in Figure 4.18 (b).

4.12 Splined Drive Shaft



(a) (b)

Figure 4.19: 3D (a) and 2D (b) Representation of Splined Drive Shaft

The splined drive shaft is the part that transmits rotary motion from the servo motor to the discharge flap with synchronized angular positioning. The splined design provides increased surface area for more grip, grip efficiency and torque energy transfer during actuation. The drive shaft in 3D and 2D can be found in Figure 4.19 (a) and (b), respectively.

4.13 Design Components and Material

Table 4.1: List of Components and materials

Sl. No	Name of the Components	Material	Quantity
1.	Hopper	Mild Steel	1
2.	Hopper Frame	Mild Steel	1
3.	Rectangular Bar Legs	Mild Steel	4
4.	Discharge Channel	Mild Steel	2
5.	Rubber Foot	Rubber	4
6.	Triangular Bracket	Mild Steel	2

7.	Rectangular Slotted Panel	Mild Steel	2
8.	Side Cover Plate	Mild Steel	1
9.	Discharge Flap	PLA	2
10.	Splined Drive Shaft	PLA	2

CHAPTER 5

AUTOMATED SYSTEM MODELLING

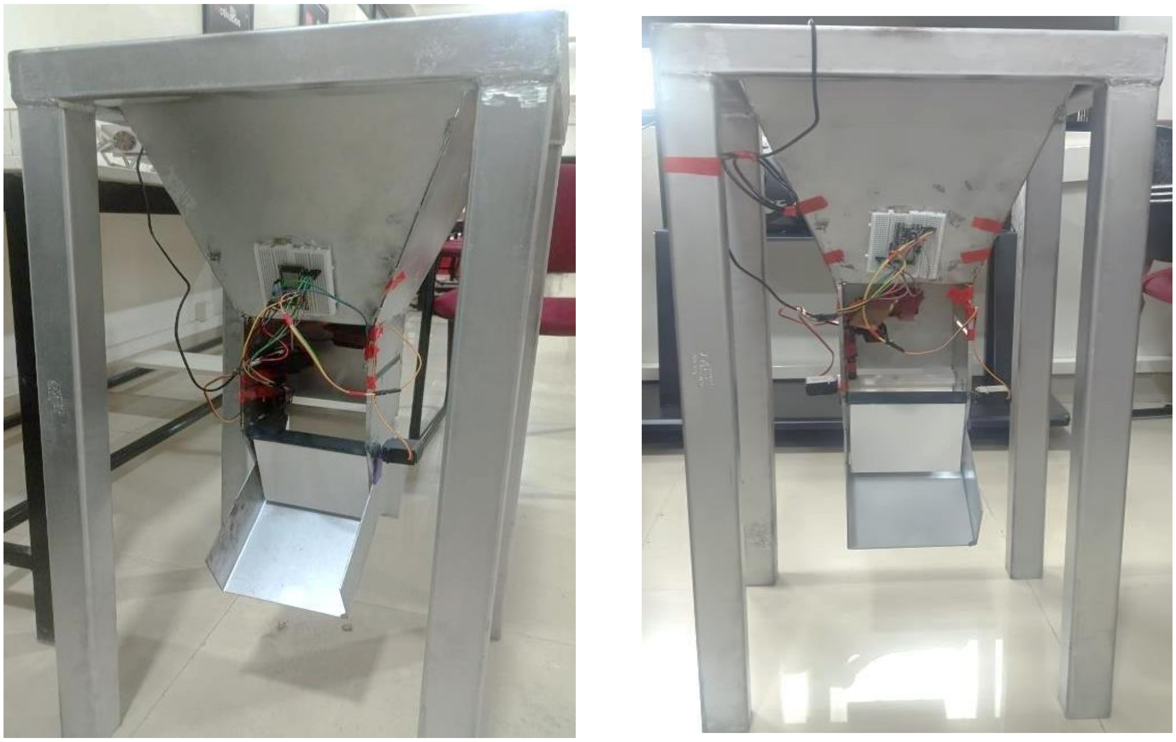
5.1 Automated System Modelling Overview

Automated system modelling involves the conceptualization, design and simulation of a system that performs a sequence of operations with minimal human involvement. When applied to the Automated Areca Sorting System, the modelling integrates mechanical, electronic and AI components to create an operational and intelligent sorting unit. The modelling concept initiates with the identification of physical and functional requirements including handling of arecanuts, division of ripe and unripe arecanuts and actuation of sorting flaps corresponding to arecanut classification.

Modelling in Autodesk Fusion 360® allows for a precise 3D modelling of desired mechanical components including the hopper, frame, discharge channel, and sorting mechanism. The models can then be used to verify the components structurally and functionally by performing simulation of part motion and verification of assembly.

(a) (b)

Figure 5.1: Complete Fabricated System (a - b)



(a) (b)

Figure 5.2: Automated Areca Sorting System Integrated with Electronics Devices

The physical object depicted in Figure 5.2 represents the finished prototype of the system design and modelling already described in previous stages of the project. The picture shows the completed prototype of the automated arecanut sorting system or integrated machine which consists of mechanical, electrical, and software components assembled into a unified system. The figure 5.1 frame is formed using mild-steel to ensure strength and durability and supports a hopper mechanism that directs the arecanuts down towards the sorting compartment. The components inside the integrated machine include an ESP32 microcontroller for actuation control that is connected to servo motors that actuate the sorting flaps. The Raspberry Pi connects to the webcam and controls the image acquisition and image classification tasks using a pre-trained deep learning model that was validated on a PC as part of the pre-design stage. The actuators and controlling electronics are powered directly from an external 24V supply inside the enclosure. The hopper and discharge channel guided the nuts towards the sorting compartment where the machine-made real-time classification decisions allowing the separation of ripe and unripe nuts. The integrated machine demonstrates a clear conversion from software-based CNN model validation to a fully functional hardware prototype recognizing that the original design approach was effective and that the prototype was ready for a real-world testing phase to determine performance in real-world agricultural sorting applications.

5.2 Electronic System Components

The electronic subsystem is at the core of control, data processing and actuation in the sorting system. The electronic subsystem is the center of control, data processing and actuation for automated arecanut sorting. Raspberry Pi, which is operates as the computer to run the deep learning classification model and obtain images from a connected webcam. the ESP32 microcontroller operates as the actuator controller, taking classification results received from the Raspberry Pi using serial communication and sending information needed to achieve the right mechanical action, Servo motors which are responsible for moving the sorting flaps to properly change the direction of nuts into their respective bins, webcam and the images it captures are vital for continuous visual input needed for classification and the external regulated power supply, which is used to create stable and regulated voltage and current to all electronic and actuation components for constant electronic control. Collectively these components create an integrated electronic system needed to link high-level data processing to low-level mechanical control in the required real-time environment for sorting.

5.2.1 Esp 32 Microcontroller

Figure 5.3: ESP32 Microcontroller [121]

The Figure 5.3 ESP32 microcontroller is a fundamental component in both embedded and automation systems including with AI models for sorting. It is low-cost, low-power and very adaptable microcontroller with Wi-Fi and Bluetooth capabilities making it suitable for wireless communication and IoT applications. In a sorting system, the microcontroller will obtain the classification results from the AI processing unit (through serial communication), processes the classification results and generates control signals for the actuators (servo motors) via PWM (Pulse Width Modulation) to produce the physical sorting based on the AI prediction.

Its capability in real time, multitasking, and peripheral communications makes it a crucial component for coordinated mechanical operations in an automated sorting context.

Specifications:

Table 5.1: Specifications of ESP32 Microcontroller

Sl. No	Parameter	Features
1.	Microcontroller	32-bit LX6 with 600 DMIPS
2.	Frequency (MHz)	160 to 240
3.	Bluetooth	Bluetooth 4.2 and BLE
4.	GPIO Pins	38 (32 Usable Pins)
5.	PWM	16 Channels
6.	ADC	12-bit
7.	TWAI	Yes
8.	Power Consumption	Approx. 80~90mA
9.	Working Temperature	-40ºC to 125ºC
10.	SPI Flash	32Mbit
11.	Current	80 mA
12.	Supply Voltage	2.2 V ~ 3.6 V
13.	Dimensions	5 x 4 x 3cms
14.	Weight	15 grams

5.2.2 MG995 Servo Motor

Figure 5.4: MG995 Servo Motor [121]

The Figure 5.3 MG995 servo motor is a metal gear servo that has high torque and is widely used in robotics and automation projects because of it is durable, precise and reliable. The MG995 servo motor works by receiving a PWM (Pulse Width Modulation) signal which is sent from the controller (ESP32) to find the desired angle which is usually represented in the 0° to 180° region. The potentiometer inside of the servo motor monitors the position of the shaft and sends feedback to the internal control circuit and provides a way for the internal control circuit to control how the servo motor is moving to reach its target angle. In my project I propose the implementation of a MG995 servo motor as an important actuator to control the mechanical components such as discharge flap in the automated sorting system in order to direct items such as arecanuts into the allocated bins based on the outcome of the classification process. Also, because of the high torque capability of the MG995 these servos have the ability to carry heavier loads or make more rapid movements when compared to cheaper or non-metal servos. In this way the system can continuously move and operate the sorting mechanism accurately and with maximum efficiency.

Specifications:

Table 5.2: Specifications of MG995 Servo Motor

Sl. No	Parameter	Features
1.	Item Type	Tower Pro Servo Motor
2.	Rotation	180°
3.	Torque Rating (kg-cm):	10 to 12
4.	Model No	MG995
5.	Operating Voltage (VDC)	4.8 ~ 7.2
6.	Operating Speed @4.8V	0.20sec/60°
7.	Operating Speed @6.6V	0.16sec/60°
8.	Stall Torque @ 4.8V (Kg-Cm)	10
9.	Stall Torque @6.6V (Kg-Cm)	12
10.	Operating Temperature (°C)	-30 to 60
11.	Gear Type	Semi-Metal

12.	Length(mm)	40.5
13.	Width (mm)	20
14.	Height (mm)	44
15.	Weight (g)	55

5.2.3 Web Camera

Figure 5.5: HD Webcam [121]

The 2K high-definition USB webcam is the image capturing device, capturing real-time images of arecanuts on the Automated Sorting System. The camera's large field of vision covers all arecanuts in the entire sorting area and integrated noise-reducing microphones and plug-and-play compatibility for seamless integration into the processing unit, therefore allowing for the reliable collection of high-quality input data for accurate classification and sorting. The Figure 5.5 describes the 2K Webcam is a modern high-definition USB camera optimized for use on desktops or laptops that features sharp QHD (2560×1440) video at 30 fps image quality utilizing a 4MP CMOS sensor with a 4-layer optical lens. The camera has an extra-wide 126° field of view, so it's a great option to capture detailed visuals in an online meeting, live stream or video recording. The 2K Webcam has dual omnidirectional microphones with noise reduction to further enhance your capture and finely engineered physical privacy shutter to ensure your gaze is secure before you press 'play.' The USB makes it a plug-and-play camera and Webcam is compatible with Windows, macOS and Linux platforms, so using the 2K Webcam is super easy. The 2K Webcam is a great webcam to use if appreciable professional communication is required, and we can confidently state that the 2K Webcam is capable of delivering both high quality video and audio in a compact, streamlined user-friendly experience.

Specifications:

Table 5.3: Specification of Webcam

Sl. No	Parameter	Features
1.	Driving Type	Plug and play without driver
2.	Output Format	MJPG/YUY2/H264
3.	Sensor Type	CMOS
4.	Scintillation Control	50HZ
6.	USB Cable Length	140cm
7.	Resolving Power	1920×1080
8.	Interface Type	USB 2.0
9.	Microphone	Built-in Microphone
10.	Product Size	75×53×49mm
11.	Support Device	Windows (2000), Mac OS, Linux operating system

5.2.4. Buck Converter

Figure 5.6: 5V 5A Buck Converter [121]

The Buck Converter module consists of high-frequency switching and an inductor to help keep the voltage steady and consistent under load. The Figure 5.6 buck converter that reduces a higher input voltage (up to 24V) to a stable 5V output. The Buck Converter module is switching regulated, which minimizes heat loss and also increases efficiency. Buck Converter is a very efficient power module that provides a even higher current 5 amps which is excellent to power micro controllers, sensors, servos and other devices. The Buck Converter module also safe and reliable with built-in overcurrent, overheating and short-circuit protection. This module is very compact and has an important function to regulate voltage for many embedded programmable systems or robotic motion systems.

Specifications:

Table 5.4: Specifications of Buck Converter

Sl. No	Parameter	Features
1.	Height (mm):	10
2.	Weight (g):	22
3.	Shipping Weight	0.025 kg
4.	Shipping Dimensions	7× 3 × 1 cm

5.2.5 Power Supply

The power supply system is intended to deliver a stable and trustworthy supply of electrical energy to all the subsystems in the Automated Areca Sorting System. The main power source is a Rechargeable Lithium-ion (Li-ion) Battery Pack that is rated at a nominal voltage of 11.1V and a capacity of 4Ah shown in figure 5.7. The three-cell battery pack is termed a 3S (three cells in series) providing sufficient voltage to power the primary variable components; for example, the ESP32 microcontroller, MG995 servo motor and, the image acquisition camera. The high currents typical of battery pack capacities (4 Ah) enable the system to be powered for long periods of time without needing to recharge on a regular basis. The battery pack is small, light and rechargeable making it perfect for mobile or semi- mobile robotic systems operating within an agricultural setting. Safety systems such as a battery management system (BMS) are usually included within or with the battery to prevent overcharging, deep discharge and short circuit. The power is wired with color- coded terminals (red is positive; black is negative) ensuring no confusion when connecting with the control circuit.

Figure 5.7: Rechargeable ion Battery

Specifications:

Table 5.5: Specification of Rechargeable ion Battery

Sl. No	Parameter	Features
1.	Battery Type	Lithium-Ion (Li-Ion) Rechargeable Battery
2.	Nominal Voltage	11.1v (3s Configuration – 3 Cells in Series)
3.	Nominal Capacity	4 Ah (Ampere-Hour)
4.	Energy Storage	44.4 Wh
5.	Model Number	1114352p
6.	Output Terminals	2-Wire (Red: Positive, Black: Negative)
7.	Charge Cycle Life	Charge Cycle Life
8.	Output Current Capability	Output Current Capability
9.	Use Case	Use Case
10.	Rechargeability	Rechargeability
11.	Protection Features	Protection Features

5.2.6 Raspberry Pi 4

Figure 5.8: Raspberry pi 4 Model B

The AI and vision components of the Automated Arecanut Sorting System are performed on the Raspberry Pi 4 Model B. The Figure 5.8 illustrates the Raspberry Pi 4 Model Characteristics. Raspberry Pi 4 has a quad-core ARM Cortex-A72 processor and 4 GB RAM that allows it to have enough computing power for lightweight inference using TensorFlow Lite. It is running on Ubuntu 20.04 LTS, which is stable and flexible in the Linux environment, ideal for the Robot Operating System (ROS) Noetic. ROS is a flexible and powerful middleware framework to allow modular development and communication between the robotics platform's subsystems. The Raspberry Pi is running only ROS and ROS is handling real- time image stream acquisition from the connected USB webcam, running Gaussian filtering and virtual background removal using the OpenCV and the Rembg libraries and forwarding the images through a TFLite based CNN image classification model, then classifying arecanut as ripe or unripe. The resulting classification is passed via serial connection to the ESP32 microcontroller where mechanical actuation is initiated and process is completed. The integration of the Raspberry Pi 4 with Ubuntu and ROS not only kept the software development platform robust, but also allowed the data and communication to be in-sync, provided numerous options for modular expansion in add-on capabilities; and but most importantly provided real-time strong performance tailoring it to an edge-computing platform for affordable agricultural automation solutions.

Specifications:

Table 5.6: Specification of Raspberry Pi 4

SI. No	Parameter	Features
1.	Item Type	Development Board
2.	Model Type	Pi 4 B
3.	RAM Capacity	4GB
4.	Video and Sound	2 × microHDMI ports, 2- Camera Port, 2- Lane Display Port, 4- Pole Stereo Audio and Composite Video Port
5.	Clock Speed	1.5 GHz
6.	Micro SD Card Slot	32G Micro SD Card Memory
7.	Operating Temperature	0 to 50 ℃
8.	Weight	52 gm

5.3 AI Implementation

The introduction to Artificial Intelligence into the domain of automated system modelling has shifted the paradigm of working of automated sorting systems like the Arecanut Sorting System from human based intuition to intelligent decision making and action in real time. In this project, AI will play the central role within the Automatic Classification and Control architecture to enable the sorting of areca nuts autonomously based on their state of ripeness in a consistent and repeatable manner. At the heart of this system will be a deep learning model that utilizes Convolutional Neural Networks (CNNs), specifically the VGG16 architecture. This model will be trained on a unique, labeled dataset featuring images of ripe and unripe arecanuts under varying light and backgrounds for generalizability and robustness.

The dataset undergoes comprehensive pre-processing to ensure optimal quality before its use as input. This can include background removal with the rembg library to allow the nut to be isolated from background clutter, a Gaussian filter to diminish high-frequency noise, and normalization to allow for pixel values to be scaled similarly for both classes over the dataset. This important step ultimately assists the model to focus on visually important features, such as differences in texture, color and shape between ripe and unripe arecanuts. Once the model has been successfully trained and validated, it is saved in .h5 format, and is later converted to TensorFlow Lite (.tflite) model in order to achieve edge inference on resource sufficient devices.

During operation, the AI module will be run on an edge device, of a Raspberry Pi 4, for example, and configured to capture images in real-time using a USB Webcam. Image capture will provide images to the RPi configured to resize the images to the model's expected input size, then pass these to the lightweight CNN model for inference. The prediction output of either "RIPE" or "UNRIPE," is then transmitted, in real time, to a control unit of choice, for example, an ESP32 microcontroller, via serial communication. The ESP32 would be configured to receive the "RIPE" or "UNRIPE" label and make an appropriate actuation, usually a servo motor to direct the nut into the correct bin or path. The system encompasses vision-based AI, real-time image acquisition, and embedded actuation inside a closed loop automation pipeline. The real-time action provided by the system is perhaps its biggest selling point as it creates edge classification that enables faster response times with minimized latency. The implementation of input resizing, model quantization, and pruning is an effort to lower the computational burden of producing such timely responses at the detriment of model accuracy. Additionally, when an event is classified by the model, the system automatically saves each annotated image with timestamps locally or in a cloud-based database, allowing for traceability, analytics and model retraining in the future.

The system is modular to support scalability. As more data is collected including novel ripeness attributes, defects and disease, the model should be retrained to continue classifying the nut variety with different user attributes. This allows for transfer learning or fine-tuning whenever data is collected, the model can be further improved. Furthermore, if the system could integrate an external services, we could remotely monitor the system to determine its health or get batch statistics. The hardware components chosen (Raspberry Pi, ESP32, camera, and servos) are open- source options that are reasonably inexpensive and energy efficient to operate, allowing this system to fit in the generalist, small farm market as well as the industrial facility planning designs. Automating the sorting of arecanut intelligently increases throughput while simultaneously lessening dependence on manual labor, addressing issues such as worker fatigue, inconsistency, and high costs. By offering the automation of a subjective visual inspection with an objective AI data-driven decisioning process, the AI mechanized process acts as a reliable, scalable and intelligent agricultural post-harvest processing system. The closed-loop nature of the AI-mechatronics system corresponds to higher level goals of Industry 4.0 as it promotes smart agriculture with the integration of AI, IoT, and embedded control systems.

5.4 System Circuit Design

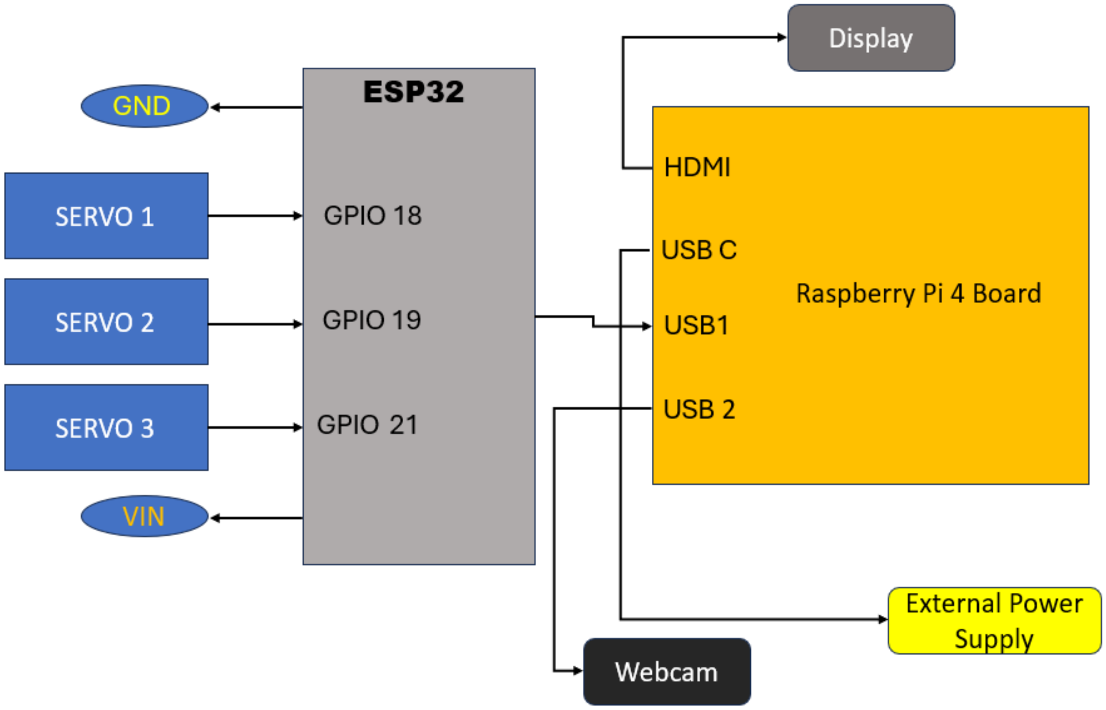


Figure 5.9: Schematic Representation of ESP32 and Raspberry Pi 4

This figure 5.10 describes the schematic of the connection between a Raspberry Pi 4 and an ESP32 microcontroller used in an automated control system. The ESP32 is connected to three servo motors through their respective GPIO pins, where those motors would be controlled or activated. The Raspberry Pi 4 is at the processing unit, connected to a webcam (for image processing) and a display (for output). The external power cable is used to ensure reliable operation of the servos and the connected components to either the Raspberry Pi 4 or ESP32.Servo Motors -

Servo 1 signal line to GPIO18 on the ESP32.

Servo 2 signal line to GPIO19 on the ESP32.

Servo 3 signal line to GPIO21 on the ESP32.

All servos share a common ground with the ESP32 and powered at a regulated 5V external power supply to give each servo sufficient current.

1. ESP32 Microcontroller -

The ESP32 serves as the actuation control unit. It accepts classification commands from the Raspberry Pi via serial connection through USB and then generates PWM signals to actuate the servo motors.

2. Raspberry Pi 4 -

The Raspberry Pi 4 is the AI processing unit and connects to:

Webcam (USB) - for live image acquisition of arecanuts.

Display (HDMI) - for monitoring and debugging.

ESP32 (USB) - for sending classification results.

The Raspberry Pi runs the TensorFlow Lite model for detecting ripe/unripe and sends the instructions for actuation to the ESP32.

3. External Power Supply -

The servo motors draw fairly high currents during operation, so power them from an external 5V supply that can supply 2A or higher. ensures that voltage drops do not cause the ESP32 or Raspberry Pi to reset. The ground for the external supply is tied to the ESP32 and Raspberry Pi ground connections as a shared reference.

4. Flow of Operation -

A webcam captures the image of an arecanut. A Raspberry Pi runs the AI model and decides whether the nut is ripe or unripe, then we pass that classification to the ESP32, which manipulates the right servo motor to sort it accordingly.

This allows for a clear delineation of tasks - the Raspberry Pi takes care of the heavy image processing and sends sorting commands to the ESP32 in order for the ESP32 to control the motor in low-latency without overwhelming the Raspberry Pi.

5.5 Multi Machine Communication

The system architecture employs a distributed multi-machine architecture based on the Robot Operating System (ROS) to accomplish accurate and efficient sorting of areca nuts. A webcam is connected to the Raspberry Pi, which provides the real-time acquisition of images of the nuts. The images were published as a ROS topic over local network to a more performant Laptop/PC. The laptop subscribed to this published topic in, order to process the incoming frames with a deep learning model, to classify all of the nuts as either ripe or unripe. Lastly, the classification result was published back to the Raspberry Pi on the classification result topic. The Raspberry Pi received the classification result, and forwarded the necessary control commands to, the ESP32 microcontroller through wired serial. The ESP32 microcontroller was interfaced directly with the servomotors and performed the sorting action, actuating the mechanical components to direct where the nuts went. This technology architecture was purposely designed to reduce processing latency with the computationally intensive AI tasks of sorting being done by Laptop, and the Raspberry Pi was relegated to handling sensors, routing communication, and controlling actuation, while providing a reliable and efficient real-time responsive performance.

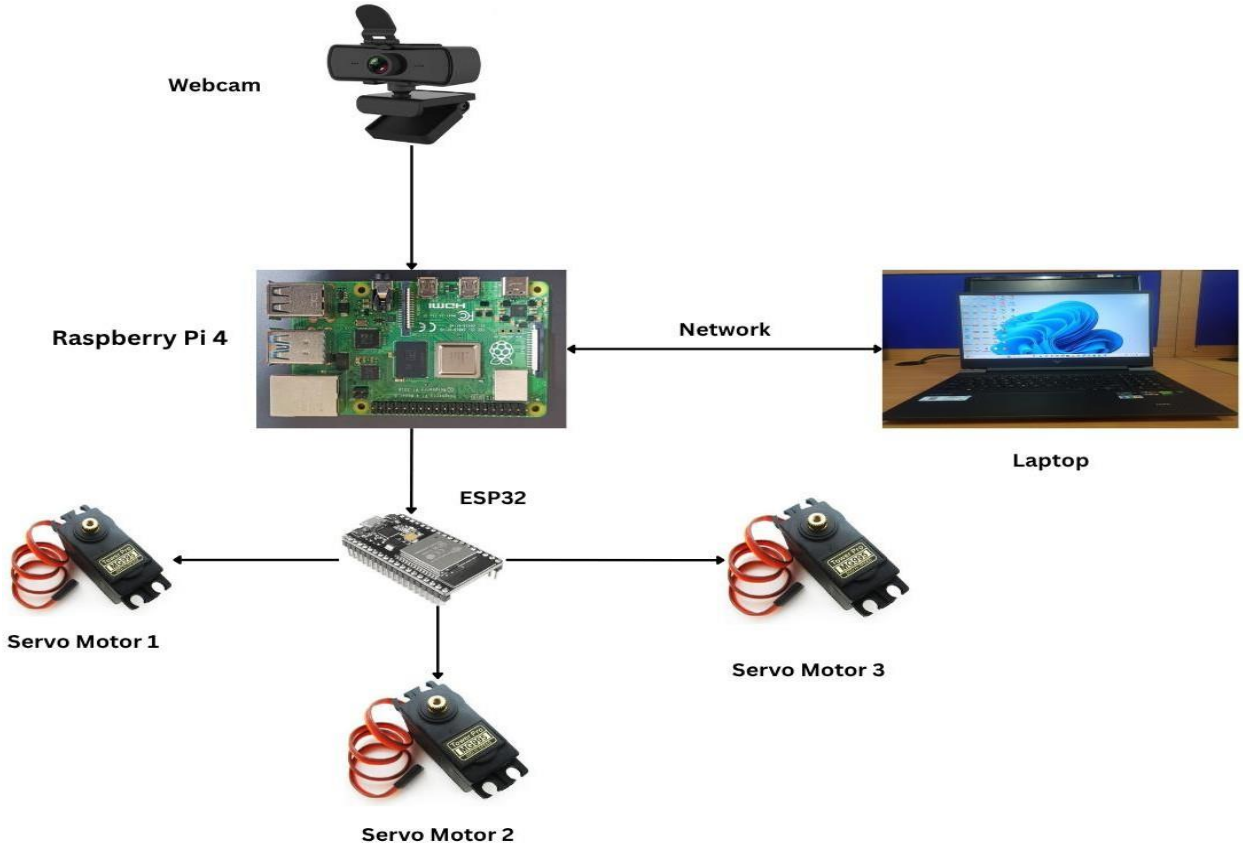


Figure 5.10 Multi Machine Communication Architecture

This Figure 5.11 illustrates a multi-device ROS based areca nut sorting system where image capture, image processing, and actuation commands have been separated across devices for increased efficiency.

1. Webcam → Raspberry Pi

The webcam is directly connected to the Raspberry Pi. It sends real time images of the areca nuts via USB, which are captured by the webcam.

2. Raspberry Pi → Laptop/PC (Network)

The Raspberry Pi publishes the raw image data for dissemination via a network (Wi-Fi) to a more capable device (the Laptop/PC) for the AI processing tasks.

3. Laptop/PC → Raspberry Pi

The Laptop/PC subscribes to camera and with a deep learning model it processes the frames and classifies each nut and publishes the result as for the Raspberry Pi to subscribe to receive.

4. Raspberry Pi → ESP32 (Serial/Wired Connection)

Based on the classification the Raspberry Pi sends direct control commands via wired serial to the ESP32.

5. ESP32 → Servomotors

The ESP32 receives and executes the commands to control servomotors via GPIO pins, which actuate the mechanical sorting system thus physically separating nuts based on the classification. Only the Raspberry Pi and the Laptop/PC will use the network for data communications.

Motor control occurs locally on the Raspberry Pi with the Raspberry Pi → ESP32 wired connection, so that actuation can occur quickly and reliably. The workload to run the AI is offloaded to the more capable Laptop/PC, and the Raspberry Pi simply deals with the sensor communication and sends actuation commands.

5.6 Automated System Design Architecture

The Automated Arecanut Sorting System brings together mechanical, imaging, artificial intelligence, control, and actuation subsystems for the classification of arecanuts. Raw arecanuts are fed into a hopper, captured by a webcam, processed through background removal and noise filtering and classified with a VGG16 classifier. The ESP32 control unit interprets the results and controls servo motors to route the nut to the ripe or unripe bins. Figure 5.11 defines the overall th systems capable of quickly, accurately and fully automating the sorting of arecanuts.

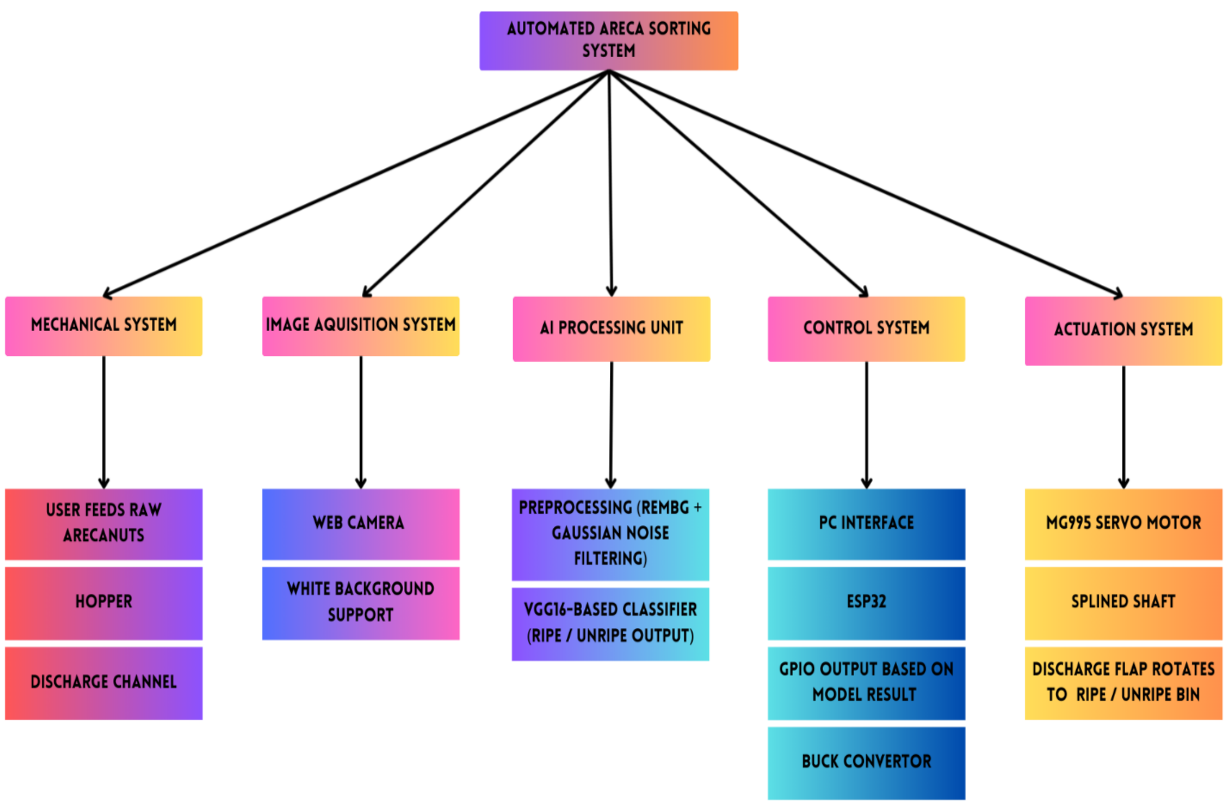


Figure 5.11: Automated System Design Architecture of Mechanical System

5.6.1 Mechanical System

The mechanical system forms the structural core of the Automated Areca Sorting System and provides guided travel of arecanuts through the classification process. The structure was made primarily from mild steel to provide strength and rigidity to the frame in order to guarantee mechanical stability through continual operation. The hopper is made from mild steel sheets which were riveted through butt doors. The hopper holds one batch of raw arecanuts and was designed for the retention of multiple nuts simultaneously.

The hopper slope takes advantage of gravity to lower the nuts one at a time into the one-in-one- out path of discharge.Controlling the discharge rate improves the overall sorting efficiency by reducing obstructions and ensuring smooth operation. Consistent picture quality was guaranteed because the stability of the acquired frames was unchanged. By giving the processing time dependable input, these design considerations enabled accurate classification.

The PLA discharge flap allowed for simple removal and modification in situations where static accumulation occurred, without necessitating whole refabrication because of wear. Accurate classification is facilitated by constant image quality, which is ensured by careful treatment of nuts. As a result, the design effectively blended operational flexibility with structural dependability.

5.6.2 Image Acquisition System

The image acquisition system is a very important subsystem for capturing high-definition images of each individual arecanut in real-time to enable accurate classification by the AI processing unit. In order to detect every arecanut as it moves through the camera's viewing area, a webcam is angled perpendicular to the arecanut flow route. In order to enable accurate alignment, flexible positioning, and prompt replacement during maintenance, the camera is installed inside a rectangular slotted panel. The splined drive shaft has a white PLA discharge flap attached to it to improve image clarity. This consistent background enhances contrast and facilitates precise identification.

The arecanut and its surrounds stand out sharply against the white backdrop, making it possible for the Remove background tool to efficiently eliminate background areas of comparable intensity. Depending on how it is set up, the camera can transmit pictures to the Raspberry Pi continually. All of these design components work together to guarantee that each arecanut is captured consistently and clearly, enabling precise inference using the AI model.

5.6.3 AI Processing Unit

The AI Processing Unit is the intelligent brain of the Automated Areca Sorting System, which is responsible for classifying the arecanuts as ripe or unripe based on visual characteristics. Once the Camera captures an image through the process pipeline, then the image The image is initially cleaned up by removing the backgrounds using the Rembg library, the Rembg is especially useful, as all images were acquired against a white background in the imaging acquisition zone which makes filtering the backgrounds quite easy. AI processing unit step eliminates unwanted disturbances caused by the camera and slightly blur pixel edges, generating cleaner and more consistent visual input for the AI model.

The VGG16 network was fine-tuned with a corrected dataset containing labelled samples of ripe and unripe arecanuts, enabling it to identify suitable visual features that differentiate the maturity levels. During interface, the model extracts relevant features from the processed image, evaluates dataset and assigns probability scores to each class. Based on each class probabilities, the Automated system determines whether the sample is ripe or unripe. The processed classification result is transmitted from Raspberry Pi to ESP32 microcontroller in the form of signal as defined by the system configuration.

5.6.4 Control System

The control system acts as the critical bridge between the AI-based decision-making unit and the mechanical actuator components of the Automated Areca Sorting System. In Automated Areca sorting system the control system acts as feedback module, which sends and receives the arecanuts classification results to mechanical actuators such as servo motors. The real time arecanuts detection like ripe and unripe results sends to ESP32 which act as control system for mechanical actuators. The Raspberry Pi 4 sends the output to ESP32 microcontroller via serial communication, which triggers the discharge flap to guide the arecanuts to their respective bins. The program is written and developed through embedded software such as Arduino IDE.

This subsystem was all about translating the classification decisions into fast, quick, and accurate physical outputs in a matter of milliseconds, while also guarding against indeterministic delays and faults, which can alter the sorting process. In my project, control system plays the vital role to perform the arecanut detection and sorting tasks very easily. The signal sends to servo through GPIO pins, which functions the motors and Also, buck converter used to reduce the excess voltage from the battery and regulates required voltage to the system.

5.6.5 Actuation System

The actuation system allows the execution of the final physical movement sorting the arecanuts according to the classification designation made by the AI processor. The system uses an MG995 high- torque servo motor because it has a great response time and decent torque for industrial automation applications. The servo motor is controlled with PWM (Pulse Width Modulation) signals sent from an ESP32 microcontroller that control the exact angular position of the discharge flap. The discharge flap is mechanically linked to the servo shaft with a splined connector to provide stability during rotation. When a ripe arecanut is recognized, the flap adjusts to direct it into the "ripe" bin with the same routine for unripe ones as well. After each nuts classification the flap returns to the neutral position to reset for the next nut. The mounting structure for the entire servo flap assembly was fabricated using 3D printed PLA parts for the flap linkage and shaft adapters, offering good strength, and flexibility. This subsystem operates in real time and is synced with the control logic allowing for trouble-free sorting.

CHAPTER 6

RESULTS AND DISSCUSSION

The procedure discussed in the previous chapter is applied to the dataset to achieve classification of real-time arecanut ripeness using deep learning techniques, distinct hyper parameters such as (horizontal flipping, vertical flipping) are applied to both the Custom Learning CNN and transfer learning using (visual geometry group) VGG16. The resultant images, obtained after employing various image processing techniques, serve as the training data for CNN models with diverse hyperparameter configurations. The focus is on detecting and classifying the ripe and unripe arecanuts, covering a comprehensive dataset ranging from the premature stages to advanced mature stages. Two approaches are utilized for model training: a deep learning model (CNN) and transfer learning with VGG16 using pre-trained leaf images. The augmented dataset comprises 2847 images distributed across two classes (Ripe and Unripe). The data is partitioned, with 80% allocated for training, 10% for validation, and 10% for testing, ensuring a robust evaluation of model performance.

6.1 CNN With Transfer Learning Using VGG16

The VGG16 architecture, pre-trained on a large dataset, is utilized for detecting ripe and unripe classification of arecanut. It consists of 13 convolutional layers and 3 fully connected layers, designed to capture essential features like color, textures, and patterns critical for classifying arecanuts as Ripe and Unripe. Convolutional layers use 3x3 filters with ReLU activation, while pooling layers reduce dimensionality. To enhance generalization and prevent overfitting, techniques like dropout, batch normalization, and data augmentation (rotation, scaling, flipping) are employed with accuracy monitored through iterative epochs.

Table 6.1 VGG16 Model Accuracy in MSE

SI. No	Model	Epoch Training	Overall Test Accuracy (MSE)
1	VGG16	15	0.984
2	VGG16	20	0.973

The above Table 6.1 provides a comparison of the VGG16-based deep learning models performance at differing training times, particularly the number of epochs and overall test accuracy utilizing Mean Squared Error (MSE). In the first example, the VGG16 model for perfected to fit data was trained for 15 epochs, achieving a highest test accuracy of 0.984 indicating an excellent classification performance with little to no predictive error. This shows the model successfully learned the distinguishing features of ripe and unripe arecanuts training time relatively short. In contrast, in the second entry, initially the same VGG16 model trained for 20 epochs, and an overall test accuracy of 0.973 The reduction in accuracy that may be explained by overfitting, where the model became too fitted on the train data and was not able to genuinely compare data it had not yet seen, that seems evident on test data. Overall, the take home message from these findings in the table is to be mindful of selecting the number of training epochs to obtain an optimal epoch number to gain an effective learning balance in a model that will generalize well. For the purposes of real time classification accuracy using the VGG16 depth neural networks age of 15 epochs seem more optimally compared to using a full 20 epochs.

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 7, 7, 512)	14,714,688
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 256)	6,422,784
dense_1 (Dense)	(None, 2)	514

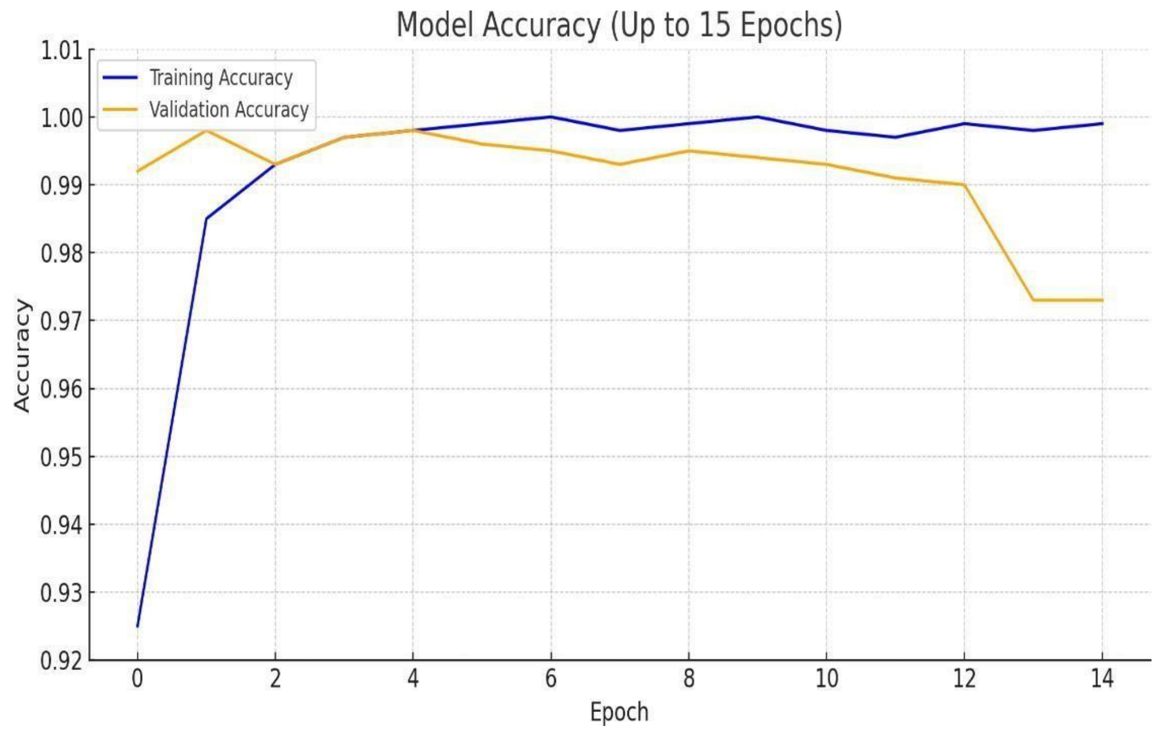
Figure 6.1: Model Architecture Overview of the Arecanut Classification Network based on VGG16

The deep learning architecture proposed for the classification of arecanut ripeness utilizes transfer learning methodology and adopts the VGG16 architecture as the weight extraction methodology. It is common to use architectures pre-trained on ImageNet as the foundation of a model, which provides a sufficient feature extraction layer but has the weights set as non-trainable to ensure features learned by VGG16 were not modified. Using the pre-trained model in this way generates an output feature map of shape (7, 7, 512) in which the 512 filters extract color, texture and shape information from the input images of arecanut. The new model's architecture adds a Flatten layer to turn the feature maps into a 1D vector of length 25,088 for input into our fully connected (dense) layers. The first dense layer of this architecture has 256 neurons, which introduces trainable parameters and adapts the features from the previous layer to the binary classification task of whether the arecanut is ripe or not. This dense layer uses a non-linear activation (usually ReLU), giving the model more capacity to learn. The last dense layer of the model has two neurons and implements a SoftMax activation to produce class probabilities associated with the two labels of ripe and unripe. In total, the model consists of 33,984,584 parameters, of which 6,423,298 can be trained and 14,714,688 are not trainable. The proposed architecture balances computational constraints with classification performance, suitable for use in operational fields as part of an automated arecanut sorting technology.

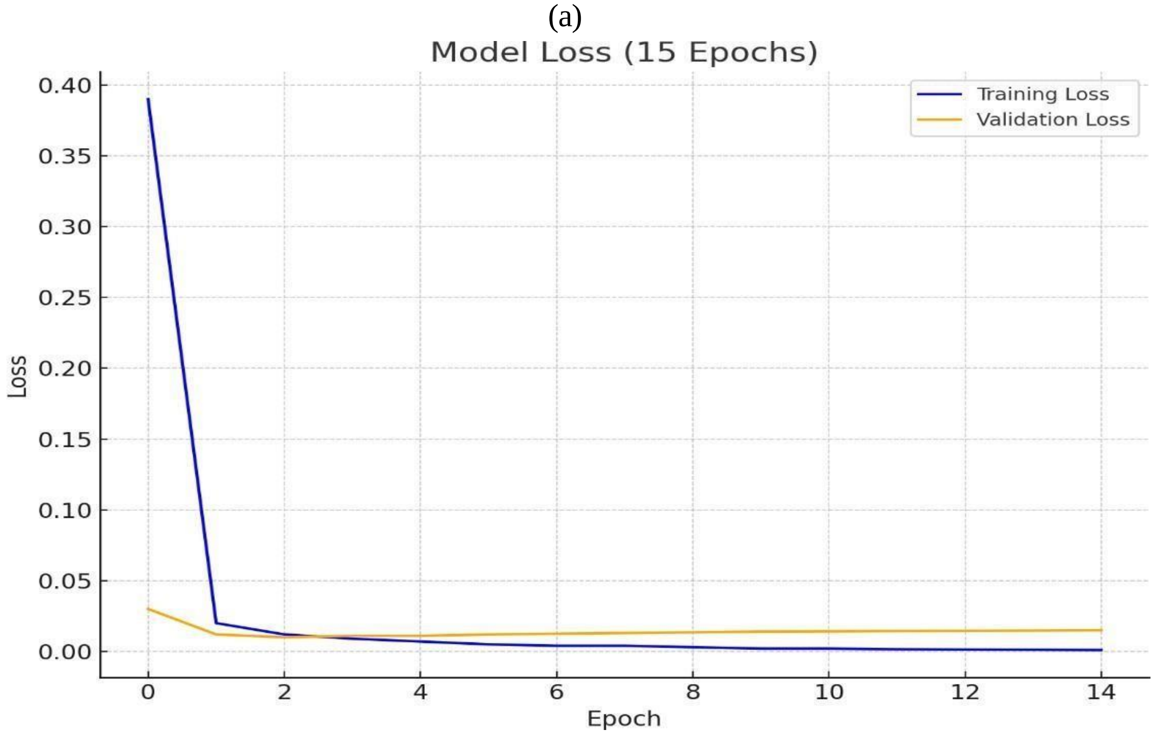
6.2 Results of Comparative Analysis

The Figure 6.2 demonstrates the increase in both training and validation accuracy across 15 epochs for the arecanut classification model. Initially, the training accuracy begins at about 92.5% and progresses rapidly to close to 100% prior to the 8th or 9th epoch. This suggests that the model is learning from the training data effectively. The validation accuracy starts at a high number of about 99.2% and remains stable during the initial epochs. However, after around 11 epochs, the validation accuracy makes a small decline, and by the 13th and into the 14th epoch, it declines to close to 97.3%. This is a classic example of using the training and validation accuracies to indicate that the model is overfitting- the model is memorizing the training data but is now unable to generalize using unseen data. The data presented indicates that there is some period of balance and optimal model performance that occurred between the 5th and 10th epochs, whereby both the training accuracies and validation accuracies are high and have a similar trend. Continuing to train the model beyond these 10 epochs leads to diminishing returns on generalization. From this analysis and the results, it seems that the optimal number of training epochs of the current model lies in the range of 8-10 epochs so as to avoid overfitting and still achieve good accuracies.

The training and validation loss trajectories of the arecanut classification model are displayed in Figure 7.2 over the 15 epochs. The loss value is on the vertical axis and the number of epochs is on the horizontal axis. The trends show that the training loss is high (0.39) at epoch 0, and indicates the predictions' initial errors, which, per epoch, show the loss decreasing sharply within the first 2-3 epochs, followed by a tapering decrease to near-zero loss values. This suggests learning of the training dataset has occurred. In comparison to the training loss, the validation loss is originally low (0.03) but still within a stable trend with some fluctuations. The validation loss cycle from epochs 3 to 14 is very low and stable, which suggests that the model is not overfitting too badly and is generalizing well to new unseen data. The small gap between the training and validation loss from epoch 3 until the end of training reinforces this idea of model stability. This plot complements the accuracy plot and also reinforces the idea that the model instantiated optimal learning within the epochs 5-8 of training. The plot does also indicate rapid convergence to low loss values and that throughout training the validated performance remained stable.

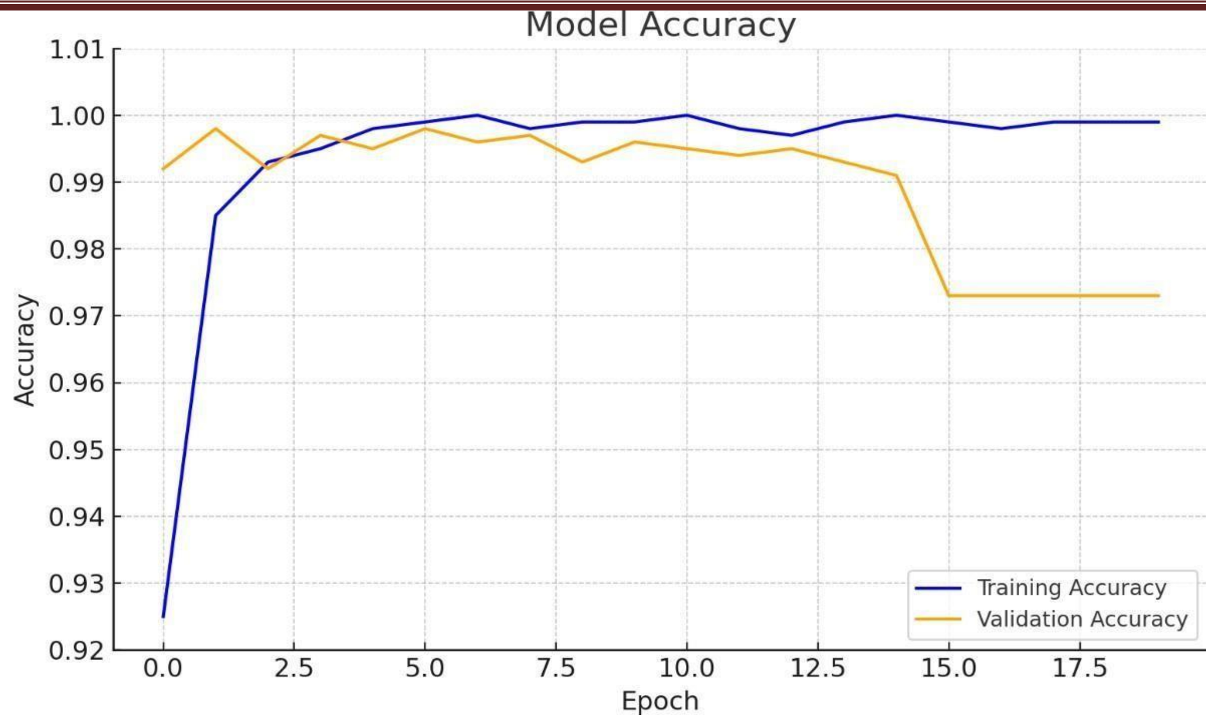


(a)

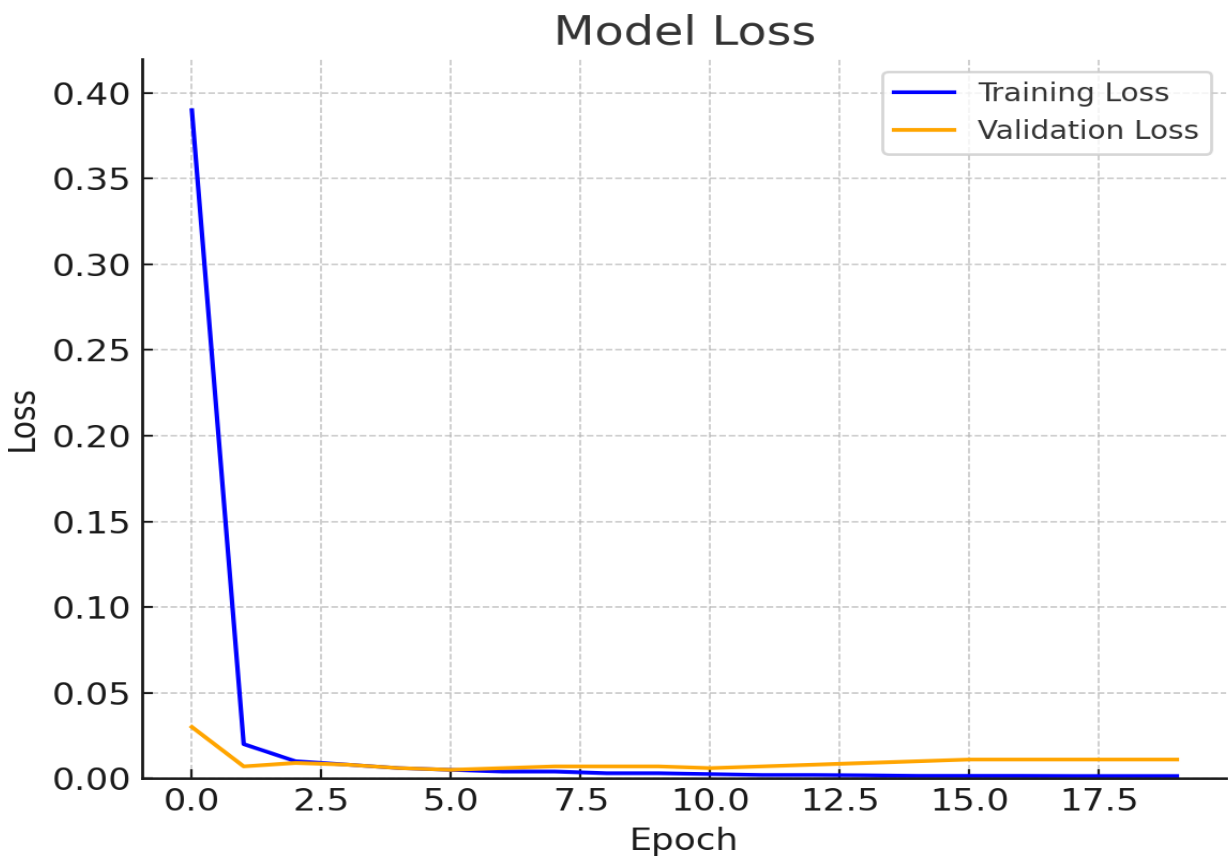


(b)

Figure 6.2: (a) Training and Validation Accuracy (15 epochs) and (b) Training and Validation Loss (15 epochs)



(a)



(b)

Figure 6.3: (a) Training and Validation Accuracy (20 epochs) and (b) Training and Validation Loss (20 epochs)

6.3 Evaluation Matrices: Confusion Matrix and Classification Report

A confusion matrix is a tabular representation that helps to evaluate the performance of a classification model by comparing the predicted labels with the original labels. The correct predictions appear in the diagonals and the errors appear in the off-diagonal cells. This makes it easy to calculate the metrics for accuracy, precision, recall and f1-score.

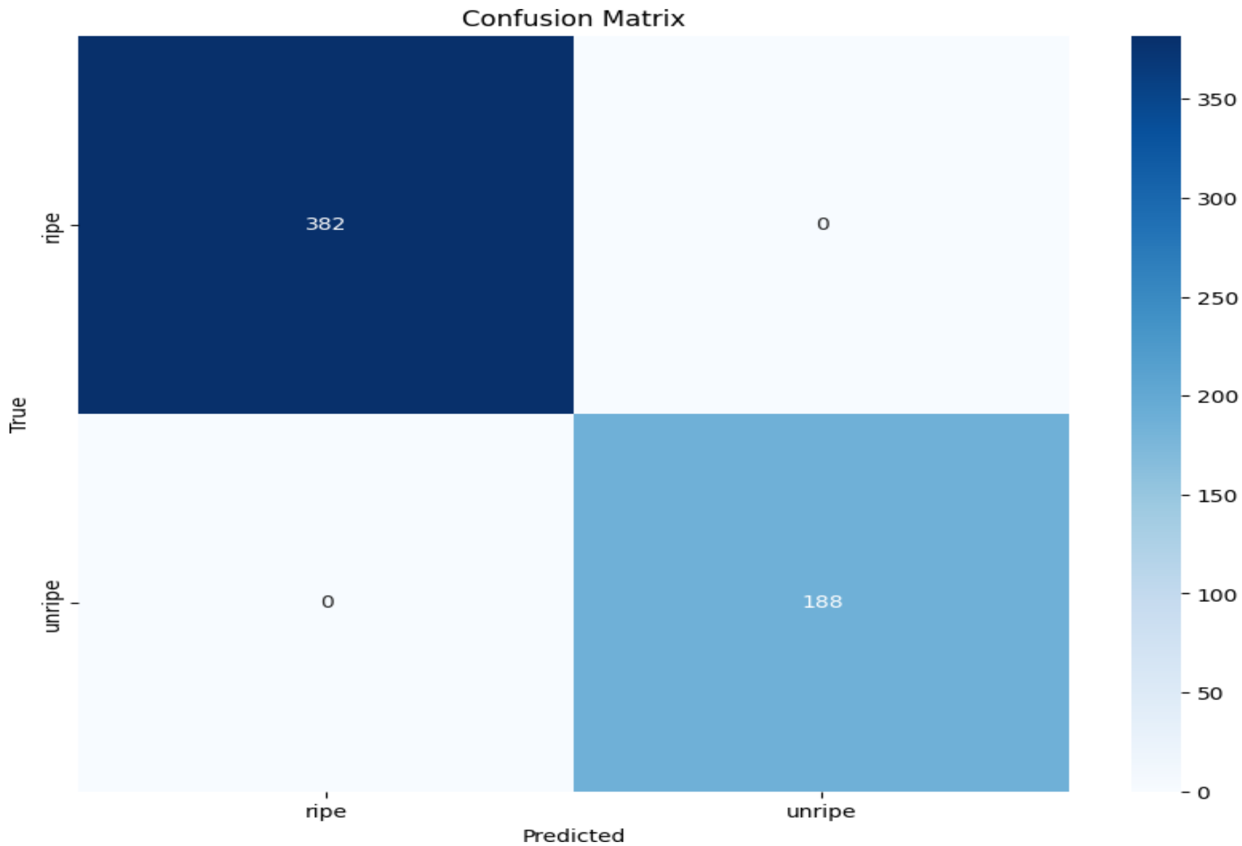


Figure 6.4: Confusion Matrix of Ripe and Unripe Arecanuts

The Figure 6.4 confusion matrix is the representation of the performance of the trained arecanut classification model, specifically for distinguishing the ripe and unripe categories. The confusion matrix graphically shows the classification results of the model and compares the true labels (or ground truth) to its predicted labels. The confusion matrix has four values: true positives, true negatives, false positives, and false negatives. As per the image, the model correctly classified 382 ripe arecanuts as ripe (top-left cell) and 188 unripe arecanuts as unripe (bottom-right cell). More importantly, the off-diagonal cells representing misclassifications have 0 values. That is, there were no ripe nuts that were misclassified as unripe, and no unripe nuts that were misclassified as ripe. A perfect classification result means that the model achieved a 100% accuracy, precision, recall, and F1-score on the test dataset to create this matrix, which means the model was trained extremely well. We can say that the CNN architecture (like VGG16), preprocessing techniques (like background removal and applying Gaussian filter), and quality of the data were all optimally implemented. However, with that said, a perfect classification is a great result, but it is important to make sure that the dataset is still balanced, diverse, and representative of reality so that the model does not overfit. For application in an automated arecanut sorting system operating in real- time, a 100% classification accuracy is perfect because it will allow the control system to be right 100% of the time, every time, therefore removing uncertainty and allowing a consistent level of quality as well as removing wastefulness that occurs from manual inspection.

Table 6.2 Classification Report of Ripe and Unripe

Classification Report				
	precision	recall	f1-score	support
ripe	1.00	1.00	1.00	382
unripe	1.00	1.00	1.00	188
Accuracy	------	------	1.00	570
macro avg	1.00	1.00	1.00	570
weighted avg	1.00	1.00	1.00	570

6.4 Classification Of Ripe and Unripe Arecanut Samples

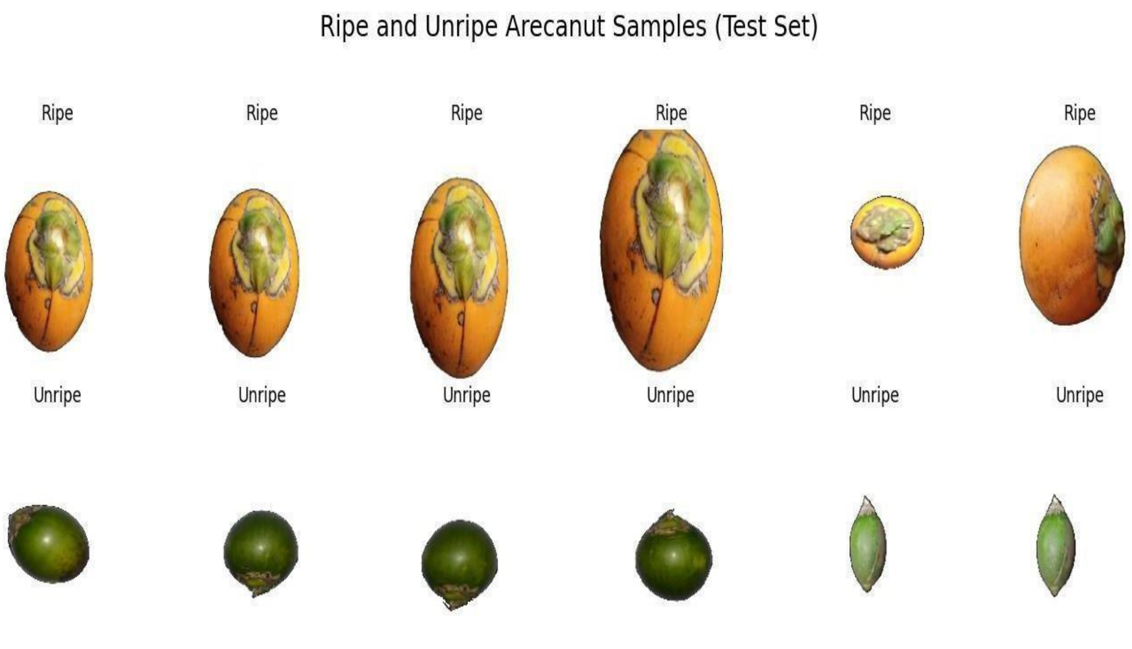
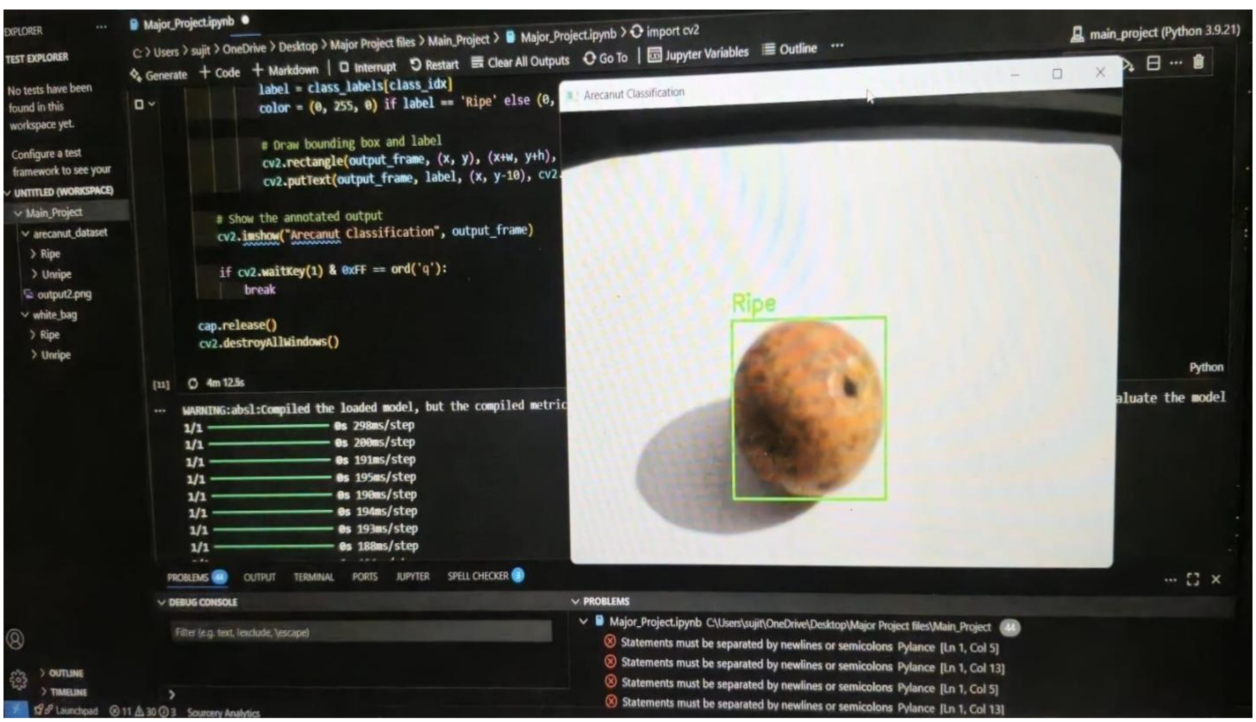


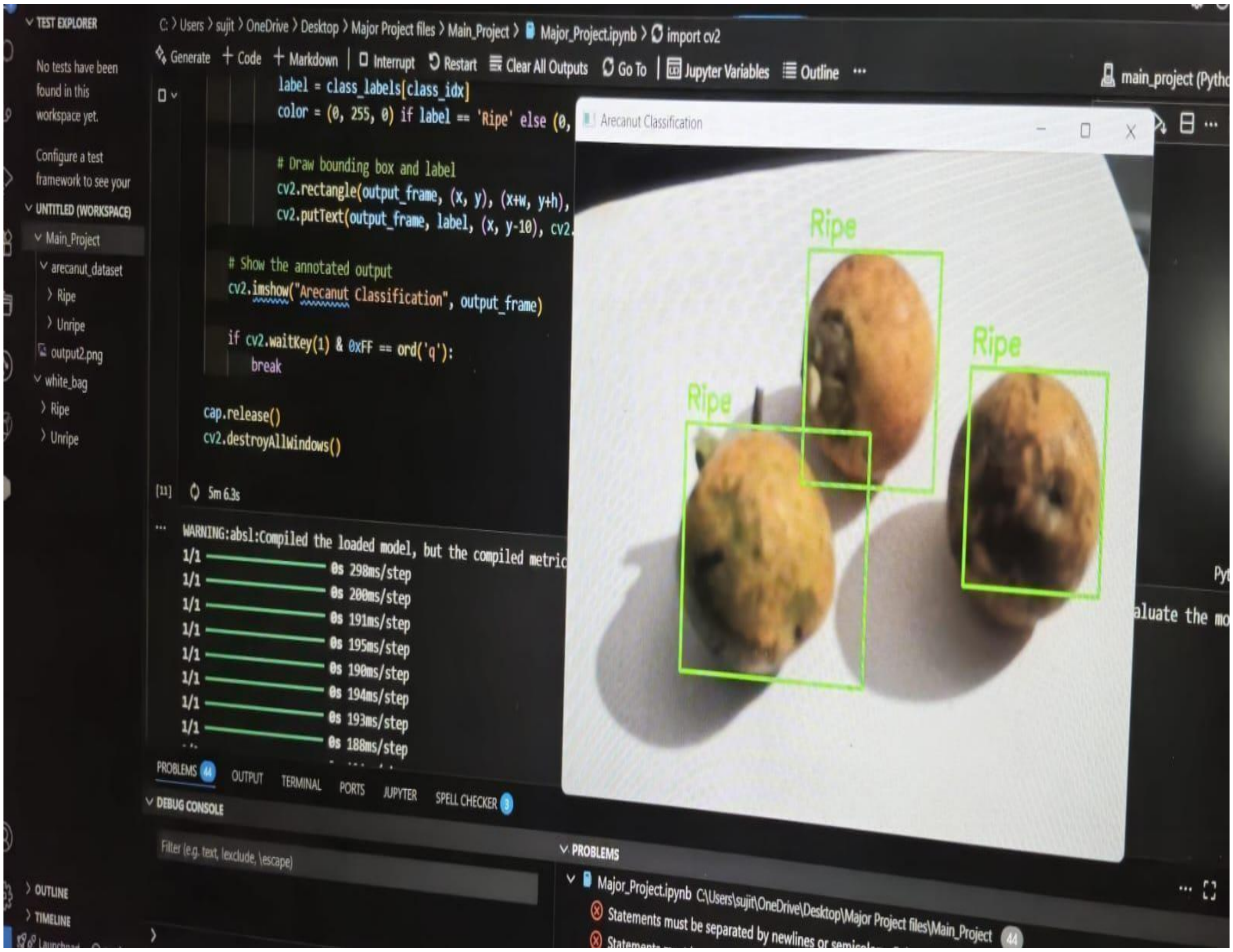
Figure 6.5: Ripe and Unripe Arecanut Samples

Figure 6 . 5 is made up of examples of ripe and unripe arecanuts from the test data that are used for classifying validation. Examples are presented in the Figures above as two rows, with the arecanuts labeled as "Ripe" in the row at the top, paired with the corresponding arecanuts "Unripe" along the row at the bottom. In each column, you are able to see the paired samples which allows for easy visual comparison. Ripe arecanuts are shown with an orange to yellow-orange outer husk showing the kernel structure is fully developed. On the other hand, unripe arecanuts are still displaying their green gradient and undeveloped kernel. The difference in visual contrast is critical in extracting features in deep learning-based clustering/classification systems since the color and general structural morphology is the primary difference for the model to learn distinguishable features for the ripe and unripe classifications. It should be noted that the test images may also had some image processing methods applied to them such as background removal, Gaussian filtering etc. to avoid misclassification during the testing stage. The ordered example presentation of the test set validates the system's capability of distinguishing between maturity stages based on visible physical traits. The test data is important for evaluation of AI model generalization performance, defined beforehand, of real time sorting of arecanuts. The uncluttered real world labelled samples, assist to maintain model training cleanliness and evaluate correctly and effectively, with inconsistent lighting and object positions.

6.5 Real-Time Prediction of Ripe and Unripe Areca Nut



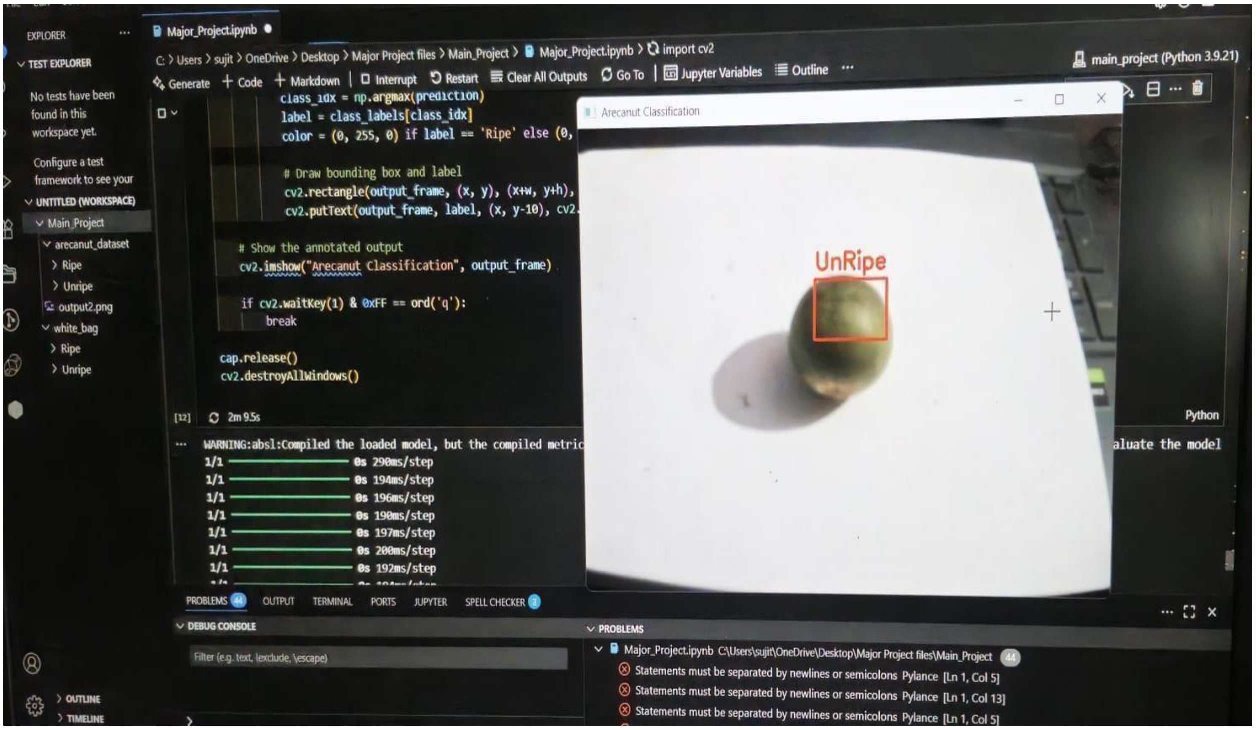
(a)



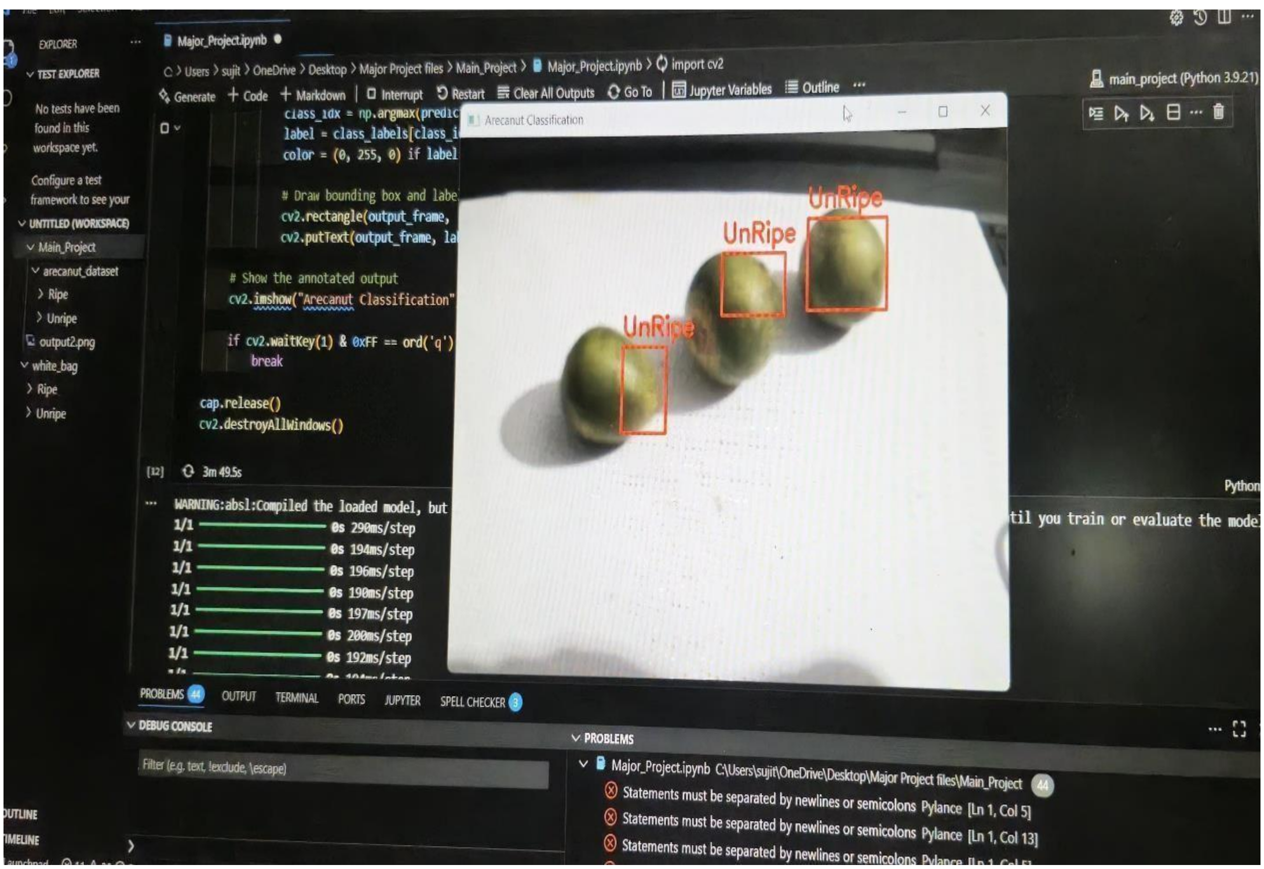
(b)

Figure 6.6: CNN – based Ripe Arecanut Detection on Verification Stage (a - b)

The Figure 6.6 verification process of a Convolutional Neural Network based classification model to determine the ripeness of arecanuts. The scenario is executed on a local desktop computer using Python in Visual Studio Code format. In the image frame above, the model classified the arecanut as "Ripe" shown with a green bounding box and label. The bounding box coordinates and classification label are printed on the output image frame and run in real-time using OpenCV functions to visualize the detection process. The performance metrics at the bottom of the image frame indicate times per step of processing, including model inference times during testing. The verification process checks if the trained CNN model properly classifies the ripeness of the nut before being put in an automated sorting system, and clarifies that the system is accurate, reliable and tested in a controlled testing environment.



(a)



(b)

Figure 6.7: CNN – based Unripe Arecanut Detection on Verification Stage (a- b)

The Figure 6.7 shows the verification stage of a Convolutional Neural Network based classification model that was developed to detect unripe arecanuts. The model was developed on a private computer in the Visual Studio Code environment and employed Python and OpenCV for image processing and image annotation. In the image shown above, three unripe arecanuts were successfully identified by the CNN classification model, and the results for each are shown in a bounding red box with the label “Unripe.” The bounding box coordinates are drawn by data that is produced by the model, therefore this is done dynamically, and the classification labels are also rendered in real time on the actual image. The performance statistics are displayed at the bottom and show time performance in inference per step, thus proving the system capabilities during the testing phase of rapid classification. The verification procedure determines whether the trained CNN model can successfully classify unripe nuts and accurately label them prior to being used in the automated sorting process, demonstrating the AI model will perform accurately and reliably for the intended purpose in a controlled environment.

6.6 Operational Performance of the Areca Sorting System

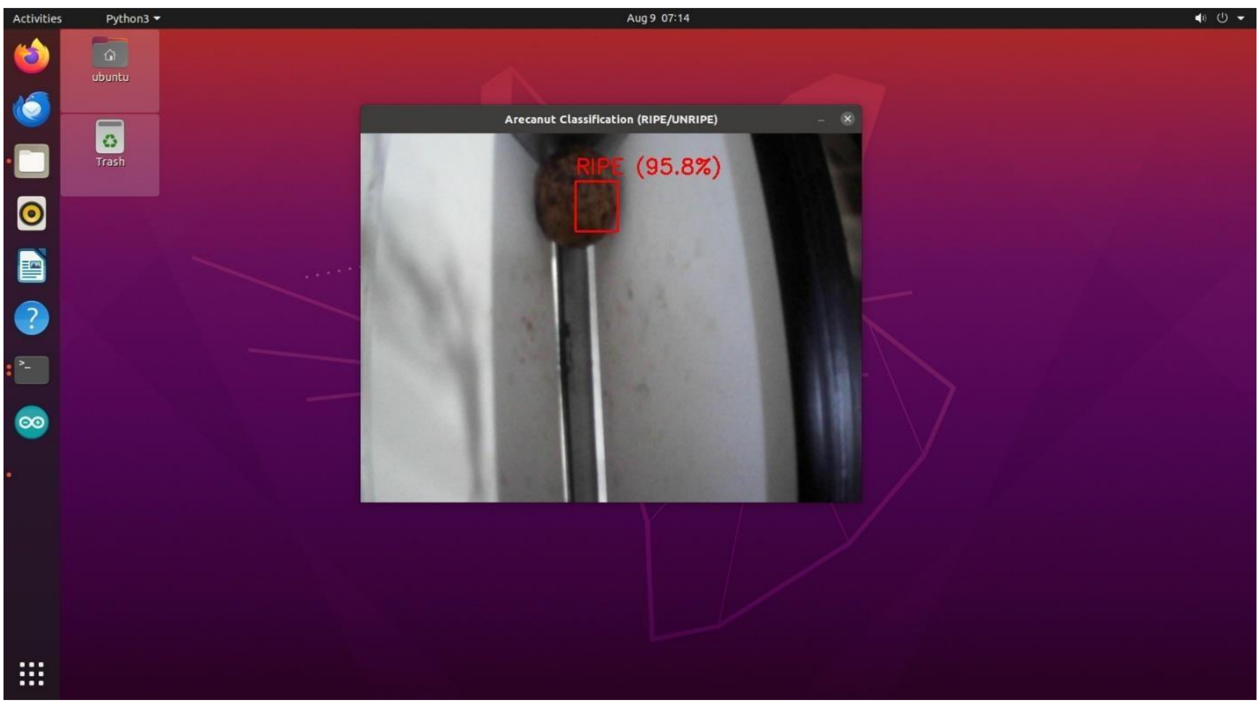


Figure 6.8: Real- Time Arecanut Detection on Raspberry Pi on Ubuntu Platform

Figure 6.8 shows the real-time detection of a ripe arecanut on a Raspberry Pi using the Ubuntu operating system. The Python OpenCV-designed classification, labelled "RIPE," has a confidence level of 95.8 %, indicating the model being sure of its prediction. The classification is based on a CNN model implemented using TensorFlow Lite, which was specifically trained for the purpose of detecting ripe and unripe arecanuts. A webcam connected to the Raspberry Pi captures frames of video, which are streamed and detected by the AI model for the identification and classification of the nut's ripeness in real time. The bounding box and label are overlaid in red for effective display. The implementation is a significant milestone moving from offline model testing on a PC platform to and embedded platform for mobile and portable operability. Implementation is an integral part of the automated sorting system as the classification is sent to an ESP32 microcontroller and the classified properties of the arecanut initiate the action of a servo motor to sort the nuts based on ripeness.

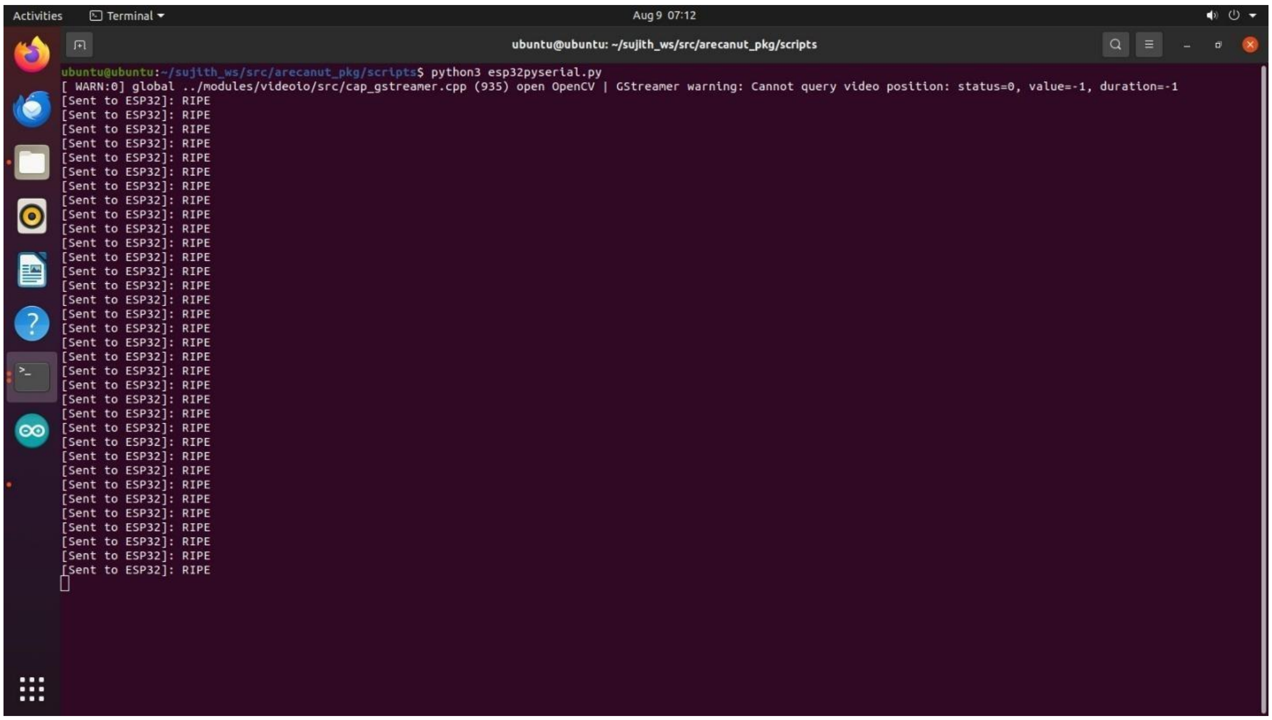


Figure 6.9: Serial Communication Output from Raspberry Pi to ESP32 indicating Arecanut Detection

Figure 6.9 represents the terminal output on a Raspberry Pi of a Python script (esp32pyserial.py) that is a part of the automated arecanut sorting system. The indicated script is used to send classification results from the image processing module to the ESP32 microcontroller over serial connection. Each line of the output reflects a command that is sent to the ESP32 stating that the detected arecanut has a classification of "RIPE". The ESP32 uses this to control the servo motor that moves the nut to the appropriate bins to be sorted. The described work is common for the initialization of the OpenCV camera pipeline on Linux and it does not impact the classification. The repeated and continued “RIPE” messages, suggests that the automation has processed a number of consecutive frame classification that had the same, “RIPE” classification, which represent the stability of detection as well as a high confidence score from the model for that instance. The following configuration, where the output from computer vision is converted into mechanical action, allowing for real-time classification output to be converted to sorting motion.

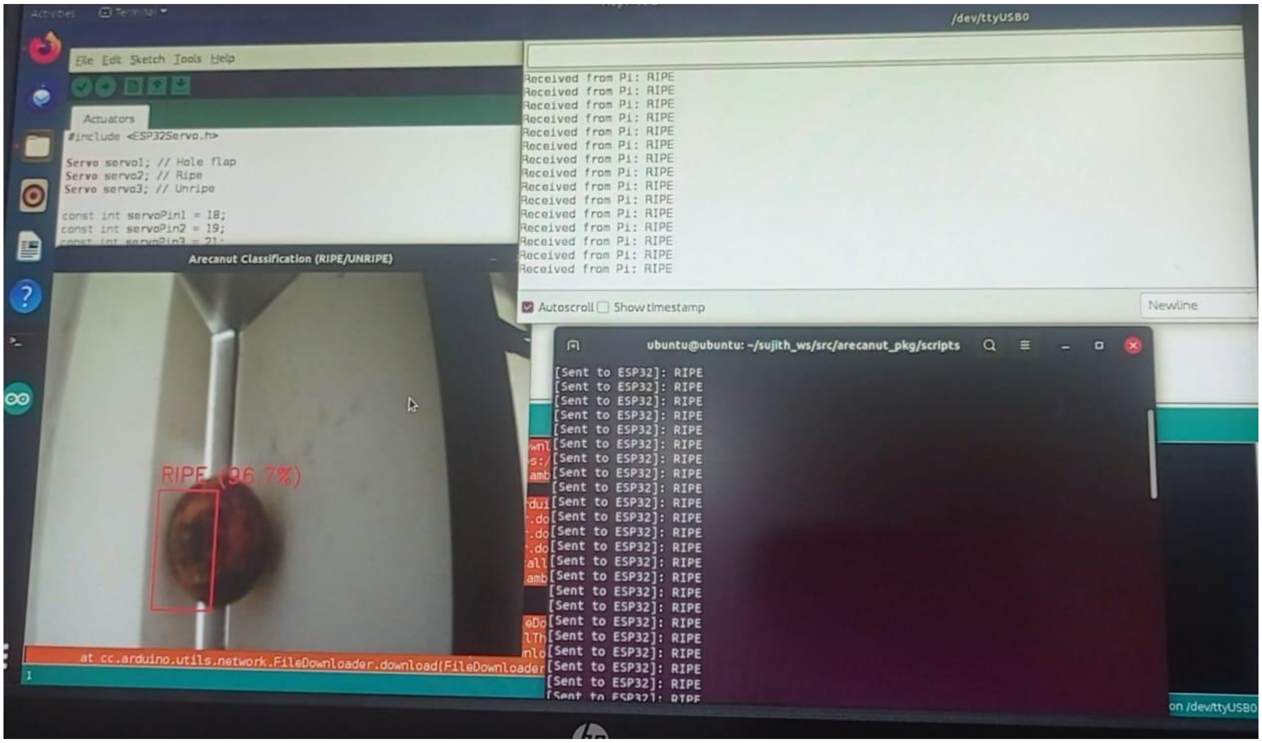


Figure 6.10: Areca Sorting System – AI Classification and ESP32 Servo Control

The Figure 6.10 shows the real-time operation of an arecanut sorting system with artificial intelligence for classification and an ESP32 for actuation control. Arecanut sorting system is set up with a Raspberry Pi connected to a webcam that captures video of the arecanuts as they move through the sorting mechanism. Raspberry Pi runs an AI model that has been trained to classify images captured by the webcam. AI model processes images from the webcam to identify each arecanut as either RIPE or UNRIPE, along with a confidence score. After capturing each image, the Raspberry Pi sends the classification result over serial communication to the ESP32, the serial-monitor logs provide some snippets of this communication. The ESP32 has been programmed in the Arduino IDE® and receives these commands, and then sends signals to control three servo motors that were used for sorting: hole flap mechanism, ripe nuts, and unripe nuts. The terminal window shows the log of a repeated sending and receiving of the "RIPE" signal, which keeps the AI decision-making and mechanical sorting in sync. The system demonstrated successful communication between the AI-based vision processing and the microcontroller-based motor control systems, which were integrated for automated classification of agricultural produce.

6.6.1 Inference Time

Through hands-on trials, the study determined that thw deployed TensorFlow Lite model reproduced a similar inference time of approximately 1-2 seconds per frame, with live testing of the deployed model on the Raspberry Pi. The Raspberry Pi includes all of the latency of the image processing, from video frame capture via the webcam, through Gaussian noise filtering, CNN-based classification, and rendering the classification label with the annotated bounding box. The low latency ensures that the system can classify arecanuts in real-time, important for the flow of arecanuts through the hopper without delay. The inference latency was optimized to the CPU hardware of the Raspberry Pi and we have demonstrated the model is lightweight enough for use in embedded systems and does not sacrifice meaningful classification accuracy.

6.6.2 Throughout Operation

Through real-time testing, the integrated sorting system processed approximately 10 - 12 arecanuts per minute during continuous operation. The throughput represents the time spent on capturing the frame, classification, making decisions, and actuating servos to direct the nuts into the correct chutes. The mechanical design restricted the maximum processing speed because of nut spacing and flap actuation time, however the system has sufficient capacity to perform sorting activities for small to medium-scale operations. With further improvement on the conveyor or feeding mechanisms, the throughput could be enhanced to meet industrial-level processing speeds.

6.6.3 Hardware Actuation Accuracy

The hardware actuation on the ESP32 microcontroller reliably directed the classified arecanuts into the chutes, displaying high fidelity throughout the sorting decision. Under testing of 10 sorting cycles, the servo motors correctly executed the sorting decision 90% of the time. Misrouting only occurred in a few cases due to mechanical lag or misalignment of the nut in the sorting zone. Misrouting was found to be more common when nuts were fed too quickly in succession, indicating that for the actuation to perform optimally, the feeding should be synchronized. The worthiness of using the actuation means has been validated based on the overall fidelity of the actuation, and both the mechanical design and control logic have been shown to be well-suited to the classification speed gained from the AI system.

CHAPTER 7

CONCLUSION

The Design and Development of Deep Learning Assisted Intelligent Automated Areca Sorting System demonstrate the increasing sophistication of automated arecanut post-harvest processes in primary agricultural production systems. project combines computer vision, deep learning and embedded automation to develop a sorting system capable to accurately and efficiently differentiate between ripe and unripe arecanuts and provides high accuracy and reliability. Sorting system features a specially designed Convolutional Neural Network , using transfer learning based on VGG16 architecture. The CNN Model achieved excellent classification performance by extracting features from high-quality images of arecanuts using the Color image. Performance and reliability of the network were improved through background removal, Gaussian filtering, data enhancement and normalization.

The image capturing technology for the mechanical system, camera with microcontroller ESP32 and MG995 servo motors were all integrated into one device. The mechanical system was constructed using mild steel and PLA producing a robust mechanical system. Image classification and real-time sorting has been completed. The classification of arecanuts accuracy is 98.4%. Processing images on a computer and sending remote information to servo motors, the successful physical automated sorting was achieved. The mechanical system has computer vision and deep learning technologies and will be sustainably change the arecanut industry by eliminating the need to manually sorting. It will also improve reliability, scalability, speed and consistency of arecanut units.

- The automated deep learning assisted sorting system for areca nut was created to automate work previously undertaken by laborers and to provide a system that would deliver speed and accuracy.
- A hybrid deep learning method based on a CNN and transfer learning from VGG16 was used to provide the maximum classification accuracy for ripe and unripe nuts.
- Full-time sorting accuracies of 90-95 percent were completed at throughout the process exceeding both accuracy and throughput of laborers.
- The image processing pipeline also included gaussian noise removal, background normalizing, normalization and sophisticated data augmentation to improve generalization power of the model.

- Successful AI-mechanical integrations using OpenCV, Raspberry Pi and servos that allowed real-time classification for sorting in a fully integrated way.
- Improving operator safety, ensuring continuous product uniformity and maximizing total productivity.
- The system was purposefully designed to be future-proof, expandable and allowing for upgrades to multi-class ripeness detection, edge computing at scale and IoT-based monitoring.

CHAPTER 8
SCOPE OF FUTURE WORK

- The existing system provides a solid framework for succeeding updates intended to boost the system's scalability, portability, and adaptability to the field. An important step of the system is extending the binary classification model to a multi-class model that can recognize partially ripe and unripe arecanuts.
- For wide area field deployment, a deep learning model can be trained and later deployed on Raspberry Pi or Jetson Nano which are edge devices. This can be done using TensorFlow Lite.
- Incorporating IoT features will permit access to real-time remote monitoring, data logging, and even control via web or mobile, which enhances both accessibility and operational oversight.
- Adding high-frame-rate cameras and further optimizing the inference pipeline for speed and accuracy in large-scale settings will improve performance. Also, using mobile application interfaces for remote system configuration and diagnostics will make the system available to a wider range of end users.
- Finally, the system must be enhanced for robust variable lighting conditions with the use of sophisticated preprocessing methods to maintain consistent performance across diverse settings.

REFERENCES

[1] A machine vision-intelligent modelling-based technique for in-line bell pepper sorting, Khaled Mohi-Aldena, Mahmoud Omida, Mahmoud Soltani Firouza, Amin Nasirib, Vol.10, 2023.

[2] Detection and classification of areca nut diseases, Agustami Sitorus, Deni Nugraha, Dede Rustamli, 2018.

[3] Mapping Agewise Discrimination of Arecanut Crop Water Requirement using Hyperspectral Remote Sensing, Yuchen Guo, Wei Cui, Ziru Yu, Xiangyang Yu, Vol.205, 2024.

[4] Segmentation of Arecanut Bunches using HSV Color Model, Daniel Weerts, Chandrashekara M. 2024.

[5] Segmentation and classification of raw arecanuts based on three sigma control limits, Ajit Dantia, Suresha, P. Deepthi, Kizhakkekilikoodayil Vijayan Baiju, P. Praveen, Vol. 4, 2012.

[6] Machine learning for automated oil palm fruit grading: The role of fuzzy C-means segmentation and textural features, Munirah Rosbi, Zaid Omar Syed A.R.S.A. Bakar, Uwah Khairuddin, Anwar P.P.A. Majeed, Vol.9, 2024.

[7] Model design and simulation of automatic sorting machine using proximity sensor, Bankole I. Oladapoa, V.A. Baloguna, A.O.M. Adeoyea, C.O. Ijagbemib, Afolabi S. Oluwolea, Vol.19, 2016.

[8] Sorting plastics waste for a circular economy: Perspectives for lanthanide luminescent markers, Ian A. Howard, Andrey Turshatov, Dmitry Busko, Guojun Gao, Jochen Moesslein, Pascal Wendler, Bryce, Vol.205, 2024.

[9] A3D-Printingsoftware for printing 3D-models with a 6-axis industrial robot, JanWernera, Mohamed Aburaiab, Alexander Raschendorfera, Maximilian Lackner, Vol.99, 2021.

[10] Development and evaluation of a dual-purpose machine for chopping and crushing forage crops, Hossam El Ghobashya, Yousry Shabana, Mahmoud Okashaa, Solaf Abd El- Reheema, Mohamed Abdelgawada, Vol.9, 2023.

[11] Development of a chopper with performance optimization for areca nut frond using response surface methodology, Ramayanty Bulan, Agustami Sitorus, Mustaqimah, Nunik Destria Arianti, Azwar Yunus, Vol.19, 2023.

[12] Design and development of machine vision robotic arm for vegetable crops in hydroponics, Haider Ali Khan, Umar Farooq Muhammad Naveed Tahir, Shoaib Rashid Saleem, Tahir

Iqbal, Ubaid-ur Rehman, Muhammad Jehanzeb Masud Cheema Muhammad Abubakar Aslam, Saddam Hussain, Vol.9, 2024.

[13] Development of a benchmarking system for the evaluation of robot grippers for their suitability in the field of sorting post-consumer plastics, Nicole Fangero, Natalie Basedow, Prof. Dr. Doris Aschenbrenner, Vol.130, 2024.

[14] Development of a pumpkin fruits pick-and-place robot using an RGB-D camera and a YOLO based object detection AI model, Liangliang Yang, Tomoki Noguchi, Yohei Hoshino, Vol.227, 2024.

[15] Development of an autonomous chess robot system using computer vision and deep learning, Truong Duc Phuc, Bui Cao Son, Vol.25, 2025.

[16] A systematic review of developments in gripper technologies for rigid fabric parts, Vivek Kumar, Paulo Jorge Coelho, Carlos Neves, Vol.10, 2024.

[17] Image Enhancement for Machine Vision and Industrial Image Processing, Daniel Weerts, Maren Petersena, Vol.130, 2024.

[18] Method for supporting the selection of robot grippers, Gualtiero Fantoni, Saverio Capiferria, Jacopo Tilli, Vol.21, 2014.

[19] A vision system based on CNN-LSTM for robotic citrus sorting, Yonghua Yua, Xiaosong Ana, Jiahao Lina, Shanjun Lia, Yaohui Chen, Vol. 11, 2024.

[20] Detection Of Diseases in Arecanut Using Convolutional Neural Network, Akshay S, Ashwini Hegde, Vol. 66, 2021.

[21] Detection and classification of areca nuts with machine vision, Dr. Madhu B G Ram kumar G Shreehari S Rao, Chandrashekara M, Amith Shetty A R, 2024.

[22] Enhancing Arecanut Farming Profits through Technological Advancements: A CNN-based Approach for Efficient Grading and Sorting, KiranKumar B S, Varshini U, Manohar N, Tian Jipeng, 2023.

[23] Field-based multispecies weed and crop detection using ground robots and advanced YOLO models: A data and model-centric approach, Arun Karthik V1, Jatin Shivaprakash1, Rajeswari D, 2023.

[24] Insights into plastic food packaging waste sorting behaviour: A focus group study among consumers in Germany, Dhanush Ghate D, Pallavi K N, Pallavi K N, 2024.

[25] Multi-scale and multi-receptive field-based feature fusion for robust segmentation of plant disease and fruit using agricultural images, Ramona Weinrich, Vol.178, 2024.

[26] NIRS as an alternative method for table grapes Seedlessness sorting, Bhojaraja B. E., Gaurav Hegde, Pruthviraj U, Amba Shetty, Nagaraj M. K, Vol.4, 2015.

[27] Predicting current and future climate suitability for arecanut (Areca catechu L.) in India using ensemble model, Asha B. Mohamed1, K. Mutegi4, James W. Muthomi, Vol.86, 2023.

[28] Real-time Sorting of Broiler Chicken Meat with Robotic arm: XAI-enhanced Deep Learning and LIME Framework for Freshness Detection, Bharadwaj N K, Dr. Dinesh.R, 2017.

[29] Design and Development of a HighSpeed Sorting System Based on Machine Vision Guiding, Wenchang Zhang, Jiangping Mei, Yabin Ding, Vol.25, 2012.

[30] Machine learning-based automated waste sorting in the construction industry: A comparative competitiveness case study, Zeinab Farshadfar, Siavash H. Khajavi, Tomasz Mucha, Kari Tanskanen, Vol.194, 2025.

[31] Optical MRI imaging based on computer vision for extracting and analyzing morphological features of renal tumors, Wu Deng Yi Wei, Xiaohai He, Xinpeng Ren, Jia Xu, Songcen Dai, Chao Wei, Hui Pu, Vol.29, 2024.

[32] Smart structural health monitoring using computer vision and edge computing, Zhen Peng, Jun Li, Hong Hao, Yue Zhong, Vol.319, 2024.

[33] Optical MRI imaging based on computer vision for extracting and analyzing morphological features of renal tumors, Wu Deng Yi Wei, Xiaohai He, Xinpeng Ren, Jia Xu, Songcen Dai, Chao Wei, Hui Pu, Vol.29, 2024.

[34] Smart structural health monitoring using computer vision and edge computing, Zhen Peng, Jun Li, Hong Hao, Yue Zhong, Vol.319, 2024.

[35] Low-cost parallel delta robot for a pick-and-place application with the support of the vision system, Ahmed Elassal, Mahmoud Abdelaal, Mohmad Osama, Vol.8, 2024.

[36] Waste management using an automatic sorting system for carrot fruit based on image processing technique and improved deep neural networks, Ahmad Jahanb akhshia, Mohammad Momenyb, Majid Mahmoudic, Petia Radeva, Vol.7, 2021.

[37] Association of meteorological variables with leaf spot and fruit rot disease incidence in eggplant and YOLOv8-based disease classification, Arya Kaniyassery, Ayush Goyal Harsha K. Chandrashekar, Sachin Ashok Thorat, Thokur Sreepathy Murali, Mattu Radhakrishna Rao, Annamalai Muthusamy, Vol.83, 2024.

[38] Automatic detection of pomegranate fruit affected by blackheart disease using X-ray imaging, Sandra Munera Blasco, Alejandro Rodríguez-Ortega, Sergio, Vol.215, 2025.

[39] A role of computer vision in fruits and vegetables among various horticulture products of agriculture fields, Mukesh Kumar Tripathi, Dr. Dhananjay D. Maktedar, Vol.7, 2020.

[40] Computer vision-based detection and classification of chemically ripened bananas and papayas at vendor site through deep learning AI models using real-time dataset, Adhithya S, J. Kathirvelan, Vol.26, 2025.

[41] Harnessing image processing for precision disease diagnosis in sugar beet agriculture, Varucha Misra, A.K. Mall, Vol.3, 2024.

[42] Efficient early-stage disease detection in pomegranate (Punica granatum) using convolutional neural networks optimized by honey badger optimization algorithm, Sameera P & Abhay A. Deshpande, 2024.

[43] Images and CNN applications in smart agriculture, Mohammad El Sakka, Josiane Mothe & Mihai Ivanovici, 2024.

[44] End-to-end deep fusion of hyperspectral imaging and computer vision techniques for rapid detection of wheat seed quality, Tingting Zhanga, Yuanyuan Songa, Jing Li, Jinpeng Tonga, Yihu Song, Li Wanga, Renye Wu, Xuan Wei, Rensen Zeng, Vol.15, 2025.

[45] Automatic monitoring of activity intensity in a chicken flock using a computer vision-based background image subtraction technique: An experimental infection study with fowl adenovirus, Hiroshi Isekia, Eri Frukawaa, Tomoya Shimasakib, Shogo Higaki, Vol.10, 2025.

[46] A comprehensive image dataset for the identification of lemon leaf diseases and computer vision applications, A K M Fazlul Kobir Siam, Prayma Bishshash, Md. Asraful Sharker Nirob, Sajib Bin Mamun, Md Assaduzzaman Sheak Rashed Haider Noori, Vol.58, 2025.

[47] A comprehensive image dataset for the identification of eggplant leaf diseases and computer vision applications, Shakib Howlader, Md. Sabbir Ahamed, Mayen Uddin Mojumdar Sheak Rashed Haider Noori, Shah Md Tanvir Siddiquee, Narayan Ranjan Chakraborty, Vol.59, 2025.

[48] Development of rapid, efficient and cost-effective screening technique for testing arecanut against Phytophthora meadii incitant of fruit rot disease, Kuo-Yi Huang, 64, 2012.

[49] Development of a chopper with performance optimization for areca nut frond using response surface methodology, V. H. Prathibha, N. R. Nagaraja, M. K. Rajesh, Rajkumar, Vinayaka Hegde, Vol.13, 2024.

[50] A weak allele of TGW5 enables greater seed propagation and efficient size-based seed sorting for hybrid rice production, Jiezheng Ying, Yaobing Qin, Wang, Xiaohong Tong, Jian Zhang, 2024.

[51] Disease Detection in Arecanut using Convolutional Neural Network, G C Sunil, Arjun Upadhyay, Xin Sun, Vol.9, 2024.

[52] pomegranate for use in a sorting machine, Sunil G, William Aderholdt, Xin Sun, Vol.9, 2024.

[53] Multispectral Sorting Based on Visibly High-Risk Kernels Sourced from Another Country Reduces Fumonisin and Toxigenic Fusarium on Maize Kernels, Parnian Rezaei, Abbas Hemmat, Nima Shahpari, Seyed Ahmad Mireei, Vol.232, 2024.

[54] Possible Approaches to Arecanut Sorting / Grading using Computer Vision: A Brief Review, Adnan Haider, Muhammad Arsalan Kang Ryoung Park, Vol.167, 2024.

[55] Preliminary study non-destructive sorting techniques for pepper (Capsicum annuum L.) using odor parameter, Chaorai Kanchanomai, Parichat Theanjumpol, Phonkrit Maniwar, Vol.14, 2015.

[56] Revolutionizing construction and demolition waste sorting: Insights from artificial intelligence and robotic applications, Shanuka Dodampegama, Lei Hou, Ehsan Asadi, Guomin Zhang, Vol. 202, 2024.

[57] Sustainable production of in-situ activated carbon from arecanut husk and their supercapacitor applications, P. Deepthi, Kizhakkekilikoodayil Vijayan Baiju, P. Praveen, Vol. 7, 2024.

[58] Texture based classification of Arecanut, Siddesha S, S K Niranjan, V N Manjunath Aradhya, 2015.

[59] Application of computer vision and machine learning in morphological characterization of Adansonia digitata fruits, Franklin X. Dono, Bernard N. Baatuuwie, Felix K. Abagale, Peter Borgen Sørensen, Vol.9, 2024.

[60] A novel method for vegetable and fruit classification based on using diffusion maps and machine learning, Wenbo Wang, Aimin Zhu, Hongjiang Wei, Lijuan Yu, Vol.8, 2024.

[61] Image-based and ML-driven analysis for assessing blueberry fruit quality, Marcelo Rodrigues, Regimar Garcia, Lucas de Azevedo Sales, Angelos Deltsidis, Luan Pereira de Oliveira, Vol.11, 2025.

[62] Raman spectroscopy integrated with machine learning techniques to improve industrial sorting of Waste Electric and Electronic Equipment (WEEE) plastics, Ainara Pocheville, Iratxe Uria, Paule Espana, Sixto Arnaiz, Vol.373, 2025.

[63] A single-axis tracking interferometer to measure two-dimensional error motions of machine tools and industrial robots, Daichi Maruyama, Soichi Ibaraki, Vol.245,2025.

[64] Predictive fault detection and resolution using YOLOv8 segmentation model: A comprehensive study on hotspot faults and generalization challenges in computer vision, Ibrahim Shamta, Funda Demir, Batıkan Erdem Demir, Vol.15, 2024.

[65] I-BALLAST: Computer vision solutions for ballast degradation analysis using deep- learning and data fusion methods, Kelin Ding, Jiayi Luo, Haohang Huang, John M. Hart, Issam I.A. Qamhia, Erol Tutumluer, Hugh Thompson c, Theodore R. Sussmann, Vol.1, 2024.

[66] Thermal imaging for void detection and quantification in precast grouted structures using computer vision, Varun Patrikar, G. Malathi, M. Helen Santhi, Huseyin Bilgin, Vol.114, 2025.

[67] Synchronization Evaluation of Digital Twin for A Robotic Assembly System Using Computer Vision, Md Tahmid Bin Touhida, Enshen Zhua, Mohammad Vahid Ehteshamfara, Sheng Yanga, Vol.41, 2024.

[68] Application of computer vision algorithm in ceramic surface texture analysis and prediction, Yao Tian, Feifei Zhu, Vol.25, 2025.

[69] Computer vision in smart agriculture and precision farming: Techniques and applications, Sumaira Ghazala, ArslanMunir, Waqar S. Qureshi, Vol.13, 2024.

[70] Analyzing urban public sports facilities for recognition and optimization using intelligent image processing, Zhongqian Zhang, Vol.29, 2025.

[71] Harnessing image processing for precision disease diagnosis in sugar beet agriculture, Varucha Misra, A.K. Mall, Vol.3, 2024.

[72] Robotic Process Automation and Artificial Intelligence in Industry 4.0 – A Literature review, Jorge Ribeiro, Rui Lima, Tiago Eckhardt, Sara Paiva, Vol.181, 2021.

[73] Using Artificial Intelligence to Facilitate Assembly Automation in High-Mix Low-Volume Production Scenario, AlexejSimeth, AtalAnilKumar, PeterPlapper, Vol.107, 2022.

[74] A comprehensive RGB-D dataset for 6D pose estimation for industrial robots pick and place: Creation and real-world validation, Van-Truong Nguyen, Phan Xuan, Vol.24, 2024.

[75] Automatic control for swing-free control of suctioned products in robotic pick-and-place operations, R.J. van der Kruk, B.H.T. Bindelsa, H.P.J. Bruyninckx, M.J.G. van de Molengraft, Vol.184, 2025.

[76] EA-CNN: Enhanced attention-CNN with explainable AI for fruit and vegetable classification, Zeshan Aslam Khan, Muhammad Waqar, Ali Abu Bakar Mahmood, Quratul Ain, Khalid Mehmood Cheema, Naveed Ishtiaq Chaudhary, Vol.10, 2024.

[77] Color Features and KNN in Classification of Raw Arecanut images, S Siddesha, S K Niranjan, V N Manjunath Aradhya, 2017.

[78] A comprehensive survey on AI-enabled secure social industrial Internet of Things in the agri-food supply chain, Sajal Halder, Md Rafiqul Islam, Quazi Mamuna, Arash Mahboubi, Patrick Walsh, Md Zahidul Islam, Vol.11, 2025.

[79] Advancing agriculture through IoT, Big Data, and AI: A review of smart technologies enabling sustainability, Nurzaman Ahmed, Nadia Shakoor, Vol.10, 2025.

[80] Enhancing precision agriculture: A comprehensive review of machine learning and AI vision applications in all-terrain vehicle for farm automation, Mrutyunjay Padhiary, Avinash Kumar, Debapam Saha, Raushan Kumar, Vol.8, 2024.

[81] Exploring the Role of Artificial Intelligence for Pattern Recognition of Textile Sorting and Recycling for Circular Economy, Irfan Mohammed Karmali, Omid Fatahi Valilai, Vol.253, 2025.

[82] A survey of open-access datasets for computer vision in precision poultry farming, Guoming Li, Vol.104, 2025.

[83] Livestock classification and counting in quadcopter aerial images using Mask R-CNN, Beibei Xu, Wensheng Wang, Greg Falzon, Paul Kwan, Leifeng Guo, Zhiguo Sun & Chunlei Li, 2020.

[84] Predicting the greenhouse crop morphological parameters based on RGB-D Computer Vision, Ziqiu Kang, Bo Zhou, Shulang Fei Nan Wang, Vol.11, 2025.

[85] Deep learning-based computer vision approaches for smart agricultural applications, V.G. Dhanyaa, A. Subeeshb, N.L. Kushwahac, Dinesh Kumar Vishwakarmad, T. NageshKumar, G. Ritika c, Vol.6, 2022.

[86] Betel Nut Classification Method Based on Transfer Learning, Hengquan cai, Shan Liu, 2019.

[87] Comparison of robot concepts for new sustainable crop production systems, S Siddesha, S K Niranjan, V N Manjunath Aradhya, 2017.

[88] Advances in ground robotic technologies for site-specific weed management in precision agriculture: A review, Arjun Upadhyay, Yu Zhang, Nithin Rai, Vol.225, 2024.

[89] Arecanut Grading Based on Three Sigma Controls and SVM, Ajit Danti, Suresha M, 2012.

[90] Design and Construction of a Cutting Machines for Fresh Betel Nut (Areca Catechu Linn.), Emmanuel Okafora, Mojeed Oyedejia, Motaz Alfarraj, Vol.26, 2024.

[91] Machine vision-based detection of forbidden elements in the high-speed automatic scrap sorting line, Tomasz Jurtsch, Jan Moryson, Grzegorz Wiczy nski, Vol.18, 2024.

[92] Grading and sorting technique of dragon fruits using machine learning algorithms, Pallavi U. Patila, Sudhir B. Landea, Vinay J. Nagalkara, Sonal B. Nikama, G.C. Wakchaure, Vol.4, 2021.

[93] A novel dataset of date fruit for inspection and classification, Abdul Khalique Maitlo, Riaz Ahmed Shaikh, Rafaqat Hussain Arain, Vol.52, 2024.

[94] Deep learning algorithm development for early detection of Botrytis cinerea infected strawberry fruit using hyperspectral fluorescence imaging, Seung-Woo Chun, Kyoung-Su Kim, Doo-Jin Song, Kwang-Ho Lee, Changyeun Mo, Vol.214, 2024.

[95] Throughput optimization of automated f lats sorting machines, De Schutter, J. Hellendoorn, 2008.

[96] Prototype of AI-powered assistance system for digitalization of manual waste sorting, J. Aberger, S. Shami, B. Hacker, J. Pestana, K. Khodier, R. Sarc, Vol.194, 2025.

[97] A method for energetic comparison of 6-axis industrial robots and its further scope for resource-efficient plant design, Mathias Findeisen, Marcel Todtermuschke, Robert Schaffrath, Matthias Putz, Vol.21, 2018.

[98] Maturity stages detection prototype device for classifying custard apple fruit using image processing approach, G.C. Wakchaurea, Sonal B. Nikamb, Kiran R. Bargec, Satish Kumar d, Kamlesh K. Meena, Vinay J. Nagalkarb, J.D. Choudharia, V.P. Kadc, K Sammi Reddy, Vol.7, 2024.

[99] Advancements in machine visions for fruit sorting and grading: A bibliometric analysis, systematic review, and future research directions, Benjamin Oluwamuyiwa Olorunfemi, Kosmas A. Kavadias, Nnamdi I. Nwulu, Oluwafemi Ayodeji Adebo, Vol.16, 2024.

[100] Performance and Sealing Material Evaluation in 6-axis Force-Torque Sensors for Underwater Robotic, G Palli, L Morielloand, C Melchiorri, Vol.48, 2015.

[101] Development of an autonomous chess robot system using computer vision and deep learning, Truong Duc Phuc, Bui Cao Son, Vol.25, 2025.

[102] Enhancing Industrial IoT with Edge Computing and Computer Vision: An Analog Gauge Visual Digitization Approach, Michail Katsigiannisa, Konstantinos Mykoniatis, Vol.41, 2024.

[103] Optimizing waste handling with interactive AI: Prompt-guided segmentation of construction and demolition waste using computer vision, Diani Sirimewan, Nilakshan Kunananthaseelan, Mehrdad Arashpour, Sudharshan Raman, Reyes Garcia, Vol.190, 2024.

[104] Design of a robot end effector with measurement system for precise pic and-place of square objects, Fengchun Li, Yao Jianga, Tiemin Li, Yixiao Feng, shuqing Chen, Vol.48, 2020.

[105] Dynamic impact mechanical damage analysis and tomato robotic post-picking crating optimization based on multiscale finite element model, Guohang Lu, Zinuo Wang Georgios Papadakis, Nianzu Dai, Jin Yuan, Xuemei Liu, Vol.10, 2025.

[106] A comprehensive standardized dataset of numerous pomegranate fruit diseases for deep learning, Pakruddin B, Hemavathy R, Vol.54, 2024.

[107] Location of seed spoilage in mango fruit using X-ray imaging and convolutional neural networks, Francisca Aba Ansah, Mark Amo-Boateng, Ebenezer K. Siabi, Paa Kwesi Bordoh, Vol.20, 2023.

[108] A stereoscopic video computer vision system for weed discrimination in rice field under both natural and controlled light conditions by machine learning, Mojtaba Dadashzadeh Sajad Sabzi, Yousef Abbaspour-Gilandeh, Juan Ignacio Arribas, Tarahom Mesri- Gundo, Vol.237, 2024.

[109] Computer vision-based prototype robotic picking cum grading system for fruits, Meer Hannan Dairath M. Zuhaib Akram, M. Waqar Akram, M. Mubashar Omar, M. Ahmad Mehmood, M. Faheem, H. Umair Sarwar, Vol.4, 2023.

[110] Performance and Sealing Material Evaluation in 6-axis Force-Torque Sensors for Underwater Robotic, G Palli, L Morielloand, C Melchiorri, Vol.48, 2015.

[111] Development of an autonomous chess robot system using computer vision and deep learning, Truong Duc Phuc, Bui Cao Son, Vol.25, 2025.

[112] Enhancing Industrial IoT with Edge Computing and Computer Vision: An Analog Gauge Visual Digitization Approach, Michail Katsigiannisa, Konstantinos Mykoniatis, Vol.41, 2024.

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[115] Location of seed spoilage in mango fruit using X-ray imaging and convolutional neural networks, Francisca Aba Ansah, Mark Amo-Boateng, Ebenezer K. Siabi, Paa Kwesi Bordoh, Vol.20, 2023.

[116] A comprehensive standardized dataset of numerous pomegranate fruit diseases for deep learning, Pakruddin B, Hemavathy R, Vol.54, 2024.

[117] Dynamic impact mechanical damage analysis and tomato robotic post-picking crating optimization based on multiscale finite element model, Guohang Lu, Zinuo Wang Georgios Papadakis, Nianzu Dai, Jin Yuan, Xuemei Liu, Vol.10, 2025.

[118] A stereoscopic video computer vision system for weed discrimination in rice field under both natural and controlled light conditions by machine learning, Mojtaba Dadashzadeh Sajad Sabzi, Yousef Abbaspour-Gilandeh, Juan Ignacio Arribas, Tarahom Mesri- Gundosh, Vol.237, 2024.

[119] Computer vision-based prototype robotic picking cum grading system for fruits, Meer Hannan Dairath M. Zuhaib Akram, M. Waqar Akram, M. Mubashar Omar, M. Ahmad Mehmood, M. Faheem, H. Umair Sarwar, Vol.4, 2023.

[120] https://agritech.tnau.ac.in/bio-tech/biotech_arecanut

[121] https://plantskingdom.in/

[122] https://robocraze.com/

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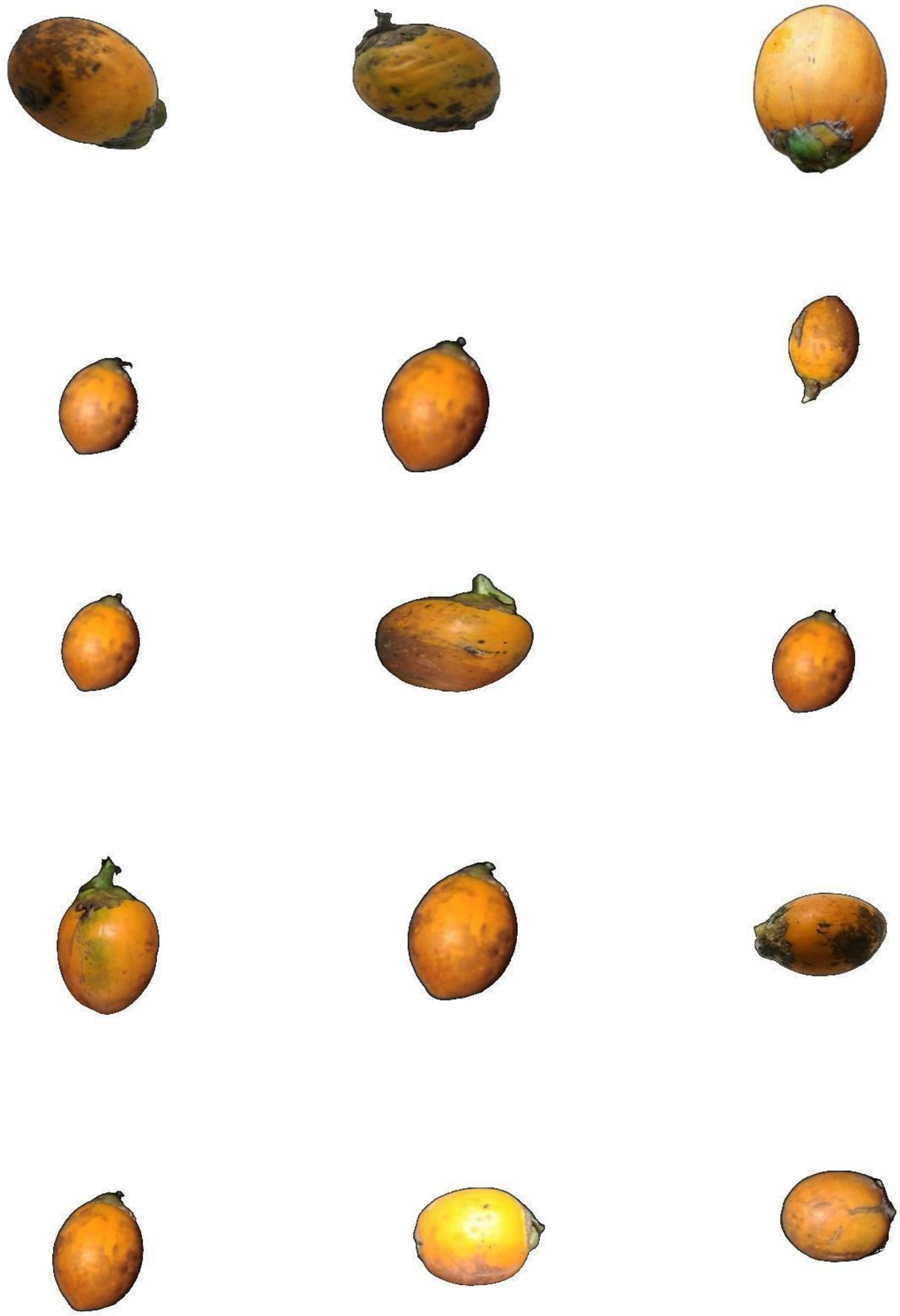

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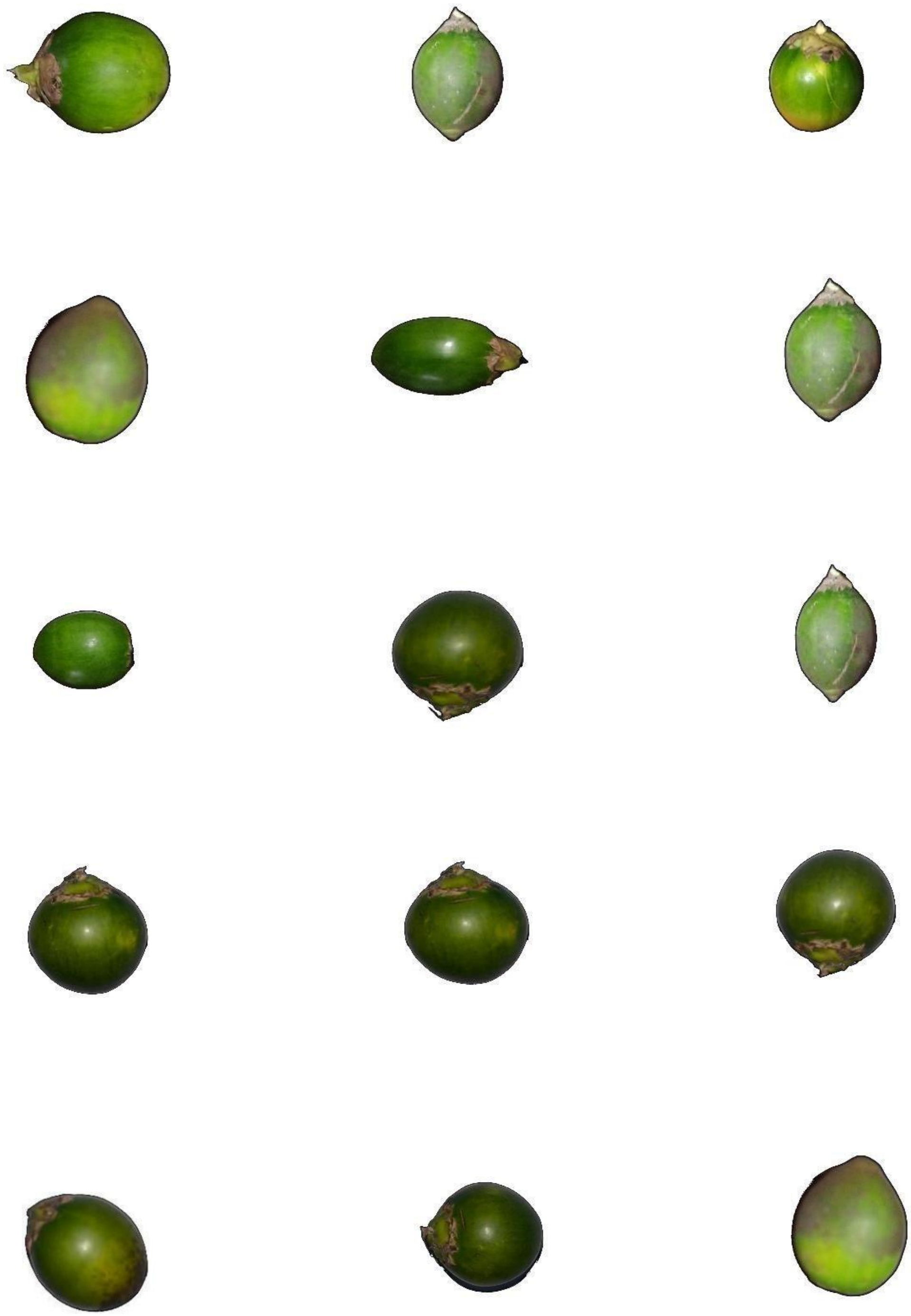

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DR PRASHANTH N	6429, NITTE MEENAKSHI COLLEGE RD, BSF CAMPUS, YELAHANKA, BENGALURU, KARNATAKA 560064-560064	08/08/2025

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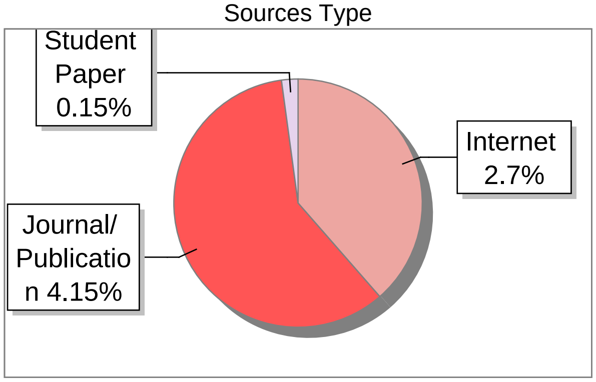

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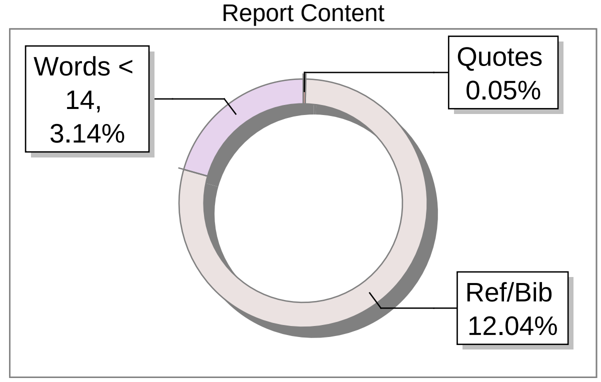

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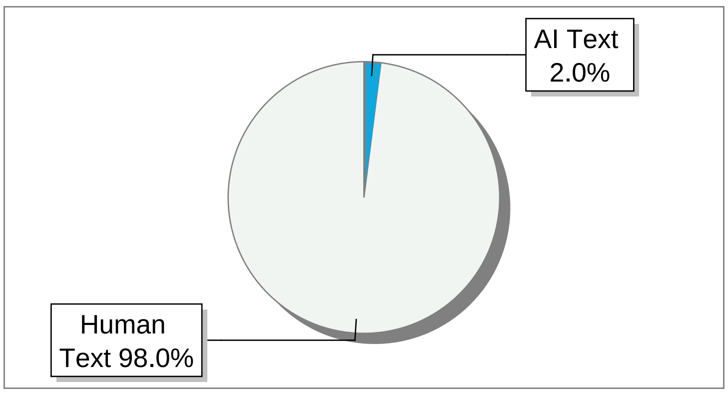

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